

Advanced Calibration Methods and VIX Derivatives

Julien Guyon

École nationale des ponts et chaussées
Institut polytechnique de Paris
NYU

**Joint lecture of the BNP Paribas Chair
“Futures of Quantitative Finance”**

BNP Paribas - Ecole des Ponts - LPSM (Univ. Paris Cité, Sorbonne Univ.)

February-March 2026

julien.guyon@enpc.fr, julien.guyon@nyu.edu

Contents

- 1 The particle method for smile calibration
- 2 The multi-asset case: calibrating local correlation models
- 3 VIX and VIX derivatives
- 4 Robust hedging of derivatives: martingale optimal transport
- 5 Extension to VIX: dispersion-constrained martingale optimal transport
- 6 Exact joint calibration of SPX and VIX smiles via minimum entropy: dispersion-constrained martingale Schrödinger problems

Contents

- 1 The particle method for smile calibration
- 2 The multi-asset case: calibrating local correlation models
- 3 VIX and VIX derivatives
- 4 Robust hedging of derivatives: martingale optimal transport
- 5 Extension to VIX: dispersion-constrained martingale optimal transport
- 6 Exact joint calibration of SPX and VIX smiles via minimum entropy: dispersion-constrained martingale Schrödinger problems

Nonlinear Stochastic Differential Equations in the sense of McKean

Traditional SDEs

- Traditional Stochastic Differential Equation:

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dW_t \quad (1.1)$$

- The pdf $p(t, \cdot)$ of X_t is solution to the Fokker-Planck PDE:

$$-\partial_t p - \sum_{i=1}^n \partial_i \left(b^i(t, x) p(t, x) \right) + \frac{1}{2} \sum_{i,j=1}^n \partial_{ij} \left(\sum_{k=1}^d \sigma_k^i(t, x) \sigma_k^j(t, x) p(t, x) \right) = 0$$

$$\lim_{t \rightarrow 0} p(t, x) = \delta(x - x_0)$$

- The PDE for $p(t, \cdot)$ is **linear**, so we sometimes speak of **linear** SDEs
- Uniqueness and existence hold if drift and volatility are Lipschitz or locally Lipschitz

Calibration of SLV models to market smiles

- Stochastic Local Volatility (SLV) model (deterministic rate and div yield):

$$\frac{dS_t}{S_t} = (r_t - q_t)dt + \sigma(t, S_t)a_t dW_t$$

- The model is exactly calibrated to market smile iff for all t, K

$$\sigma(t, K)^2 \mathbb{E}[a_t^2 | S_t = K] = \sigma_{\text{loc}}(t, K)^2$$

- Calibrated model follows a **nonlinear** SDE in the sense of McKean:

$$\frac{dS_t}{S_t} = (r_t - q_t)dt + \frac{\sigma_{\text{loc}}(t, S_t)}{\sqrt{\mathbb{E}[a_t^2 | S_t]}} a_t dW_t \quad (*)$$

- The local volatility function depends on the joint pdf $p(t, S, a)$ of (S_t, a_t) :

$$\sigma(t, S, \textcolor{red}{p}) = \sigma_{\text{loc}}(t, S) \sqrt{\frac{\int p(t, S, a') da'}{\int a'^2 p(t, S, a') da'}}$$

- Equation (*) is an example of **McKean SDE**.

McKean SDEs (Henry McKean, 1966)

- McKean SDE: $X_t \in \mathbb{R}^n$

$$dX_t = b(t, X_t, \mathbb{P}_t) dt + \sigma(t, X_t, \mathbb{P}_t) dW_t, \quad \mathbb{P}_t = \text{Law}(X_t) \quad (1.2)$$

- One-factor SLV: $n = 2$, $X_t = (S_t, a_t)$
- The pdf $p(t, \cdot)$ of X_t is solution to the Fokker-Planck PDE:

$$-\partial_t p - \sum_{i=1}^n \partial_i (b^i(t, x, \mathbb{P}_t) p(t, x)) + \frac{1}{2} \sum_{i,j=1}^n \partial_{ij} \left(\sum_{k=1}^d \sigma_k^i(t, x, \mathbb{P}_t) \sigma_k^j(t, x, \mathbb{P}_t) p(t, x) \right) = 0$$

$$\lim_{t \rightarrow 0} p(t, x) = \delta(x - X_0)$$

- It is **nonlinear** because $b^i(t, x, \mathbb{P}_t)$ and $\sigma_k^i(t, x, \mathbb{P}_t)$ depend on the unknown $p \Rightarrow$ We speak of **nonlinear** SDEs
- Uniqueness and existence can be proved if drift and volatility coefficients are Lipschitz-continuous functions of $\mathbb{P}_t$, with respect to the Wasserstein metric (see Sznitman, Méléard, Villani)

McKean SDEs (Henry McKean, 1966)

- McKean SDE: $X_t \in \mathbb{R}^n$

$$dX_t = b(t, X_t, \mathbb{P}_t) dt + \sigma(t, X_t, \mathbb{P}_t) \cdot dW_t, \quad \mathbb{P}_t = \text{Law}(X_t)$$

- One-factor SLV: $n = 2$, $X_t = (S_t, a_t)$
- The pdf $p(t, \cdot)$ of X_t is solution to the Fokker-Planck PDE:

$$-\partial_t p - \sum_{i=1}^n \partial_i (b^i(t, x, \mathbf{p}(t, x)) p(t, x)) + \frac{1}{2} \sum_{i,j=1}^n \partial_{ij} \left(\sum_{k=1}^d \sigma_k^i(t, x, \mathbf{p}(t, x)) \sigma_k^j(t, x, \mathbf{p}(t, x)) p(t, x) \right) = 0 \quad (1.3)$$

$$\lim_{t \rightarrow 0} p(t, x) = \delta(x - X_0)$$

- It is **nonlinear** because $b^i(t, x, \mathbb{P}_t)$ and $\sigma_k^i(t, x, \mathbb{P}_t)$ depend on the unknown $p \Rightarrow$ We speak of **nonlinear** SDEs
- Uniqueness and existence can be proved if drift and volatility coefficients are Lipschitz-continuous functions of $\mathbb{P}_t$, with respect to the Wasserstein metric (see Sznitman, Méléard, Villani)

Wasserstein Metric

The Monge-Kantorovich Distance (or Wasserstein metric) between two probability measures $\mathbb{P}_1$ and $\mathbb{P}_2$ is defined as:

$$d_{\text{MK}}(\mathbb{P}_1, \mathbb{P}_2)^p = \inf_{\tau \in \mathcal{P}(\mathbb{R}^n, \mathbb{R}^n) \text{ with marginals } \mathbb{P}_1 \text{ and } \mathbb{P}_2} \left(\int_{\mathbb{R}^n \times \mathbb{R}^n} d(x, y)^p \tau(dx, dy) \right)$$

Optimal transport interpretation

Financial interpretation ($n = 1$):

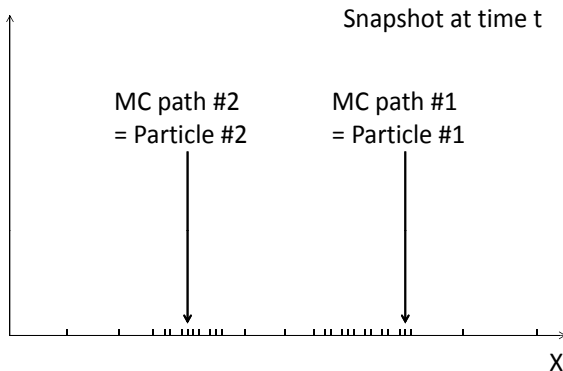
- Consider a European option on two assets $X = S_T^1$ and $Y = S_T^2$ paying $d(X, Y)^p$
- A model τ is calibrated to the vanilla options on S_T^1 and S_T^2 if and only if the marginal distributions of τ are the risk-neutral distributions $\mathbb{P}_1$ and $\mathbb{P}_2$
- The pricing model means choosing the joint distribution τ that minimizes the fair value of the option:

$$\mathbb{E}^\tau[d(X, Y)^p]$$

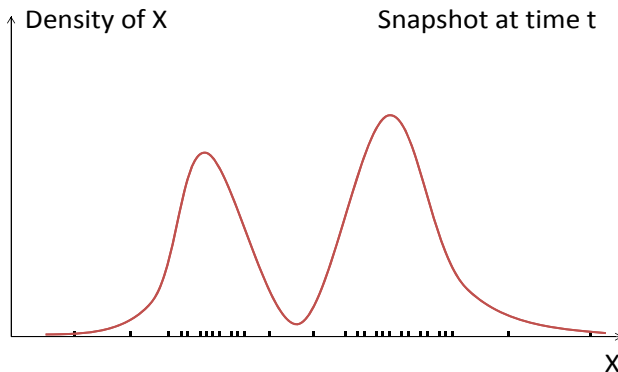
- This is precisely the Monge-Kantorovich distance between $\mathbb{P}_1$ and $\mathbb{P}_2$

How to simulate a McKean SDE?

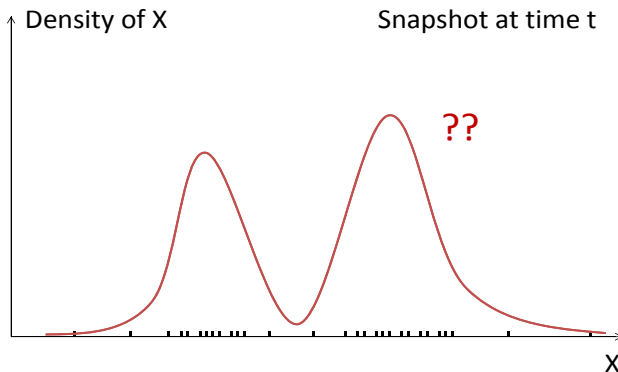
$$dX_t = b(t, X_t, \mathbb{P}_t) dt + \sigma(t, X_t, \mathbb{P}_t) \cdot dW_t, \quad \mathbb{P}_t = \text{Law}(X_t)$$



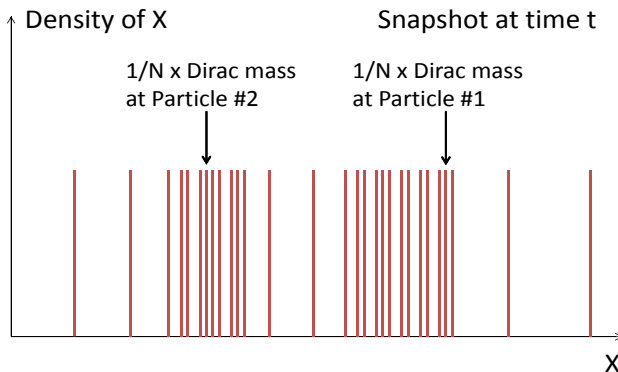
How to simulate a McKean SDE?



How to simulate a McKean SDE?



How to simulate a McKean SDE?



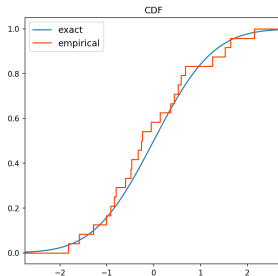
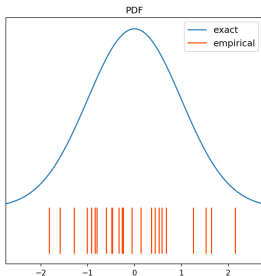
Empirical Measure

Definition (Empirical measure)

Let $X_1, \dots, X_N$ be i.i.d. random variables with law μ . The empirical measure associated to the configuration $(X_1, \dots, X_N)$ is

$$\hat{\mu}^N = \frac{1}{N} \sum_{i=1}^N \delta_{X_i}, \quad \text{or} \quad \hat{\mu}^N(A) = \frac{1}{N} \sum_{i=1}^N \mathbf{1}_A(X_i)$$

Note that the empirical measure $\hat{\mu}^N$ is a random probability measure with expectation $\mathbb{E}_\mu[\hat{\mu}^N] = \mu$, meaning that for all events A , $\mathbb{E}_\mu[\hat{\mu}^N(A)] = \mu(A)$.



Particle Method

- Principle: replace the law $\mathbb{P}_t$ by the empirical distribution

$$\mathbb{P}_t^N = \frac{1}{N} \sum_{i=1}^N \delta_{X_t^{i,N}}$$

where the $X_t^{i,N}$ are solution to the $(\mathbb{R}^n)^N$ -dimensional **classical** SDE

$$dX_t^{i,N} = b\left(t, X_t^{i,N}, \mathbb{P}_t^N\right) dt + \sigma\left(t, X_t^{i,N}, \mathbb{P}_t^N\right) dW_t^i, \quad \text{Law}\left(X_0^{i,N}\right) = \mathbb{P}_0$$

- $\{X_t^i\}_{1 \leq i \leq N}$ = system of N **interacting particles**. The interaction with the other $N - 1$ particles comes from $\mathbb{P}_t^N$
- Propagation of chaos (Sznitman, Méléard, Villani) implies the convergence of the particle method:

$$\frac{1}{N} \sum_{i=1}^N f(X_t^{i,N}) \xrightarrow[N \rightarrow \infty]{L^1} \int_{\mathbb{R}^d} f(x) p(t, x) dx$$

- In the large N limit, the $(\mathbb{R}^n)^N$ -dimensional **linear** Fokker-Planck PDE approximates the **nonlinear** low-dimensional (n -dimensional) Fokker-Planck PDE

An archetypal example: the McKean-Vlasov SDE

- For $1 \leq k \leq n$ and $1 \leq l \leq d$

$$\begin{aligned} b_k(t, x, \mathbb{P}_t) &= \int b_k(t, x, y) \mathbb{P}_t(dy) = \mathbb{E}[b_k(t, x, X_t)] \\ \sigma_{kl}(t, x, \mathbb{P}_t) &= \int \sigma_{kl}(t, x, y) \mathbb{P}_t(dy) = \mathbb{E}[\sigma_{kl}(t, x, X_t)] \end{aligned} \quad (1.4)$$

- Particle method:

$$dX_t^{i,N} = \left(\int b(t, X_t^{i,N}, y) d\mathbb{P}_t^N(y) \right) dt + \left(\int \sigma(t, X_t^{i,N}, y) d\mathbb{P}_t^N(y) \right) dW_t^i$$

- This is equivalent to

$$dX_t^{i,N} = \frac{1}{N} \sum_{j=1}^N b\left(t, X_t^{i,N}, \mathbf{X}_t^{j,N}\right) dt + \frac{1}{N} \sum_{j=1}^N \sigma\left(t, X_t^{i,N}, \mathbf{X}_t^{j,N}\right) dW_t^i \quad (1.5)$$

Particle Method

A CLASS OF MARKOV PROCESSES ASSOCIATED WITH NONLINEAR PARABOLIC EQUATIONS

BY H. P. McKEAN, JR.*

MASSACHUSETTS INSTITUTE OF TECHNOLOGY AND THE ROCKEFELLER UNIVERSITY

Communicated by Mark Kac, August 31, 1966

Introduction.—The familiar connection between the Brownian motion and the differential operator $f \rightarrow f''/2$, based upon the fact that the Brownian transition function $(2\pi t)^{-1/2} \exp[-(b-a)^2/2t]$ is also the elementary solution of $\partial u/\partial t = (1/2)\partial^2 u/\partial a^2$, is the simplest and most fruitful instance of the connection between Markov processes with constant transition mechanism and linear parabolic equations. The purpose of this paper is to explain a similar connection between a wider class of Markov processes and certain *nonlinear* parabolic equations. Boltzmann's equation from statistical mechanics is a special case, as is Burgers' equation: $\partial u/\partial t = (1/2)\partial^2 u/\partial a^2 - u\partial u/\partial a$.

1. *Markov Processes with Constant Transition Mechanism.*—Given a nice topological space Q , let $x: [0, \infty) \rightarrow Q$ be a *sample path* with coordinates $x(t) = x(t+) \in Q (t \geq 0)$, let Z be the associated *universal field* measuring $x(t)$ for each $t \geq 0$, and to each $a \in Q$ attach probabilities $P_a(B)$, defined for $B \in Z$, such that

(a) for each $B \in Z$, $P_a(B)$ is a (topological) Borel function of a ,

(b) for each $a \in Q$, $P_a[x(0) \in db]$ is the unit mass at $b = a$.



Particle Method

1908

MATHEMATICS: H. P. MCKEAN, JR.

PROC. N. A. S.

Markovian in the sense that for any $t \geq s \geq 0$ and $B \in \mathcal{Z}$, $P_f[x_t^+ \in B|x(s): s \leq t]$ depends upon $a = x(t)$ alone, but do not suppose that it coincides with $P_a(B)$ as in §1; (3) instead of that, let

$$P_f[x_t^+ \in B|x(s): s \leq t] = P_u[B|x(0) = a]$$

with $a = x(t)$ and $u = P_f[x(t) \in db]$.

According to the customary nomenclature, $M = [Q, Z, P]$ is a Markov process with nonconstant transition mechanism, but of a very special class. Roughly speaking, the transition mechanism depends upon t via the distribution $u = P_f[x(t) \in db]$, only. The map $f \rightarrow u$ is still multiplicative, but now the infinitesimal operator $D^*f = \lim_{t \downarrow 0} t^{-1}[u - f]$ can be nonadditive, as will be illustrated below in actual examples. u is still a solution of $\partial u / \partial t = D^*u$, but the exponential formula $u = e(tD^*)f$ is not correct any more; a replacement will be indicated in §5.

3. *A Simple Example: 2-State Maxwellian Gas.*—Consider the space $Q: \pm 1$, take independent e_1, e_2 , etc. $= \pm 1$ with common distribution f , and let $n = n(t)$ be the population of an independent standard branching process with $n(0) = 1$ and distribution $P_f[n(t) = j] = e^{-t}(1 - e^{-t})^{j-1}$ ($j \geq 1$), subject to the rule that, conditional on $n(s): s \leq t_1$, $n(t_1 + t_2)$ should be identical in law to the sum of $n(t_1)$ independent copies of $n(t_2)$. Define $P_f(B)$ to be the probabilities induced on the space of sample paths $x: [0, \infty) \rightarrow Q$ by the recipe $x(t) = e_1 e_2 \dots e_n$ ($t \geq 0$).

$M = [Q, Z, P]$ satisfies all the conditions of §2, and after a little computation, it develops that

$$\partial u / \partial t = u(-1)^2 + u(+1)^2 - u(+1) - u(-1), \quad e_s = \pm 1$$

Particle Method

1910

MATHEMATICS: H. P. McKEAN, JR.

PROC. N. A. S.

$$D_n = \sum_{i \leq n} D(i) + \binom{n}{1}^{-1} \sum_{i < j \leq n} D(ij) + \binom{n}{2}^{-1} \sum_{i < j < k \leq n} D(ijk) + \dots + D(12\dots n).$$

[A *backward* operator D is the dual of a *forward* infinitesimal operator D^* as described in §1.] $D(1)$ is an (additive) 1-molecule backward infinitesimal operator acting on the first coordinate of Q^n . $D(i)$ is a copy of $D(1)$ acting on the i th coordinate ($i \leq n$). $D(12) = D(21)$ is a 2-molecule operator acting on the pair of coordinates 12. $D(ij)$ is a copy of $D(12)$ acting on the pair ij ($i < j \leq n$), etc. $D(1)$ governs the motion of molecule number 1 in isolation, $D(12)$ governs the interaction (double collisions) between molecules 1 and 2, $D(123)$ the (triple) collisions between molecules 1, 2, 3, etc. D_n^* commutes with permutations of the coordinates so that if the initial n -molecule distribution f^n is symmetrical, then so is $v = e(tD_n^*)f^n$ at any later time.

Given a 1-molecule distribution function f , let f^n be the outer product $f \otimes \dots \otimes f$ so that the molecules are independent at time $t = 0$ with common distribution f . A formal computation indicates that as $n \uparrow \infty$, $v = e(I D_n^*) f^n$ tends to the infinite outer product $u^\infty = u \otimes u \otimes \dots$, etc., of the 1-molecule distribution function $u = \lim_{n \uparrow \infty} P_n[x_1(t) \in db]$. Kac,⁴ describing the first known instance of this phenomenon, called it the *propagation of chaos*. u is the (formal) solution of $\partial u / \partial t = \text{D}_1 \cdot \text{D}_1 u$ and $u|_{t=0} = f$.

The Particle Method

- Convergence of the particle method is guaranteed when the propagation of chaos property holds:
- **Propagation of chaos:** If at $t = 0$, the $X_0^{i,N}$ are independent particles, then as $N \rightarrow \infty$, for any fixed $t > 0$, the $X_t^{i,N}$ are asymptotically independent and their empirical measure $\mathbb{P}_t^N$ converges in distribution toward the true measure $\mathbb{P}_t$.
- This means that, in the space of probabilities over the space of probabilities, the distribution of the random measure $\mathbb{P}_t^N$ converges toward a Dirac mass at the deterministic measure $\mathbb{P}_t$. Practically, it means that for all functions $\varphi \in C_b(\mathbb{R}^n)$

$$\frac{1}{N} \sum_{i=1}^N \varphi(X_t^{i,N}) \xrightarrow[N \rightarrow \infty]{L^1} \int_{\mathbb{R}^n} \varphi(x) p(t, x) dx$$

where $p(t, \cdot)$ is the fundamental solution to the nonlinear Fokker-Planck PDE (1.3). Hence, if propagation of chaos holds, the particle method is convergent.

μ -chaotic distributions

Definition (μ -chaotic distribution)

Let $\{\mu^N\}_{N \in \mathbb{N}}$ be a sequence of *symmetric* probabilities on $(\mathbb{R}^n)^N$. Let μ be a probability measure on $\mathbb{R}^n$. We say that μ^N is μ -chaotic if for each integer $k \geq 1$ and for all test functions $\varphi_1, \dots, \varphi_k \in C_b(\mathbb{R}^n)$ (i.e., for all bounded continuous functions), we have

$$\int \varphi_1(x_1) \cdots \varphi_k(x_k) d\mu^N(x_1, \dots, x_N) \xrightarrow{N \rightarrow \infty} \int \varphi_1 d\mu \cdots \int \varphi_k d\mu$$

Stated otherwise, k particles (within N) are **asymptotically independent** and **identically distributed with distribution μ** as $N \rightarrow \infty$ (k being fixed). We say that μ^N is chaotic if there exists μ such that μ^N is μ -chaotic.

Definition (Propagation of chaos)

Let us consider an N -dimensional SDE flow that associates to an initial probability measure μ_0^N a probability μ_t^N at time t . We say that this flow propagates the chaos if, for any initial chaotic measure μ_0^N and any $t > 0$, μ_t^N is chaotic.

Intuition

What the propagation of chaos essentially says is that as the number of particles becomes very large:

- A **fixed subset of particles** becomes independent as the sample increases
- The empirical measure converges to the law of a single particle
- It is related to the idea that in the large N -limit a linear Fokker-Planck PDE of high dimension approximates a non-linear low dimensional Fokker-Planck PDE

Some Theory

Theorem

Let $\{\mu^N\}_{N \in \mathbb{N}}$ be a sequence of symmetric probabilities on $(\mathbb{R}^n)^N$, and μ be a probability measure on $\mathbb{R}^n$. The following four properties are equivalent:

- (i) $\{\mu^N\}_{N \in \mathbb{N}}$ is μ -chaotic.
- (ii) For all test functions $\varphi_1, \varphi_2 \in C_b(\mathbb{R}^n)$:

$$\int \varphi_1(x_1) \varphi_2(x_2) d\mu^N(x_1, \dots, x_N) \xrightarrow{N \rightarrow \infty} \int \varphi_1 d\mu \int \varphi_2 d\mu$$

- (iii) Let $X_1, \dots, X_N$ be random variables such that $\text{Law}(X_1, \dots, X_N) = \mu^N$. Then, for all $\varphi \in C_b(\mathbb{R}^n)$,

$$\frac{1}{N} \sum_{i=1}^N \varphi(X_i) \xrightarrow[N \rightarrow \infty]{L^1} \int \varphi d\mu$$

- (iv) Let $\hat{\mu}^N$ be the empirical measure associated to μ^N . Then

$$\mathbb{E}_{\mu^N} \left[\left| \int \varphi d\hat{\mu}^N - \int \varphi d\mu \right| \right] \xrightarrow{N \rightarrow \infty} 0$$

for all $\varphi \in C_b(\mathbb{R}^n)$.

Some Theory

(i) $\implies$ (ii) follows from the definition with $k = 2$.

(ii) $\implies$ (iii). We have

$$\begin{aligned}
 & \mathbb{E}_{\mu^N} \left[\left(\frac{1}{N} \sum_{i=1}^N \varphi(X_i) - \int \varphi d\mu \right)^2 \right] \\
 = & \mathbb{E}_{\mu^N} \left[\frac{1}{N^2} \sum_{i,j=1}^N \varphi(X_i) \varphi(X_j) - 2 \int \varphi d\mu \frac{1}{N} \sum_{i=1}^N \varphi(X_i) + \left(\int \varphi d\mu \right)^2 \right] \\
 = & \frac{1}{N} \mathbb{E}_{\mu^N} [\varphi(X_1)^2] + \frac{N-1}{N} \mathbb{E}_{\mu^N} [\varphi(X_1) \varphi(X_2)] \\
 & - 2 \int \varphi d\mu \mathbb{E}_{\mu^N} [\varphi(X_1)] + \left(\int \varphi d\mu \right)^2 \\
 & \xrightarrow{N \rightarrow \infty} 0 + \left(\int \varphi d\mu \right)^2 - 2 \left(\int \varphi d\mu \right)^2 + \left(\int \varphi d\mu \right)^2 = 0
 \end{aligned}$$

as from (ii) $\mathbb{E}_{\mu^N} [\varphi(X_1) \varphi(X_2)] \xrightarrow{N \rightarrow \infty} \left(\int \varphi d\mu \right)^2$ and $\mathbb{E}_{\mu^N} [\varphi(X_1)] \xrightarrow{N \rightarrow \infty} \int \varphi d\mu$.

(iii) $\implies$ (iv) follows from the definition of convergence in mean.

Some Theory

(iv) $\implies$ (i). Let $k \geq 1$ and $\varphi_1, \dots, \varphi_k \in C_b(\mathbb{R}^n)$. We have (from the triangle inequality):

$$\begin{aligned} & \left| \mathbb{E}_{\mu^N} [\varphi_1(X_1) \cdots \varphi_k(X_k)] - \int \varphi_1 d\mu \cdots \int \varphi_k d\mu \right| \\ & \leq \left| \mathbb{E}_{\mu^N} [\varphi_1(X_1) \cdots \varphi_k(X_k)] - \mathbb{E}_{\mu^N} \left[\int \varphi_1 d\hat{\mu}^N \cdots \int \varphi_k d\hat{\mu}^N \right] \right| \\ & \quad + \left| \mathbb{E}_{\mu^N} \left[\int \varphi_1 d\hat{\mu}^N \cdots \int \varphi_k d\hat{\mu}^N \right] - \int \varphi_1 d\mu \cdots \int \varphi_k d\mu \right| \end{aligned}$$

The second term on the r.h.s. converges to zero thanks to hypothesis (iv). The first term on the r.h.s. reads

$$\left| \mathbb{E}_{\mu^N} [\varphi_1(X_1) \cdots \varphi_k(X_k)] - \frac{1}{N^k} \sum_{i_1, \dots, i_k=1}^N \mathbb{E}_{\mu^N} [\varphi_1(X_{i_1}) \cdots \varphi_k(X_{i_k})] \right|$$

In the above sum, $\frac{N!}{(N-k)!}$ terms are such that the indices $i_1, \dots, i_k$ are all different. By symmetry, they are equal to $\mathbb{E}_{\mu^N} [\varphi_1(X_1) \cdots \varphi_k(X_k)]$ and can be added to the first term. The other terms can be bounded by M^k with $M = \sum_j \|\varphi_j\|_\infty$, so the first term is bounded by

$$\begin{aligned} & \mathbb{E}_{\mu^N} [\varphi_1(X_1) \cdots \varphi_k(X_k)] \left(1 - \frac{1}{N^k} \frac{N!}{(N-k)!} \right) + \left(1 - \frac{1}{N^k} \frac{N!}{(N-k)!} \right) M^k \\ & \leq 2M^k \left(1 - \frac{1}{N^k} \frac{N!}{(N-k)!} \right) \xrightarrow{N \rightarrow \infty} 0 \end{aligned}$$

Propagation of Chaos - McKean-Vlasov SDE

Theorem

The propagation of chaos property holds for the McKean-Vlasov SDE (1.5)

The intuition of the proof relies on a sort of “coupling method”, which consists of introducing a process Y_t^m , $1 \leq m \leq N$, defined as:

$$dY_t^m = b(Y_t^m, \mathbb{P}_t) dt + \sigma(Y_t^m, \mathbb{P}_t) dW_t^m, \quad Y_0^m = X_0$$

with $b(y, \mathbb{P}_t) := \int b(y, z) \mathbb{P}_t(dz)$, $\sigma(y, \mathbb{P}_t) := \int \sigma(y, z) \mathbb{P}_t(dz)$, and $\mathbb{P}_t := \text{Law}(X_t)$. These are standard SDEs which admit a strong solution as b and σ are Lipschitz-continuous functions. The density $q_m(t, x)$ of Y_t^m satisfies the *linear* Fokker-Planck equation

$$\begin{aligned} - \partial_t q_m(t, x) - \sum_{i=1}^n \partial_i (b^i(t, x, \mathbb{P}_t) q_m(t, x)) \\ + \frac{1}{2} \sum_{i,j=1}^n \partial_{ij} \left(\sum_{k=1}^d \sigma_k^i(t, x, \mathbb{P}_t) \sigma_k^j(t, x, \mathbb{P}_t) q_m(t, x) \right) = 0 \end{aligned}$$

with Dirac initial condition. From (1.3), the density $p(t, x)$ of X_t is also a solution. By uniqueness, $\mathbb{P}_t = \text{Law}(Y_t^m)$.

Propagation of Chaos - McKean-Vlasov SDE

Proposition

Let $(X^{i,N})_{1 \leq i \leq N}$ be defined by the particle method. Then

$$\mathbb{E}[|X_t^{1,N} - Y_t^1|] \leq \frac{C(t)}{\sqrt{N}}$$

where $C(t)$ is a smooth function of time independent of N .

Proof of Chaos Propagation Theorem for McKean-Vlasov SDE

Let us denote by μ_t the law of the solution X_t of the McKean SDE (1.4). Then from Theorem 4 (iii), it is enough to show that for all $\varphi \in C_b(\mathbb{R}^n)$,

$$\frac{1}{N} \sum_{i=1}^N \varphi(X_t^{i,N}) \xrightarrow[N \rightarrow \infty]{L^1} \int \varphi d\mu_t \quad (1.6)$$

where the $X_t^{i,N}$ are defined by (1.5).

Propagation of Chaos - McKean-Vlasov SDE

$$\begin{aligned} & \mathbb{E}_{\mu^N} \left[\left| \frac{1}{N} \sum_{i=1}^N \varphi(X_t^{i,N}) - \int \varphi d\mu_t \right| \right] \\ & \leq \mathbb{E}_{\mu^N} \left[\left| \frac{1}{N} \sum_{i=1}^N \left(\varphi(X_t^{i,N}) - \varphi(Y_t^i) \right) \right| \right] + \mathbb{E}_{\mu^N} \left[\left| \frac{1}{N} \sum_{i=1}^N \varphi(Y_t^i) - \int \varphi d\mu_t \right| \right] \end{aligned}$$

As by construction the processes $\{Y_t^i\}_{1 \leq i \leq N}$ are i.i.d. with law μ_t , the second term above goes to zero as $N \rightarrow \infty$ from the law of large numbers.

For all $\epsilon > 0$, there exists a Lipschitz-continuous function φ_ϵ such that $|\varphi - \varphi_\epsilon| \leq \epsilon$. The first term is then bounded by

$$\begin{aligned} & 2\epsilon + \mathbb{E}_{\mu^N} \left[\left| \frac{1}{N} \sum_{i=1}^N \left(\varphi_\epsilon(X_t^{i,N}) - \varphi_\epsilon(Y_t^i) \right) \right| \right] \\ & \leq 2\epsilon + \|\varphi_\epsilon\|_{\text{Lip}} \mathbb{E}[\|X_t^{1,N} - Y_t^1\|] \leq 2\epsilon + \|\varphi_\epsilon\|_{\text{Lip}} \frac{C(t)}{\sqrt{N}} \end{aligned}$$

from the proposition above.

Calibration of models to market smiles

The tools and concepts that will be used: A bit of history

- **Concept of propagation of chaos:** Kac, 1960's
- **Nonlinear stochastic differential equations:** McKean, 1966
A Class of Markov Processes Associated with Nonlinear Parabolic Equations (Proc Natl Acad Sci U S A. 1966 December; 56(6): 1907–1911)
- **Particle method and propagation of chaos:** Sznitman, Méléard, Chorin, Puckett, Talay, Bossy, Jourdain, etc., 1980's and 1990's
- **Application to various smile calibration problems in quantitative finance:** G. and Henry-Labordère, 2011
Being Particular About Calibration (Risk, January 2012)

Local Volatility Model (LVM)

- Local Volatility Model (deterministic rate and div yield):

$$\frac{dS_t}{S_t} = (r_t - q_t) dt + \sigma_{\text{loc}}(t, S_t) dW_t$$

where r_t and q_t are deterministic, and $\sigma_{\text{loc}}(t, S)$ is a function of t and S .

- Local volatility function from call option prices (*Dupire's Formula*)

$$\sigma_{\text{loc}}^2(T, K) = \frac{\frac{\partial C}{\partial T} + (r_t - q_t) K \frac{\partial C}{\partial K} + q_t C}{\frac{1}{2} K^2 \frac{\partial^2 C}{\partial K^2}}.$$

where

$$C(T, K) = \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_0^T r_t dt} (S_T - K)^+ \right]$$

is the price of the call option on S with maturity T and strike K .

$$\sigma_{\text{loc}}^2(T, K) := \frac{\frac{\partial C^{\text{mkt}}}{\partial T} + (r_t - q_t) K \frac{\partial C^{\text{mkt}}}{\partial K} + q_t C^{\text{mkt}}}{\frac{1}{2} K^2 \frac{\partial^2 C^{\text{mkt}}}{\partial K^2}}.$$

Local Volatility Model (LVM)

■ From Implied Volatility to Local Volatility:

$$\sigma_{\text{loc}}^2(t, x) = \frac{\frac{\partial w}{\partial T}}{1 - \frac{x}{w} \frac{\partial w}{\partial x} + \frac{1}{4} \left(-\frac{1}{4} - \frac{1}{w} + \frac{x^2}{w^2} \right) \left(\frac{\partial w}{\partial x} \right)^2 + \frac{1}{2} \frac{\partial^2 w}{\partial x^2}}$$

where log-moneyness $x = \log(K/F_T)$, $F_T = S_0 e^{\int_0^T (r_t - q_t) dt}$ is the forward with maturity T , and $w(T, x) = \sigma_{\text{imp}}^2(K, T)T$.

■ Stochastic Volatility model

$$\frac{dS_t}{S_t} = (r_t - q_t) dt + a_t dW_t$$

(where a_t is stochastic) is calibrated to the market smile if and only if

$$\mathbb{E} [a_t^2 | S_t = K] = \sigma_{\text{loc}}^2(t, K).$$

(Proof will be given later)

Stochastic Local Volatility (SLV) models

- Exotic options bear exotic risks such as volatility-of-volatility risk, forward smile risk, or spot/volatility correlation risk.
- To price and hedge these risks, one should not use the Black-Scholes model, or the local volatility model, because such models give no control on them.
- One would rather use a Stochastic Volatility (SV) model

$$\frac{dS_t}{S_t} = (r_t - q_t) dt + a_t dW_t \quad (1.7)$$

with a_t the stochastic volatility, a random process (e.g., Heston, SABR, double-lognormal) which allows us to handle exotic risks through parameters such as volatility-of-volatility, spot/volatility correlation, or mean-reversion.

Heston Stochastic Volatility Model - Example of SV Model

$$\frac{dS_t}{S_t} = (r_t - q_t) dt + \sqrt{V_t} dW_t^S$$
$$dV_t = \kappa(\theta - V_t) dt + \omega\sqrt{V_t} dW_t^V$$

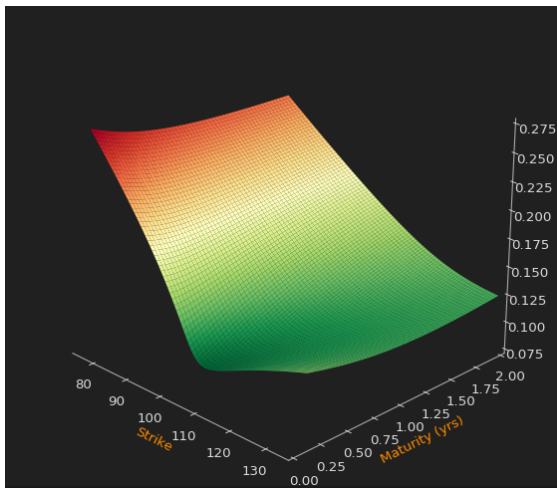
with

$$d\langle W^S, W^V \rangle_t = \rho dt.$$

Model Parameters

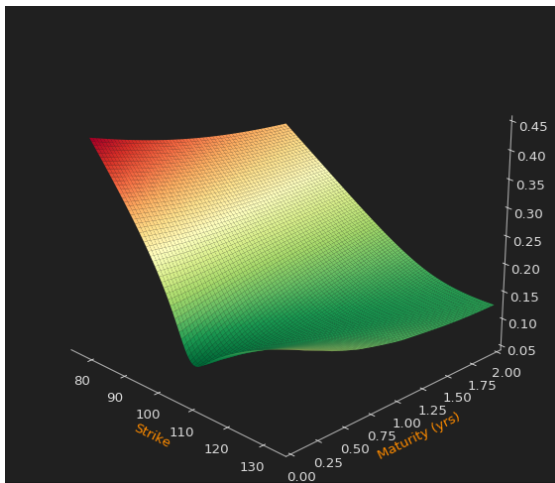
- κ : mean reversion rate
- V_0 : initial variance
- θ : Long term variance
- ρ : spot-vol correlation
- ω : volatility of variance

Heston Stochastic Volatility Model - Implied Volatility Surface



$$V_0 = 0.0165, \kappa = 1.3093, \theta = 0.0541, \omega = 0.6344, \rho = -0.744, S_0 = 100$$

Heston Stochastic Volatility Model - Local Volatility Surface



$$V_0 = 0.0165, \kappa = 1.3093, \theta = 0.0541, \omega = 0.6344, \rho = -0.744, S_0 = 100$$

Stochastic Local Volatility (SLV) models

- SV models produce a smile of implied volatilities (see the seminal papers by Hull-White, Renault-Touzi; see also Bergomi-Guyon for an analysis of the smile in general multi-factor second-generation SV models).
- Since they have a finite number of parameters (typically, 3 to 10), SV models cannot be perfectly calibrated to full (inter- and extrapolated) market smiles, indexed by all strikes and maturities (until some final maturity T).
- Now, vanilla options often provide a good hedge of exotic options. Dynamic or static trading of vanilla options often results in reducing the variance of the final P&L. In those cases, it is important that the SV model incorporates the correct initial prices of the hedging instruments—the vanilla options.

Stochastic Local Volatility (SLV) models

- How do we build an SV model that calibrates exactly to a full surface of implied volatilities?
- Due to the double infinity of constraints, indexed by strikes and maturities, one needs to introduce a double infinity of parameters.
- A natural way to do so, and a common practice in the foreign exchange market, is to embed a local volatility $\sigma(t, S)$ (also called “leverage function”) into the SV model:

$$\frac{dS_t}{S_t} = (r_t - q_t) dt + a_t \sigma(t, S_t) dW_t \quad (1.8)$$

We speak of **stochastic local volatility (SLV) models**.

- The main issue we address in these lecture notes is how to build the local volatility $\sigma(t, S)$ to ensure that the market smile is exactly calibrated.
- Note that this local volatility function differs from Dupire’s local volatility. For instance, if the smile produced by the “naked” SV model (1.7) is close to the market smile, one expects the calibrated local volatility $\sigma(t, S)$ to be uniformly close to 1.

Calibration of SLV models to market smiles

- SLV model (deterministic rate and div yield):

$$\frac{dS_t}{S_t} = (r_t - q_t)dt + \sigma(t, S_t)a_t dW_t$$

- Model exactly calibrated to market smile iff for all t, K

$$\sigma(t, K)^2 \mathbb{E}[a_t^2 | S_t = K] = \sigma_{\text{loc}}(t, K)^2$$

- $\Rightarrow$ **Nonlinear SDE:**

$$\frac{dS_t}{S_t} = (r_t - q_t)dt + \frac{\sigma_{\text{loc}}(t, S_t)}{\sqrt{\mathbb{E}[a_t^2 | S_t]}} a_t dW_t \quad (1.9)$$

- The local volatility function depends on the joint pdf $p(t, S, a)$ of (S_t, a_t) :

$$\sigma(t, S, \textcolor{red}{p}) = \sigma_{\text{loc}}(t, S) \sqrt{\frac{\int p(t, S, a') da'}{\int a'^2 p(t, S, a') da'}}$$

- Equation (1.9) is an example of **McKean SDE**

The calibration condition

Proposition. Let us consider the following dynamics for an asset S , where the volatility α_t , the interest rate r_t , and the repo q_t , inclusive of the dividend yield, are all stochastic processes:

$$\frac{dS_t}{S_t} = (r_t - q_t) dt + \alpha_t dW_t \quad (1.10)$$

Model (1.10) is exactly calibrated to the market smile of S if and only if

$$\begin{aligned} \frac{\mathbb{E}[D_{0t}\alpha_t^2|S_t = K]}{\mathbb{E}[D_{0t}|S_t = K]} &= \sigma_{\text{loc}}(t, K)^2 \\ &- \frac{\mathbb{E}[D_{0t}(r_t - q_t - (r_t^0 - q_t^0)) \mathbf{1}_{S_t > K}]}{\frac{1}{2}K\partial_K^2\mathcal{C}(t, K)} + \frac{\mathbb{E}[D_{0t}(q_t - q_t^0)(S_t - K)^+]}{\frac{1}{2}K^2\partial_K^2\mathcal{C}(t, K)} \end{aligned} \quad (1.11)$$

for all (t, K) , where $D_{0t} = \exp\left(-\int_0^t r_s ds\right)$ is the discount factor, r_t^0 and q_t^0 are deterministic rates and repos, and

$$\sigma_{\text{loc}}(t, K)^2 = \frac{\partial_t \mathcal{C}(t, K) + (r_t^0 - q_t^0)K\partial_K \mathcal{C}(t, K) + q_t^0 \mathcal{C}(t, K)}{\frac{1}{2}K^2\partial_K^2 \mathcal{C}(t, K)} \quad (1.12)$$

with $\mathcal{C}(t, K)$ the market price of the call option on S with strike K , mat. t .

Case where rates and repo are deterministic

Proposition. In the case where rates r_t and repo/dividend yield q_t are deterministic, Model (1.10) is exactly calibrated to the market smile of S if and only if

$$\mathbb{E}[\alpha_t^2 | S_t = K] = \sigma_{\text{loc}}(t, K)^2$$

for all (t, K) , where

$$\sigma_{\text{loc}}(t, K)^2 = \frac{\partial_t \mathcal{C}(t, K) + (r_t - q_t)K \partial_K \mathcal{C}(t, K) + q_t \mathcal{C}(t, K)}{\frac{1}{2} K^2 \partial_K^2 \mathcal{C}(t, K)} \quad (1.13)$$

with $\mathcal{C}(t, K)$ the market price of the call option on S with strike K and maturity t .

When $\alpha_t = \sigma(t, S_t) a_t$ the calibration condition reads

$$\sigma(t, K)^2 \mathbb{E}[a_t^2 | S_t = K] = \sigma_{\text{loc}}(t, K)^2$$

Proof of the general proposition

By applying Itô-Tanaka's formula on a discounted vanilla call payoff with maturity t and strike K , $\mathcal{P}_t \equiv D_{0t}(S_t - K)^+$, we have:

$$\begin{aligned} d\mathcal{P}_t &= -D_{0t}(S_t - K)^+ r_t dt + D_{0t} \mathbf{1}_{S_t > K} S_t ((r_t - q_t) dt + \alpha_t dW_t) \\ &\quad + \frac{1}{2} S_t^2 \alpha_t^2 D_{0t} \delta(S_t - K) dt \\ &= D_{0t} \mathbf{1}_{S_t > K} (r_t - q_t) K dt - D_{0t} q_t (S_t - K)^+ dt + D_{0t} \mathbf{1}_{S_t > K} \alpha_t S_t dW_t \\ &\quad + \frac{1}{2} K^2 \alpha_t^2 D_{0t} \delta(S_t - K) dt \end{aligned}$$

By taking the expectation $\mathbb{E}[\cdot]$ on both sides of the above equation and by assuming that $M_t = \int_0^t D_{0s} \mathbf{1}_{S_s > K} \alpha_s S_s dW_s$ is a true martingale, we get

$$\begin{aligned} \partial_t \mathcal{C}_m(t, K) &= K \mathbb{E}[D_{0t}(r_t - q_t) \mathbf{1}_{S_t > K}] - \mathbb{E}[D_{0t} q_t (S_t - K)^+] \\ &\quad + \frac{1}{2} K^2 \mathbb{E}[D_{0t} \alpha_t^2 \delta(S_t - K)] \end{aligned}$$

where $\mathcal{C}_m(t, K) = \mathbb{E}[\mathcal{P}_t]$ denotes the price of the call option in the model.

Proof of the general proposition (continued)

Then, by using that $\partial_K \mathcal{C}_m(t, K) = -\mathbb{E}[D_{0t} \mathbf{1}_{S_t > K}]$ and $\partial_K^2 \mathcal{C}_m(t, K) = \mathbb{E}[D_{0t} \delta(S_t - K)]$, we deduce that

$$\begin{aligned} \partial_t \mathcal{C}_m(t, K) &= K \mathbb{E}[D_{0t}(r_t - q_t - (r_t^0 - q_t^0)) \mathbf{1}_{S_t > K}] - (r_t^0 - q_t^0) K \partial_K \mathcal{C}_m(t, K) \\ &\quad - \mathbb{E}[D_{0t}(q_t - q_t^0)(S_t - K)^+] - q_t^0 \mathcal{C}_m(t, K) + \frac{1}{2} K^2 \partial_K^2 \mathcal{C}_m(t, K) \frac{\mathbb{E}[D_{0t} \alpha_t^2 | S_t = K]}{\mathbb{E}[D_{0t} | S_t = K]} \end{aligned}$$

with the initial condition $\mathcal{C}_m(0, K) = (S_0 - K)^+$ so by uniqueness of the solution to this PDE the model is calibrated to the market smile of S if and only if

$$\begin{aligned} \partial_t \mathcal{C}(t, K) &= K \mathbb{E}[D_{0t}(r_t - q_t - (r_t^0 - q_t^0)) \mathbf{1}_{S_t > K}] - (r_t^0 - q_t^0) K \partial_K \mathcal{C}(t, K) \\ &\quad - \mathbb{E}[D_{0t}(q_t - q_t^0)(S_t - K)^+] - q_t^0 \mathcal{C}(t, K) + \frac{1}{2} K^2 \partial_K^2 \mathcal{C}(t, K) \frac{\mathbb{E}[D_{0t} \alpha_t^2 | S_t = K]}{\mathbb{E}[D_{0t} | S_t = K]} \end{aligned}$$

From the definition of $\sigma_{\text{loc}}(t, K)$, this is equivalent to (1.11), which completes the proof.

Back to the calibration of SLV models

- In the SLV model, the approximated conditional expectation is

$$\mathbb{E}^{\mathbb{P}_t^N}[a_t^2|S_t = S] = \frac{\int a'^2 p_N(t, S, a') da'}{\int p_N(t, S, a') da'} = \frac{\sum_{i=1}^N (a_t^{i,N})^2 \delta(S_t^{i,N} - S)}{\sum_{i=1}^N \delta(S_t^{i,N} - S)}$$

- Instead of the Dirac delta function $\delta(\cdot)$, we use a kernel $\delta_{t,N}(\cdot)$ and approximate $\mathbb{E}^{\mathbb{P}_t^N}[a_t^2|S_t = S]$ by (simple nonparametric regression!)

$$\bar{a}(t, S)^2 = \frac{\sum_{i=1}^N \left(a_t^{i,N}\right)^2 \delta_{t,N}\left(S_t^{i,N} - S\right)}{\sum_{i=1}^N \delta_{t,N}\left(S_t^{i,N} - S\right)} \quad (1.14)$$

- Then we define

$$\sigma_N(t, S) = \frac{\sigma_{\text{loc}}(t, S)}{\bar{a}(t, S)}$$

and simulate

$$\frac{dS_t^{i,N}}{S_t^{i,N}} = (r_t - q_t)dt + \sigma_N(t, S_t^{i,N}) a_t^{i,N} dW_t^i \quad (1.15)$$

- $O(N^2)$ operations at each discretization date $\Rightarrow$ **it is crucial to design acceleration techniques**

How to make the particle method efficient in practice

- Computing $\sigma_N(t, S_t^{i,N})$ for all i is useless $\Rightarrow$ **Compute $\sigma_N(t, S)$ for only a grid of values $G_{S,t}$ of S** , of a size much smaller than N , and then inter- and extrapolate
- In the sums in (1.14), a large number of terms make a negligible contribution $\Rightarrow$ **Disregard $S_t^{i,N}$ when it is far from S** , say, when $\delta_{t,N}(S_t^{i,N} - S) < \eta$
 $\Rightarrow$ **Sort particles according to spot value**. Cost of sorting, $O(N \ln N)$, is more than compensated by the acceleration in the evaluations of (1.14)
- Alternative methods for estimating such conditional expectations include B-spline techniques (Corlay, 2013), neural networks, random forests, nearest neighbors, etc.
- Calibration and pricing can be achieved in the course of the same Monte-Carlo simulation
- If we need more paths for pricing, we can reuse the calibration paths, and simulate new paths using the already calibrated local volatility function

Choice of kernel and bandwidth

- It is natural to take $\delta_{t,N}(x) = \frac{1}{h_{t,N}} K\left(\frac{x}{h_{t,N}}\right)$ where K is a fixed, symmetric kernel with a bandwidth $h_{t,N}$ that tends to zero as N grows to infinity.
- Typical examples:

Exponential kernel:
$$K(x) = \frac{1}{\sqrt{2\pi}} \exp(-x^2/2)$$

Epanechnikov kernel:
$$K(x) = \frac{3}{4} (1 - x^2) \mathbf{1}_{\{|x| \leq 1\}}$$

Quartic kernel:
$$K(x) = \frac{15}{16} (1 - x^2)^2 \mathbf{1}_{\{|x| \leq 1\}}$$

The last two are faster to compute than the exponential.

Choice of kernel and bandwidth

- We take

$$h_{t,N} = \kappa S_0 \sigma_{VS,t} \sqrt{\max(t, t_{\min})} N^{-\frac{1}{5}}$$

with $\sigma_{VS,t}$ the variance swap volatility at maturity t .

- The factor $N^{-\frac{1}{5}}$ comes from the minimization of the asymptotic mean integrated squared error of the Nadaraya-Watson estimator, which is the sum of two terms: bias and variance. The smaller the bandwidth, the smaller the bias, but the larger the variance. The critical bandwidth that minimizes the sum of bias and variance decreases as $N^{-\frac{1}{5}}$ for large N .
- Following Silverman's rule of thumb, the prefactor $\kappa S_0 \sigma_{VS,t} \sqrt{\max(t, t_{\min})}$ is on the order of the standard deviation of the regressor S_t , if κ is of the order of magnitude of 1. In practice, we take $\kappa \in [1, 3]$, $t_{\min} = 0.1$.

Particle algorithm (G. and Henry-Labordère, 2011)

- 1 Initialize $k = 1$ and set $\sigma_N(t, S) = \frac{\sigma_{\text{loc}}(0, S)}{\alpha}$ for all $t \in [t_0 = 0, t_1]$
- 2 Simulate the N processes $\{S_t^{i,N}, a_t^{i,N}\}_{1 \leq i \leq N}$ from t_{k-1} to t_k using a discretization scheme for (1.15) - say a log-Euler scheme
- 3 Sort particles according to spot value. For $S \in G_{S, t_k}$, find the smallest index $\underline{i}(S)$ and the largest index $\bar{i}(S)$ for which $\delta_{t_k, N} \left(S_{t_k}^{i,N} - S \right) > \eta$, and compute the local volatility

$$\sigma_N(t_k, S) = \sigma_{\text{loc}}(t_k, S) \sqrt{\frac{\sum_{i=\underline{i}(S)}^{\bar{i}(S)} \delta_{t_k, N} \left(S_{t_k}^{i,N} - S \right)}{\sum_{i=\underline{i}(S)}^{\bar{i}(S)} \left(a_{t_k}^{i,N} \right)^2 \delta_{t_k, N} \left(S_{t_k}^{i,N} - S \right)}}$$

Interpolate the square root using cubic splines, and extrapolate (e.g., flat).

Set $\sigma_N(t, S) \equiv \sigma_N(t_k, S)$ for all $t \in [t_k, t_{k+1}]$

- 4 Set $k := k + 1$. Iterate steps 2 and 3 up to the maturity date T

Adding stochastic rates

- SIR-SLV model:

$$\frac{dS_t}{S_t} = r_t dt + \sigma(t, S_t) a_t dW_t$$

- Model exactly calibrated to market smile iff for all t, K

$$\sigma(t, K)^2 \frac{\mathbb{E}[D_{0t} a_t^2 | S_t = K]}{\mathbb{E}[D_{0t} | S_t = K]} = \sigma_{\text{loc}}(t, K)^2 - \frac{\mathbb{E}[D_{0t} (r_t - r_t^0) 1_{S_t > K}]}{\frac{1}{2} K \partial_K^2 \mathcal{C}(t, K)} \quad (1.16)$$

where r_t^0 is a deterministic rate and

$$\sigma_{\text{loc}}(t, K)^2 = \frac{\partial_t \mathcal{C}(t, K) + r_t^0 K \partial_K \mathcal{C}(t, K)}{\frac{1}{2} K^2 \partial_K^2 \mathcal{C}(t, K)}$$

- Equivalently, using the t -forward measure $\mathbb{P}^t$,

$$\sigma(t, K)^2 \mathbb{E}^{\mathbb{P}^t}[a_t^2 | S_t = K] = \sigma_{\text{loc}}(t, K)^2 - P_{0t} \frac{\mathbb{E}^{\mathbb{P}^t}[(r_t - r_t^0) 1_{S_t > K}]}{\frac{1}{2} K \partial_K^2 \mathcal{C}(t, K)}$$

- Usually $r_t^0 = -\partial_t \ln P_{0t} = \mathbb{E}^{\mathbb{P}^t}[r_t]$
- Does the r.h.s. of (1.16) stay nonnegative?

Adding stochastic rates

- $\Rightarrow$ The calibrated model follows the nonlinear SDE

$$\frac{dS_t}{S_t} = r_t dt + \sigma(t, S_t, \mathbb{P}_t) a_t dW_t$$

where

$$\sigma(t, K, \mathbb{P}_t)^2 = \left(\sigma_{\text{loc}}(t, K)^2 - \frac{\mathbb{E}[D_{0t} (r_t - r_t^0) 1_{S_t > K}]}{\frac{1}{2} K \partial_K^2 \mathcal{C}(t, K)} \right) \frac{\mathbb{E}^{\mathbb{P}}[D_{0t} | S_t = K]}{\mathbb{E}^{\mathbb{P}}[D_{0t} a_t^2 | S_t = K]} \quad (1.17)$$

- Equivalently:

$$\frac{df_t}{f_t} = \sigma(t, P_{tT} f_t, \mathbb{P}_t^T) a_t dW_t^T - \sigma_P^T(t) . dB_t^T$$

where

$$\sigma(t, K, \mathbb{P}_t^T)^2 = \left(\sigma_{\text{loc}}(t, K)^2 - P_{0T} \frac{\mathbb{E}^{\mathbb{P}^T}[P_{tT}^{-1} (r_t - r_t^0) 1_{S_t > K}]}{\frac{1}{2} K \partial_K^2 \mathcal{C}(t, K)} \right) \frac{\mathbb{E}^{\mathbb{P}^T}[P_{tT}^{-1} | S_t = K]}{\mathbb{E}^{\mathbb{P}^T}[P_{tT}^{-1} a_t^2 | S_t = K]} \quad (1.18)$$

Particle algorithm (G. and Henry-Labordère) - Case I

If we use (1.17) we define

$$\sigma_N(t, S)^2 = \left(\sigma_{\text{loc}}(t, S)^2 - \frac{\frac{1}{N} \sum_{i=1}^N D_{0t}^{i,N} \left(r_t^{i,N} - r_t^0 \right) 1_{S_t^{i,N} > S}}{\frac{1}{2} S \partial_K^2 \mathcal{C}(t, S)} \right) \times \frac{\sum_{i=1}^N D_{0t}^{i,N} \delta_{t,N} \left(S_t^{i,N} - S \right)}{\sum_{i=1}^N D_{0t}^{i,N} \left(a_t^{i,N} \right)^2 \delta_{t,N} \left(S_t^{i,N} - S \right)}$$

and simulate

$$\frac{dS_t^{i,N}}{S_t^{i,N}} = r_t^{i,N} dt + \sigma_N \left(t, S_t^{i,N} \right) a_t^{i,N} dW_t^i$$

Particle algorithm (G. and Henry-Labordère) - Case II

In many commonly used short rate models, P_{tT} bond has a closed form formula, we can use (1.18), define

$$\sigma_N(t, S)^2 = \left(\sigma_{\text{loc}}(t, S)^2 - P_{0T} \frac{\frac{1}{N} \sum_{i=1}^N \left(P_{tT}^{i,N} \right)^{-1} \left(r_t^{i,N} - r_t^0 \right) 1_{S_t^{i,N} > S}}{\frac{1}{2} S \partial_K^2 \mathcal{C}(t, S)} \right) \\ \times \frac{\sum_{i=1}^N \left(P_{tT}^{i,N} \right)^{-1} \delta_{t,N} \left(S_t^{i,N} - S \right)}{\sum_{i=1}^N \left(P_{tT}^{i,N} \right)^{-1} \left(a_t^{i,N} \right)^2 \delta_{t,N} \left(S_t^{i,N} - S \right)}$$

and simulate

$$df_t^{i,N} = f_t^{i,N} \sigma_N \left(t, f_t^{i,N} P_{tT}^{i,N} \right) a_t^{i,N} dW_t^i - f_t^{i,N} \sigma_P^{T,i,N}(t) . dB_t^i$$

where W^i and B^i are $\mathbb{P}^T$ -Brownian motions.

Adding stochastic repo/dividend yield (G., 2013)

- SIR-SDY-SLV model:

$$\frac{dS_t}{S_t} = (r_t - q_t) dt + \sigma(t, S_t) a_t dW_t$$

- Model exactly calibrated to market smile iff for all (t, K)

$$\begin{aligned} \sigma(t, K)^2 \frac{\mathbb{E}[D_{0t} a_t^2 | S_t = K]}{\mathbb{E}[D_{0t} | S_t = K]} &= \sigma_{\text{loc}}(t, K)^2 - \frac{\mathbb{E}[D_{0t} (r_t - q_t - (r_t^0 - q_t^0)) 1_{S_t > K}]}{\frac{1}{2} K \partial_K^2 \mathcal{C}(t, K)} \\ &\quad + \frac{\mathbb{E}[D_{0t} (q_t - q_t^0) (S_t - K)^+]}{\frac{1}{2} K^2 \partial_K^2 \mathcal{C}(t, K)} \end{aligned}$$

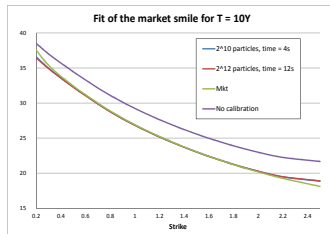
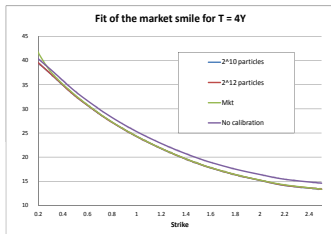
where r_t^0 and q_t^0 are deterministic rates and repos and

$$\sigma_{\text{loc}}(t, K)^2 = \frac{\partial_t \mathcal{C}(t, K) + (r_t^0 - q_t^0) K \partial_K \mathcal{C}(t, K) + q_t^0 \mathcal{C}(t, K)}{\frac{1}{2} K^2 \partial_K^2 \mathcal{C}(t, K)}$$

- The calibrated model follows a McKean SDE
- Does $\sigma(t, K)^2$ stay nonnegative?

LV + Ho-Lee

- DAX market smiles (30-May-11)
- Ho-Lee parameters: $\sigma_r = 1\%$, $\rho = 40\%$
- Particle algorithm: $\Delta = 1/100$, $N = 2^{10}$ particles. MC pricing: $N = 2^{15}$ paths



LV + Ho-Lee

Strike	0.5	0.7	0.8	0.9	1	1.1	1.2	1.3	1.5	1.8
with Malliavin	14	10	10	9	8	7	6	5	3	1
without Malliavin	16	8	7	4	1	1	1	3	3	5

Table: DAX (30-May-11) Implied volatilities $T = 10Y$. Errors in bps using the particle method with $N = 2^{10}$ particles

LV + Bergomi

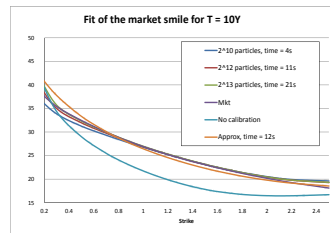
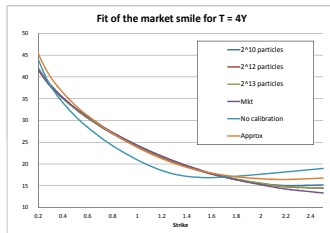
$$\begin{aligned}
df_t &= f_t \sigma(t, f_t) \sqrt{\xi_t^T} dW_t \\
\xi_t^T &= \xi_0^T f^T(t, x_t^T) \\
f^T(t, x) &= \exp(2\nu x - 2\nu^2 h(t, T)) \\
x_t^T &= \alpha_\theta \left((1 - \theta) e^{-k_X(T-t)} X_t + \theta e^{-k_Y(T-t)} Y_t \right) \\
\alpha_\theta &= \left((1 - \theta)^2 + \theta^2 + 2\rho_{XY}\theta(1 - \theta) \right)^{-1/2} \\
dX_t &= -k_X X_t dt + dW_t^X \\
dY_t &= -k_Y Y_t dt + dW_t^Y
\end{aligned}$$

where

$$\begin{aligned}
h(t, T) &= \text{Var}(x_t^T) = \alpha_\theta^2 \left((1 - \theta)^2 e^{-2k_X(T-t)} \mathbb{E}[X_t^2] + \theta^2 e^{-2k_Y(T-t)} \mathbb{E}[Y_t^2] \right. \\
&\quad \left. + 2\theta(1 - \theta) e^{-(k_X + k_Y)(T-t)} \mathbb{E}[X_t Y_t] \right) \\
\mathbb{E}[X_t^2] &= \frac{1 - e^{-2k_X t}}{2k_X}, \quad \mathbb{E}[Y_t^2] = \frac{1 - e^{-2k_Y t}}{2k_Y}, \quad \mathbb{E}[X_t Y_t] = \rho_{XY} \frac{1 - e^{-(k_X + k_Y)t}}{k_X + k_Y}
\end{aligned}$$

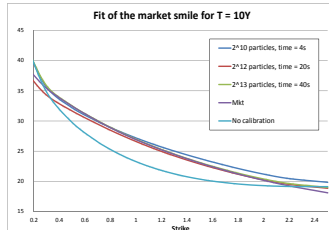
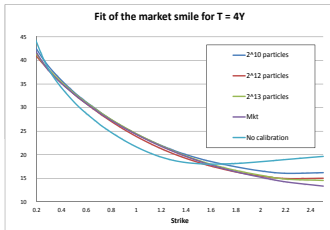
LV + Bergomi

- Parameters of Bergomi model: $\nu = 200\%$, $\theta = 22.65\%$, $k_X = 4$, $k_Y = 12.5\%$, $\rho = 30\%$, $\rho_{SX} = -50\%$, $\rho_{SY} = -50\%$



LV + Bergomi + Ho-Lee

- Ho-Lee parameters: $\sigma_r = 1\%$, $\rho_{Sr} = 40\%$
- Parameters of Bergomi model: $\nu = 200\%$, $\theta = 22.65\%$, $k_X = 4$, $k_Y = 12.5\%$, $\rho = 30\%$, $\rho_{SX} = -50\%$, $\rho_{SY} = -50\%$. Correlation Vol/Rate = 0
- Requires solving a 4d-PDE!

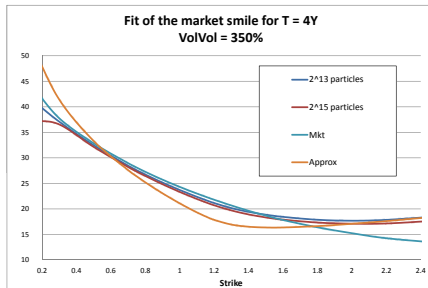


No optimization at all!

- Classical calibration procedures involve optimization over parameters: minimize the mean squared error between prices in the model and market prices.
- Time consuming, there could be local minima...
- And calibration to whole smile is imperfect (due to the finite number of parameters)
- It is remarkable that the particle method involves **no optimization at all!** And it gives perfect calibration.

Existence?

- Parameters of Bergomi model: $\nu = 350\%$, $\theta = 22.65\%$, $k_X = 4$, $k_Y = 12.5\%$, $\rho = 30\%$, $\rho_{SX} = -50\%$, $\rho_{SY} = -50\%$.



Existence?

$$\sigma(t, S, p) = \sigma_{\text{loc}}(t, S) \sqrt{\frac{\int p(t, S, a) da}{\int a^2 p(t, S, a) da}}$$

- Lipschitz condition w.r.t. $\mathbb{P}$ not satisfied $\implies$ **uniqueness and existence results for calibrated SLV models are not at all obvious**
- Partial result by Abergel and Tachet (2010): the calibration problem for a SLV model is proved to be well posed (a) until some maturity T^* , (b) if the vol-of-vol is small enough, and (c) for suitably regularized initial conditions - not for a Dirac mass!
- Partial result by Jourdain and Zhou (2016): Regime-switching LV model = when the SV (a_t) is a pure jump process taking finitely many values, with jump intensities depending on the spot level. Range of (a_t^2) not too large $\implies$ existence of a weak solution to the McKean SDE.
- Our numerical experiments with SLV show that the calibration fails for large vol-of-vol, whatever the algorithm used: PDE, particle method.
Numerical issue? Non-existence?

Contents

- 1 The particle method for smile calibration
- 2 The multi-asset case: calibrating local correlation models
- 3 VIX and VIX derivatives
- 4 Robust hedging of derivatives: martingale optimal transport
- 5 Extension to VIX: dispersion-constrained martingale optimal transport
- 6 Exact joint calibration of SPX and VIX smiles via minimum entropy: dispersion-constrained martingale Schrödinger problems

Calibration of local correlation models to market smiles

Another example: calibration of local correlation (LC) models

- LC model for a triangle of FX rates (deterministic rates):

$$\begin{aligned} dS_t^1 &= (r_t^d - r_t^1) S_t^1 dt + \sigma_1(t, S_t^1) S_t^1 dW_t^1 \\ dS_t^2 &= (r_t^d - r_t^2) S_t^2 dt + \sigma_2(t, S_t^2) S_t^2 dW_t^2 \\ d\langle W^1, W^2 \rangle_t &= \rho(t, S_t^1, S_t^2) dt \end{aligned}$$

- Example: $S^1 = \text{EUR/USD}$, $S^2 = \text{GBP/USD}$ and $S^{12} = \text{EUR/GBP}$
- Model calibrated to market smile of cross rate $S^{12} := S^1/S^2$ iff for all t

$$\mathbb{E}_\rho^{\mathbb{Q}^f} \left[\sigma_1^2(t, S_t^1) + \sigma_2^2(t, S_t^2) - 2\rho(t, S_t^1, S_t^2) \sigma_1(t, S_t^1) \sigma_2(t, S_t^2) \left| \frac{S_t^1}{S_t^2} \right. \right] = \sigma_{12}^2 \left(t, \frac{S_t^1}{S_t^2} \right) \quad (2.1)$$

- $\mathbb{Q}^f$ = risk-neutral measure associated to the foreign currency in S^2 (GBP):

$$\frac{d\mathbb{Q}^f}{d\mathbb{Q}} = \frac{S_T^2}{S_0^2} \exp \left(\int_0^T (r_t^2 - r_t^d) dt \right)$$

$$\frac{\mathbb{E}_\rho^{\mathbb{Q}} \left[\sigma_1^2(t, S_t^1) + \sigma_2^2(t, S_t^2) - 2\rho(t, S_t^1, S_t^2) \sigma_1(t, S_t^1) \sigma_2(t, S_t^2) \right] \left| \frac{S_t^1}{S_t^2} \right.}{\mathbb{E}_\rho^{\mathbb{Q}} \left[\left. \frac{S_t^2}{S_t^2} \right| \frac{S_t^1}{S_t^2} \right]} = \sigma_{12}^2 \left(t, \frac{S_t^1}{S_t^2} \right)$$

Local in cross correlation

- Assume ρ is **local in cross**: $\rho\left(t, \frac{S_t^1}{S_t^2}\right)$ (Reghai, G. and Henry-Labordère, 2011)
- Calibration condition reads

$$\mathbb{E}_\rho^{\mathbb{Q}^f} \left[\sigma_1^2(t, S_t^1) + \sigma_2^2(t, S_t^2) \left| \frac{S_t^1}{S_t^2} \right. \right] - 2\rho\left(t, \frac{S_t^1}{S_t^2}\right) \mathbb{E}_\rho^{\mathbb{Q}^f} \left[\sigma_1(t, S_t^1) \sigma_2(t, S_t^2) \left| \frac{S_t^1}{S_t^2} \right. \right] = \sigma_{12}^2\left(t, \frac{S_t^1}{S_t^2}\right)$$

- Calibrated model follows the **McKean SDE**:

$$\begin{aligned} dS_t^1 &= (r_t^d - r_t^1) S_t^1 dt + \sigma_1(t, S_t^1) S_t^1 dW_t^1 \\ dS_t^2 &= (r_t^d - r_t^2) S_t^2 dt + \sigma_2(t, S_t^2) S_t^2 dW_t^2 \\ d\langle W^1, W^2 \rangle_t &= \frac{\mathbb{E}^{\mathbb{Q}^f} \left[\sigma_1^2(t, S_t^1) + \sigma_2^2(t, S_t^2) \left| \frac{S_t^1}{S_t^2} \right. \right] - \sigma_{12}^2\left(t, \frac{S_t^1}{S_t^2}\right)}{2\mathbb{E}^{\mathbb{Q}^f} \left[\sigma_1(t, S_t^1) \sigma_2(t, S_t^2) \left| \frac{S_t^1}{S_t^2} \right. \right]} dt \\ &= \frac{\mathbb{E}^{\mathbb{Q}} \left[S_t^2 (\sigma_1^2(t, S_t^1) + \sigma_2^2(t, S_t^2)) \left| \frac{S_t^1}{S_t^2} \right. \right] - \sigma_{12}^2\left(t, \frac{S_t^1}{S_t^2}\right) \mathbb{E}^{\mathbb{Q}} \left[S_t^2 \left| \frac{S_t^1}{S_t^2} \right. \right]}{2\mathbb{E}^{\mathbb{Q}} \left[S_t^2 \sigma_1(t, S_t^1) \sigma_2(t, S_t^2) \left| \frac{S_t^1}{S_t^2} \right. \right]} dt \end{aligned}$$

- Does the r.h.s. of last equation stay in $[-1, 1]$?

Particle method for local in cross correl (G. and Henry-Labordère, 2011)

- 1 Set $k = 1$ and $\rho(t, S^1, S^2) = \frac{\sigma_1^2(0, S^1) + \sigma_2^2(0, S^2) - \sigma_{12}^2(0, \frac{S^1}{S^2})}{2\sigma_1(0, S^1)\sigma_2(0, S^2)}$ for $t \in [t_0, t_1]$
- 2 Simulate $(S_t^{1,i}, S_t^{2,i})_{1 \leq i \leq N}$ from t_{k-1} to t_k using a discretization scheme
- 3 For all S^{12} in a grid G_{t_k} of cross rate values, compute non-parametric kernel estimates $E_{t_k}^{\text{num}}(S^{12})$ and $E_{t_k}^{\text{den}}(S^{12})$ of $\mathbb{E}_\rho^{\mathbb{Q}^f} \left[\sigma_1^2 + \sigma_2^2 \left| \frac{S_t^1}{S_t^2} \right| \right]$ and $\mathbb{E}_\rho^{\mathbb{Q}^f} \left[\sigma_1 \sigma_2 \left| \frac{S_t^1}{S_t^2} \right| \right]$ at date t_k , define

$$\rho(t_k, S^{12}) = \frac{E_{t_k}^{\text{num}}(S^{12}) - \sigma_{12}^2(t_k, S^{12})}{2E_{t_k}^{\text{den}}(S^{12})}$$

interpolate $\rho(t_k, \cdot)$, e.g., using cubic splines, extrapolate, and, for all $t \in [t_k, t_{k+1}]$, set $\rho(t, S^1, S^2) = \rho(t_k, \frac{S^1}{S^2})$

- 4 Set $k := k + 1$. Iterate steps 2 and 3 up to the maturity date T

A new family of local correlation models (G., 2013)

- ρ local in cross/index = a particular modeling choice only guided by computational convenience
- “Admissible correlation” (FX) = any $\rho : [0, T] \times \mathbb{R}_+^* \times \mathbb{R}_+^* \rightarrow [-1, 1]$ satisfying the calibration condition (2.1). How to build all admissible ρ ?
- Let ρ be admissible. Pick two functions $a(t, S^1, S^2)$ and $b(t, S^1, S^2)$ such that b does not vanish and $a + b\rho$ is local in cross:

$$a(t, S^1, S^2) + b(t, S^1, S^2)\rho(t, S^1, S^2) \equiv f\left(t, \frac{S^1}{S^2}\right)$$

- We can always do so, e.g., $b \equiv 1$, $a(t, S^1, S^2) = f\left(t, \frac{S^1}{S^2}\right) - \rho(t, S^1, S^2)$
- Local in cross correl: $a \equiv 0$ and $b \equiv 1$ (Reghai, G. and Henry-Labordère)
- Local in cross volatility of the cross: $a = \sigma_1^2 + \sigma_2^2$ and $b = -2\sigma_1\sigma_2$
(Langnau, Kovrizhkin): $\rho^* = \frac{\sigma_1^2 + \sigma_2^2 - \sigma_{12}^2}{2\sigma_1\sigma_2}$

$$\begin{aligned} \sigma_{12}^2\left(t, \frac{S_t^1}{S_t^2}\right) &= \mathbb{E}_\rho^{\mathbb{Q}^f} \left[\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2 \left| \frac{S_t^1}{S_t^2} \right. \right] \\ &= \mathbb{E}_\rho^{\mathbb{Q}^f} \left[\sigma_1^2 + \sigma_2^2 + 2\frac{a}{b}\sigma_1\sigma_2 \left| \frac{S_t^1}{S_t^2} \right. \right] - 2(a + b\rho) \left(t, \frac{S_t^1}{S_t^2} \right) \mathbb{E}_\rho^{\mathbb{Q}^f} \left[\frac{\sigma_1\sigma_2}{b} \left| \frac{S_t^1}{S_t^2} \right. \right] \end{aligned}$$

A new family of local correlation models (G., 2013)

$$\sigma_{12}^2 \left(t, \frac{S_t^1}{S_t^2} \right) = \mathbb{E}_{\rho}^{\mathbb{Q}^f} \left[\sigma_1^2 + \sigma_2^2 + 2 \frac{a}{b} \sigma_1 \sigma_2 \left| \frac{S_t^1}{S_t^2} \right. \right] - 2(a + b\rho) \left(t, \frac{S_t^1}{S_t^2} \right) \mathbb{E}_{\rho}^{\mathbb{Q}^f} \left[\frac{\sigma_1 \sigma_2}{b} \left| \frac{S_t^1}{S_t^2} \right. \right]$$

$$\implies \rho_{(a,b)} = \frac{1}{b} \left(\frac{\mathbb{E}_{\rho_{(a,b)}}^{\mathbb{Q}^f} \left[\sigma_1^2 + \sigma_2^2 + 2 \frac{a}{b} \sigma_1 \sigma_2 \left| \frac{S_t^1}{S_t^2} \right. \right] - \sigma_{12}^2}{2 \mathbb{E}_{\rho_{(a,b)}}^{\mathbb{Q}^f} \left[\frac{\sigma_1 \sigma_2}{b} \left| \frac{S_t^1}{S_t^2} \right. \right]} - a \right)$$

- Calibrated model follows the **McKean SDE**:

$$\begin{aligned} dS_t^1 &= (r_t^d - r_t^1) S_t^1 dt + \sigma_1(t, S_t^1) S_t^1 dW_t^1 \\ dS_t^2 &= (r_t^d - r_t^2) S_t^2 dt + \sigma_2(t, S_t^2) S_t^2 dW_t^2 \\ d\langle W^1, W^2 \rangle_t &= \left(\frac{\mathbb{E}^{\mathbb{Q}} \left[S_t^2 \left(\sigma_1^2 + \sigma_2^2 + 2 \frac{a}{b} \sigma_1 \sigma_2 \right) \left| \frac{S_t^1}{S_t^2} \right. \right] - \sigma_{12}^2 \left(t, \frac{S_t^1}{S_t^2} \right) \mathbb{E}^{\mathbb{Q}} \left[S_t^2 \left| \frac{S_t^1}{S_t^2} \right. \right]}{2 \mathbb{E}^{\mathbb{Q}} \left[S_t^2 \frac{\sigma_1 \sigma_2}{b} \left| \frac{S_t^1}{S_t^2} \right. \right]} \right. \\ &\quad \left. - a(t, S_t^1, S_t^2) \right) dt / b(t, S_t^1, S_t^2) \end{aligned}$$

Local in index correl for stock indices (G. and Henry-Labordère, 2011)

- Consider an index $I_t = \sum_{i=1}^N \alpha_i S_t^i$ made of N weighted stocks, each modeled using a local volatility (rates are deterministic):

$$dS_t^i = r_t S_t^i dt + S_t^i \sigma_i(t, S_t^i) dW_t^i, \quad d\langle W^i, W^j \rangle_t = \rho_{ij}(t, S_t^1, \dots, S_t^N) dt$$

- Model calibrated to basket smile iff for all t, I

$$I^2 \sigma_{\text{loc}}^I(t, I)^2 = \sum_{i,j=1}^N \alpha_i \alpha_j \mathbb{E}[\rho_{ij}(t, S_t^1, \dots, S_t^N) \sigma_i(t, S_t^i) \sigma_j(t, S_t^j) S_t^i S_t^j | I_t = I]$$

- Assume that correl is **local in index** (e.g., $\rho^0 = \rho^{\text{hist}}$, $\rho^1 = \mathbf{1}$):

$$\rho(t, I) = (1 - \lambda(t, I)) \rho^0 + \lambda(t, I) \rho^1, \quad \lambda(t, I) \in [0, 1]$$

- The calibrated model then follows the **McKean SDE**

$$\begin{aligned} dS_t^i &= r_t S_t^i dt + S_t^i \sigma_i(t, S_t^i) dW_t^i, & d\langle W^i, W^j \rangle_t &= \rho_{ij}(t, I_t) dt \\ \lambda(t, I) &= \frac{I^2 \sigma_{\text{loc}}^I(t, I)^2 - \sum_{i,j=1}^N \alpha_i \alpha_j \rho_{ij}^0 \mathbb{E}[\sigma_i(t, S_t^i) \sigma_j(t, S_t^j) S_t^i S_t^j | I_t = I]}{\sum_{i,j=1}^N \alpha_i \alpha_j (\rho_{ij}^1 - \rho_{ij}^0) \mathbb{E}[\sigma_i(t, S_t^i) \sigma_j(t, S_t^j) S_t^i S_t^j | I_t = I]} \end{aligned}$$

- Does λ stay in $[0, 1]$?

Extensions (G., 2013)

- Same trick works for building a family of local correlations that calibrate to the market smile of a **stock index**, by assuming that

$$a(t, S^1, \dots, S^N) + b(t, S^1, \dots, S^N) \lambda(t, S^1, \dots, S^N) \equiv f(t, I_t)$$

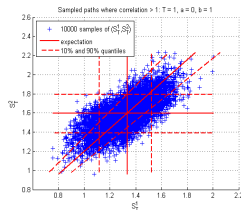
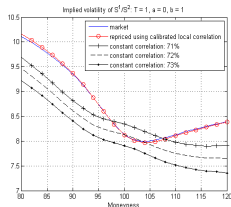
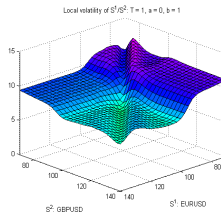
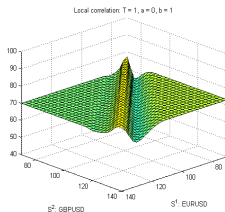
where $\rho = (1 - \lambda)\rho^0 + \lambda\rho^1$, $\lambda \in [0, 1]$

- Another possible application: calibration to options on **interest rate spreads**
- The particle method and the construction of the new class of local correlations have been generalized (see G., 2013) to handle
 - LV+LC+stoch rates
 - LV+stoch vol+LC
 - LV+stoch vol+LC+stoch rates
 - LV+stoch vol+LC+stoch rates+stoch dividend yield

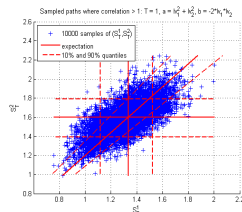
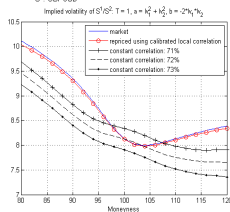
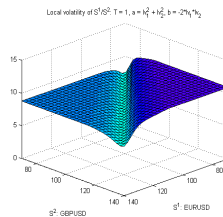
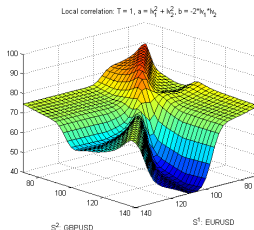
Numerical examples

- $S^1 = \text{EURUSD}$, $S^2 = \text{GBPUSD}$, $S^{12} = S^1/S^2 = \text{EURGBP}$ (March 2012)
- $T = 1$
- $N = 10,000$ particles
- $\Delta t = \frac{1}{80}$
- $K(x) = (1 - x^2)^2 1_{\{|x| \leq 1\}}$
- Bandwidth $h = \kappa \bar{\sigma}^{12} S_0^{12} \sqrt{\max(t, t_{\min})} N^{-\frac{1}{5}}$, $\bar{\sigma}^{12} = 10\%$, $t_{\min} = 0.25$ and $\kappa \approx 3$
- The constant correlation that fits ATM implied volatility of cross rate at maturity = 72%

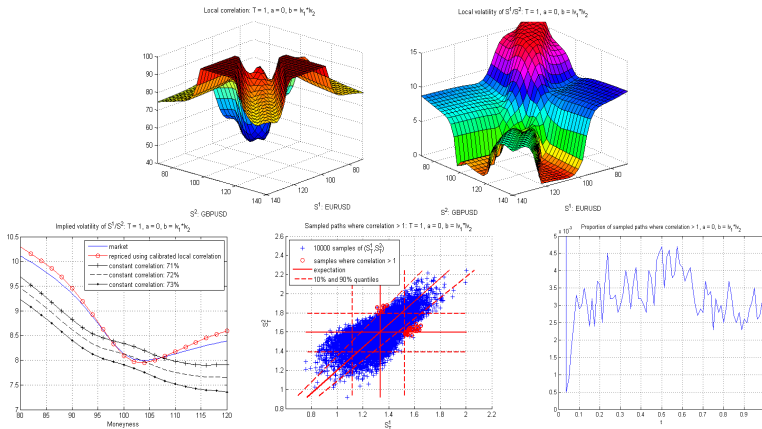
$a = 0, b = 1$ (local in cross correlation)



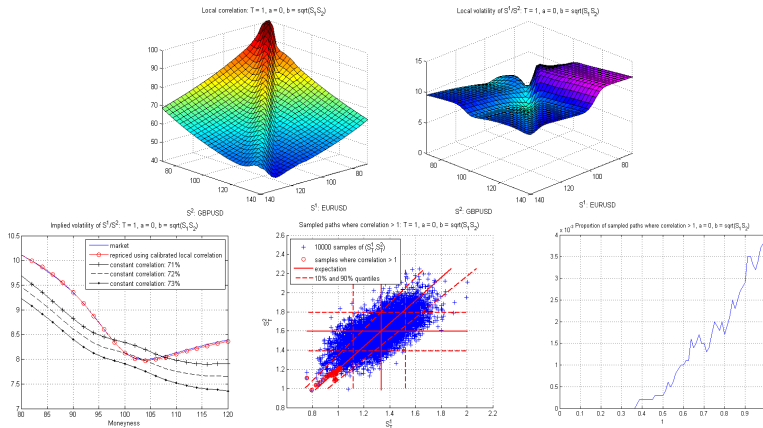
$$a = \sigma_1^2 + \sigma_2^2, \quad b = -2\sigma_1\sigma_2 \text{ (local in cross volatility of the cross)}$$



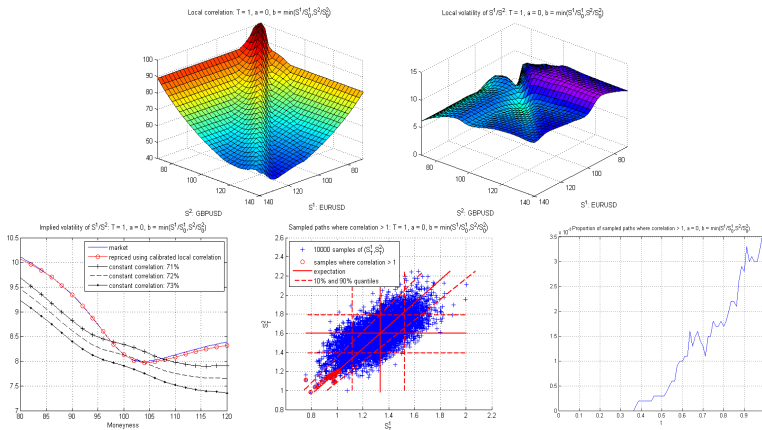
$a = 0, b = \sigma_1 \sigma_2$ (local in cross covariance)



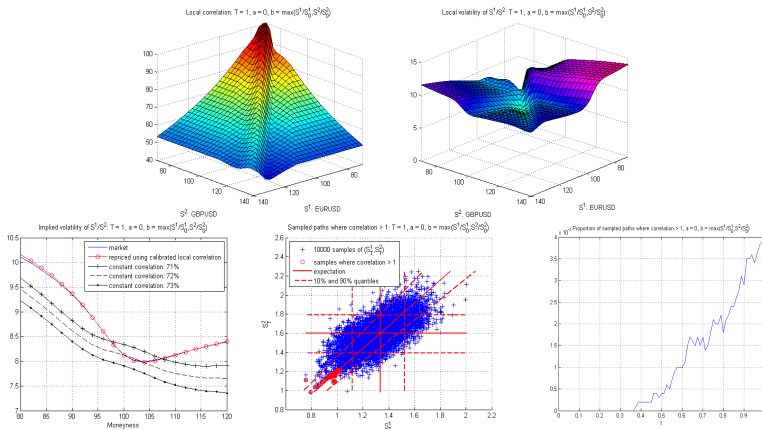
$$a = 0, b = \sqrt{S^1 S^2}$$



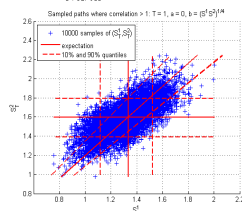
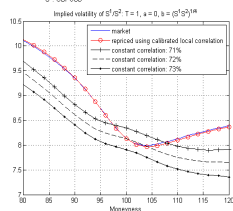
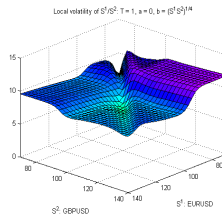
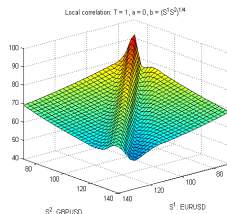
$$a = 0, b = \min(S^1, S^2)$$



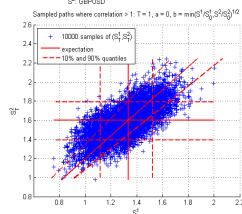
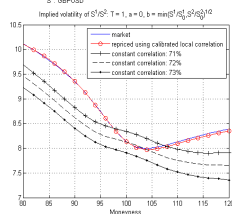
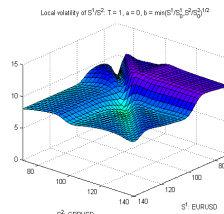
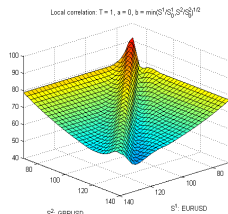
$$a = 0, b = \max(S^1, S^2)$$



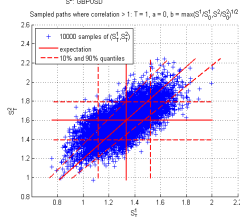
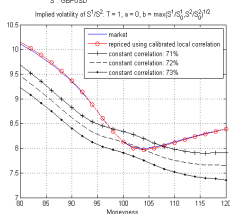
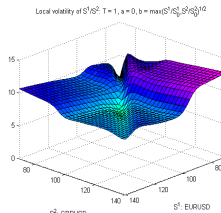
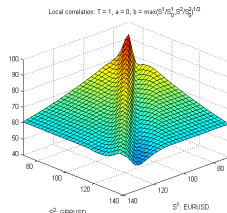
$$a = 0, b = (S^1 S^2)^{1/4}$$



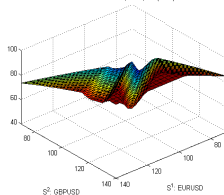
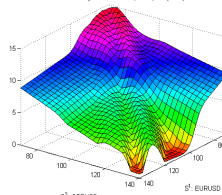
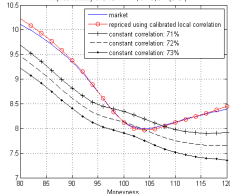
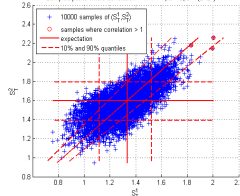
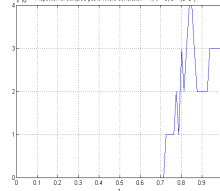
$$a = 0, b = \sqrt{\min(S^1, S^2)}$$



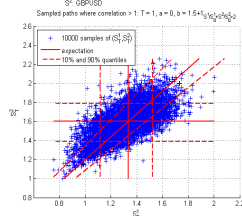
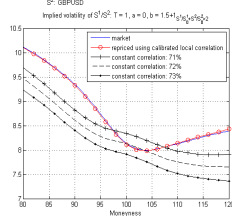
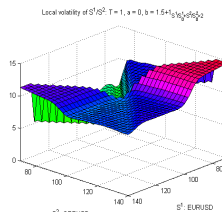
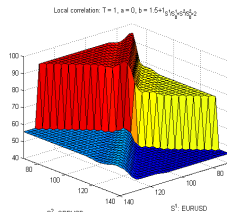
$$a = 0, b = \sqrt{\max(S^1, S^2)}$$



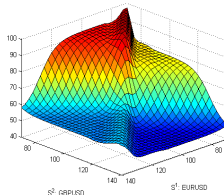
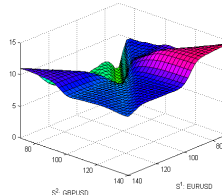
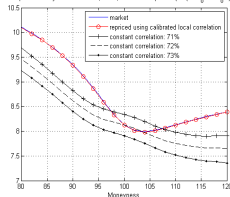
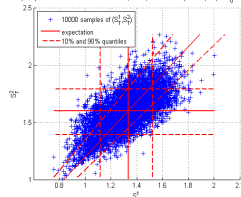
$$a = 0, b = \frac{1}{\sqrt{S^1 S^2}}$$

Local correlation: $T = 1, a = 0, b = (S^1 S^2)^{-1/2}$ Local volatility of S^1/S^2 : $T = 1, a = 0, b = (S^1 S^2)^{-1/2}$ Implied volatility of S^1/S^2 : $T = 1, a = 0, b = (S^1 S^2)^{-1/2}$ Sampled paths where correlation > 1: $T = 1, a = 0, b = (S^1 S^2)^{-1/2}$  $\times 10^{-4}$ Proportion of sampled paths where correlation > 1: $T = 1, a = 0, b = (S^1 S^2)^{-1/2}$ 

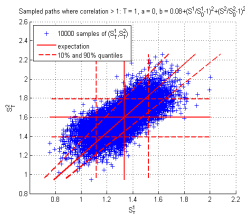
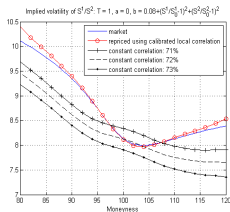
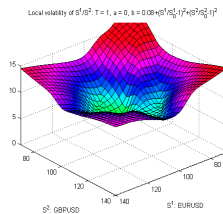
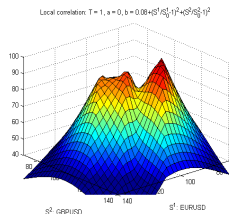
$$a = 0, b = 1.5 + 1 \left\{ \frac{S_1^1}{S_0^1} + \frac{S_2^2}{S_0^2} > 2 \right\}$$



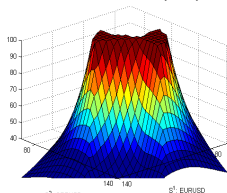
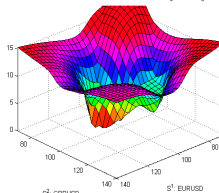
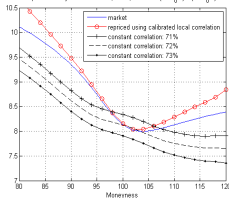
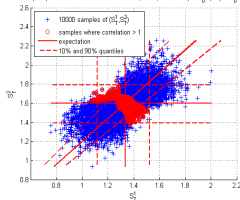
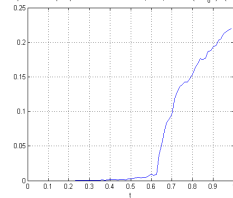
$$a = 0, b = 1.5 + \frac{1}{2} \left(1 + \tanh \left(10 \left(\frac{S_1^1}{S_0^1} + \frac{S_2^2}{S_0^2} - 2 \right) \right) \right)$$

Local correlation: $T = 1, a = 0, b = 1.5 + 0.5 \cdot (1 + \tanh(10 \cdot (S_1^1/S_0^1 + S_2^2/S_0^2 - 2)))$ Local volatility of S^1/S^2 : $T = 1, a = 0, b = 1.5 + 0.5 \cdot (1 + \tanh(10 \cdot (S_1^1/S_0^1 + S_2^2/S_0^2 - 2)))$ Implied volatility of S^1/S^2 : $T = 1, a = 0, b = 1.5 + 0.5 \cdot (1 + \tanh(10 \cdot (S_1^1/S_0^1 + S_2^2/S_0^2 - 2)))$ Sampled paths where correlation > 1 : $T = 1, a = 0, b = 1.5 + 0.5 \cdot (1 + \tanh(10 \cdot (S_1^1/S_0^1 + S_2^2/S_0^2 - 2)))$ 

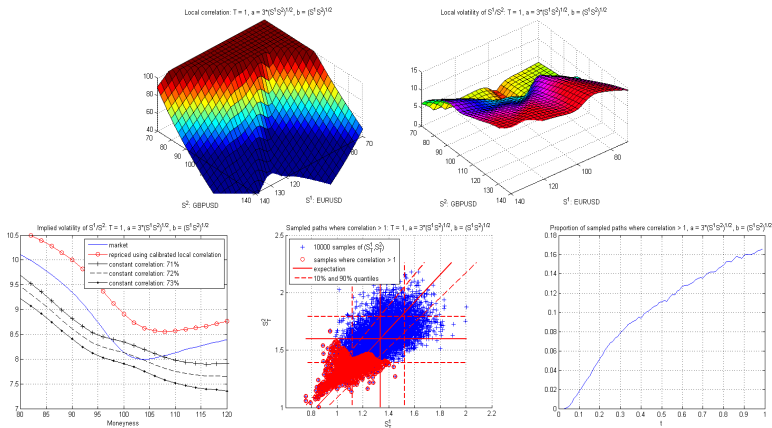
$$a = 0, b = 0.08 + \left(\frac{S^1}{S_0^1} - 1\right)^2 + \left(\frac{S^2}{S_0^2} - 1\right)^2$$



$$a = 0, b = 0.02 + \left(\frac{S^1}{S_0^1} - 1\right)^2 + \left(\frac{S^2}{S_0^2} - 1\right)^2 \text{ (bad choice of } (a, b))$$

Local correlation: $T = 1, a = 0, b = 0.02 + (S^1/S_0^1 - 1)^2 + (S^2/S_0^2 - 1)^2$ Local volatility of $S^1/S^2, T = 1, a = 0, b = 0.02 + (S^1/S_0^1 - 1)^2 + (S^2/S_0^2 - 1)^2$ Implied volatility of $S^1/S^2, T = 1, a = 0, b = 0.02 + (S^1/S_0^1 - 1)^2 + (S^2/S_0^2 - 1)^2$ Sampled paths where correlation $> 1, T = 1, a = 0, b = 0.02 + (S^1/S_0^1 - 1)^2 + (S^2/S_0^2 - 1)^2$ Proportion of sampled paths where correlation $> 1, a = 0, b = 0.02 + (S^1/S_0^1 - 1)^2 + (S^2/S_0^2 - 1)^2$ 

$$a = 3\sqrt{S^1 S^2}, b = \sqrt{S^1 S^2} \text{ (bad choice of } (a, b) \text{)}$$



Path-dependent volatility (G., 2013)

- The $a + b\rho$ (or $a + b\lambda$) trick also works for path-dep correlation models:

$$a(t, S^1, S^2, \mathbf{X}) + b(t, S^1, S^2, \mathbf{X})\rho(t, S^1, S^2, \mathbf{X}) \equiv f\left(t, \frac{S^1}{S^2}\right)$$

- X can be **any** path-dep variable: running/moving averages, maximums/mimumums, realized correlations/variances (cf GARCH), etc.
- What works for correl works for vol! Path-dep vol model:

$$\frac{dS_t}{S_t} = (r_t - q_t) dt + \sigma(t, S_t, \mathbf{X}_t) dW_t$$

- Determ. rates and div yield: model calibrated to the smile iff for all t

$$\mathbb{E}[\sigma(t, S_t, \mathbf{X}_t)^2 | S_t] = \sigma_{\text{loc}}^2(t, S_t)$$

- **All** calibrated path-dep vol models can be built by picking a particular $\sigma(t, S, X)$ and computing $l(t, S)$ such that

$$\mathbb{E}[\sigma(t, S_t, X_t)^2 l(t, S_t)^2 | S_t] = \sigma_{\text{loc}}^2(t, S_t)$$

using the particle method, i.e.,

$$l(t, S) = \frac{\sigma_{\text{loc}}(t, S)}{\sqrt{\mathbb{E}[\sigma(t, S_t, X_t)^2 | S_t = S]}}$$

Path-dependent volatility (G., 2013)

- Calibrated model follows the **McKean SDE**

$$\frac{dS_t}{S_t} = (r_t - q_t) dt + \sigma(t, S_t, X_t) \frac{\sigma_{\text{loc}}(t, S_t)}{\sqrt{\mathbb{E}[\sigma(t, S_t, X_t)^2 | S_t]}} dW_t$$

- Complete** model: prices are **uniquely** defined. Joint spot-vol dynamics richer than in the local vol and stochastic vol models. Captures joint historical behaviour of vol and spot returns
- Extension to stochastic vol, stochastic rates, stochastic div yield is easy:

$$\frac{dS_t}{S_t} = (r_t - q_t) dt + \sigma(t, S_t, X_t) a_t dW_t$$

Model calibrated to the smile iff for all t, K

$$\frac{\mathbb{E}[D_{0t} \sigma^2(t, S_t, X_t) a_t^2 | S_t = K]}{\mathbb{E}[D_{0t} | S_t = K]} = \sigma_{\text{loc}}^2(t, K) - \frac{\mathbb{E}[D_{0t} (r_t - q_t - (r_t^0 - q_t^0)) 1_{S_t > K}]}{\frac{1}{2} K \partial_K^2 \mathcal{C}(t, K)} + \frac{\mathbb{E}[D_{0t} (q_t - q_t^0) (S_t - K)^+]}{\frac{1}{2} K^2 \partial_K^2 \mathcal{C}(t, K)}$$

where σ_{loc} is the Dupire local volatility computed using r_t^0 and q_t^0

Conclusion

- The particle method is a **very powerful** tool for smile calibration problems
- **Single asset**: We calibrate to smile **virtually any diffusive model** with a LV component, possibly including any combination of stoch vol, path-dep vol, stoch rates, and stoch dividend yield
- **Multi-asset**: We calibrate to basket smile **virtually any diffusive model** with a LC component, possibly including any combination of LV, stoch vol, path-dep correl, stoch rates, and stoch dividend yield
- The particle method is **fast**: as fast as a standard MC
 - For low-dimensional models, it **outperforms usual PDE** implementations
 - For high-dimensional models, it is the **first** exact calibration method
- It is **easy to implement**
- It is **robust**: unlike PDE methods, it is insensitive to dimension
- + We can deal exactly with dividends (cash+yield) + Calibration and pricing can be achieved within the same MC procedure
- + There are very nice mathematics behind the scenes (McKean SDEs, propagation of chaos)

A few selected references



Abergel, F., Tachet, R. : *A nonlinear partial integrodifferential equation from mathematical finance*, Discrete Cont. Dynamical Systems, Serie A, vol. 27, 3, p. 907-917, 2010.



Bergomi, L., Guyon, J.: *Stochastic Volatility's Orderly Smiles*, Risk Magazine, May 2012.



Corlay, S., *B-spline techniques for volatility modeling*, available at <http://hal.archives-ouvertes.fr/hal-00830378>, 2013.



Dupire, B.: *Pricing with a smile*, Risk, January, 1994.



Guyon, J. and Henry-Labordère, P., *Being Particular About Calibration*, Risk, Jan. 2012. Long version in Post-Crisis Quant Finance, Risk Books, 2013, and Quant of the Year 2000-2014, Risk Books, 2014.



Guyon, J. and Henry-Labordère, P., *Nonlinear Option Pricing*, Chapman & Hall/CRC Financial Mathematics Series, Dec. 2013.



Guyon, J. *A New Class of Local Correlation Models*, Risk, February 2014.

A few selected references



Guyon, J. *Path-Dependent Volatility*, Risk, October 2014.



Hull, J., White, A.: *The pricing of options on assets with stochastic volatilities*, The Journal of Finance, 42(2):281–300, 1987.



Méléard, S. : *Probabilistic Models for Nonlinear Partial Differential Equations*, Series: Lecture Notes in Mathematics, Vol. 1627, Springer.



Renault, E., Touzi, N.: *Option pricing and implied volatilities in a stochastic volatility model*, Mathematical Finance, 6:279–302, 1996.



Sznitman, A.S. : *Topics in propagation of chaos*, Ecole d'été de probabilités de Saint-Flour XIX - 1989, volume 1464 of Lect. Notes in Math. Springer-Verlag (1991).

VIX, VIX derivatives, and joint SPX/VIX smile calibration via optimal transport

Contents

- 1 The particle method for smile calibration
- 2 The multi-asset case: calibrating local correlation models
- 3 **VIX and VIX derivatives**
- 4 Robust hedging of derivatives on SPX and/or VIX: VIX-constrained martingale optimal transport
- 5 Joint SPX/VIX arbitrages
- 6 Exact joint calibration of SPX and VIX smiles: VIX-constrained martingale Schrödinger problems/bridges

The VIX index

- VIX = **V**olatility **I**nde**X**.
- Published every 15 seconds by the Chicago Board Options Exchange.
- Indicator of short-term options-implied volatility. Known as “fear factor.”
- **Objective of CBOE: VIX is meant to reflect the 30-day implied volatility of SPX options.**
- Problem: implied vol of SPX call/put options depend on the option strike. VIX should be a strike-free measure of SPX implied vol.
- Natural choice: define VIX as the implied volatility of a 30-day variance swap on SPX.
- Problem: Variance swaps are OTC. Not listed on an exchange.
- $\Rightarrow$ **VIX is defined as the implied volatility of a 30-day log-contract on SPX:**

$$(\text{VIX}_t)^2 := -\frac{2}{\tau} \text{Price}_t \left[\ln \left(\frac{S_{t+\tau}}{F_t^{t+\tau}} \right) \right], \quad \tau = 30 \text{ days}$$

- The log-contract is not listed on an exchange but it can be replicated at t using OTM call and put options on the SPX with maturity $t + \tau$.

Static replication of a vanilla payoff by OTM puts and calls

Taylor-Lagrange:

$$\begin{aligned}
 f(y) &= f(x) + f'(x)(y - x) + \int_x^y f''(K)(y - K) dK \\
 &= f(x) + f'(x)(y - x) + \int_x^y f''(K)((y - K)_+ - (K - y)_+) dK \\
 &= f(x) + f'(x)(y - x) + \int_y^x f''(K)(K - y)_+ dK + \int_x^y f''(K)(y - K)_+ dK \\
 &= f(x) + f'(x)(y - x) + \int_0^x f''(K)(K - y)_+ dK + \int_x^\infty f''(K)(y - K)_+ dK
 \end{aligned}$$

Take $y = S_T$, $x = F_t^T$:

$$\begin{aligned}
 f(S_T) &= f(F_t^T) + f'(F_t^T)(S_T - F_t^T) + \int_0^{F_t^T} f''(K)(K - S_T)_+ dK \\
 &\quad + \int_{F_t^T}^\infty f''(K)(S_T - K)_+ dK
 \end{aligned}$$

VIX index

$$\begin{aligned}
 f(S_T) &= f(F_t^T) + f'(F_t^T)(S_T - F_t^T) + \int_0^{F_t^T} f''(K)(K - S_T)_+ dK \\
 &\quad + \int_{F_t^T}^{\infty} f''(K)(S_T - K)_+ dK
 \end{aligned}$$

Take **undiscounted** prices at t :

$$\text{Price}_t[f(S_T)] = f(F_t^T) + \int_0^{F_t^T} f''(K)P_t(T, K) dK + \int_{F_t^T}^{\infty} f''(K)C_t(T, K) dK$$

Take $f = \ln$:

$$\begin{aligned}
 \text{Price}_t \left[\ln \left(\frac{S_T}{F_t^T} \right) \right] &= - \int_0^{F_t^T} \frac{P_t(T, K)}{K^2} dK - \int_{F_t^T}^{\infty} \frac{C_t(T, K)}{K^2} dK \\
 (\text{VIX}_t)^2 &= - \frac{2}{\tau} \text{Price}_t \left[\ln \left(\frac{S_{t+\tau}}{F_t^{t+\tau}} \right) \right] \\
 &= \frac{2}{\tau} \int_0^{F_t^{t+\tau}} \frac{P_t(t+\tau, K)}{K^2} dK + \frac{2}{\tau} \int_{F_t^{t+\tau}}^{\infty} \frac{C_t(t+\tau, K)}{K^2} dK
 \end{aligned}$$

VIX index

$$(\text{VIX}_t)^2 = \frac{2}{\tau} \int_0^{F_t^{t+\tau}} \frac{P_t(t+\tau, K)}{K^2} dK + \frac{2}{\tau} \int_{F_t^{t+\tau}}^{\infty} \frac{C_t(t+\tau, K)}{K^2} dK$$

- Requires undiscounting: $P_t(t+\tau, K)$ and $C_t(t+\tau, K)$ are undiscounted put and call prices.
- Requires interpolation/extrapolation in strikes and interpolation in maturities.
- CBOE uses trapezoidal rule for integration over strikes (up to two consecutive zero bid quotes).
- CBOE uses affine interpolation in maturities, using the two closest listed maturities T_- and T_+ s.t. $T_- < t+\tau < T_+$.

VIX futures

- The VIX index cannot be traded, but VIX futures can.
- VIX future expiring at T = the instrument that pays VIX_T at T .
- VIX_T^2 can be replicated using vanilla options on SPX:

$$(VIX_T)^2 := -\frac{2}{\tau} \text{Price}_T \left[\ln \left(\frac{S_{T+\tau}}{F_T^{T+\tau}} \right) \right]$$

$\implies$ To replicate exactly $(VIX_T)^2$ at time 0: buy $-\frac{2}{\tau} \ln S_{T+\tau}$, sell $-\frac{2}{\tau} \ln F_T^{T+\tau}$.

- But its square root VIX_T cannot.
- Contrary to the price of the future on VIX^2 , the price of the VIX future cannot be inferred by arbitrage arguments from the SPX smile.
- T is 30 days before SPX option monthly maturities, i.e., 30 days before the third Friday of each month.

VIX options

- Pays $(VIX_T - K)_+$ or $(K - VIX_T)_+$ at T ; T is a VIX future maturity.
- The underlying asset of VIX options is the VIX future with the same expiry. Only one option maturity per VIX future: the VIX future expiry.
- VIX options are quoted on the CBOE and are liquidly traded, just like SPX options.
- We can replicate the payoff VIX_T^2 using OTM VIX options: take $y = VIX_T$, $x = F_t^{VIX,T}$, and $f(y) = y^2$ in the Carr-Madan replication formula

$$\begin{aligned}
 f(y) &= f(x) + f'(x)(y - x) + \int_0^x f''(K)(K - y)_+ dK \\
 &\quad + \int_x^\infty f''(K)(y - K)_+ dK \\
 \text{Price}_t [VIX_T^2] &= \left(F_t^{VIX,T}\right)^2 + 2 \int_0^{F_t^{VIX,T}} \text{Price}_t [(K - VIX_T)_+] dK \\
 &\quad + 2 \int_{F_t^{VIX,T}}^\infty \text{Price}_t [(VIX_T - K)_+] dK
 \end{aligned}$$

Motivation

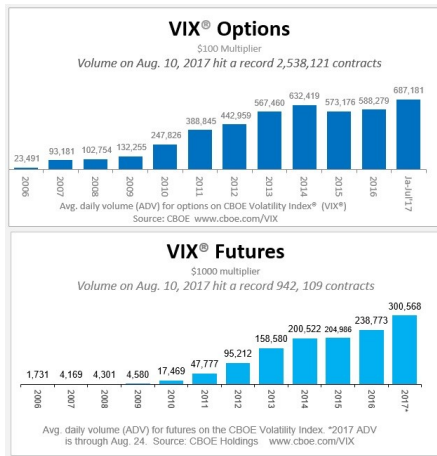


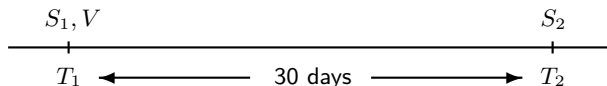
Figure: Average daily volume for VIX options and VIX futures. Source: CBOE

Contents

- 1 The particle method for smile calibration
- 2 The multi-asset case: calibrating local correlation models
- 3 VIX and VIX derivatives
- 4 Robust hedging of derivatives on SPX and/or VIX: VIX-constrained martingale optimal transport
- 5 Joint SPX/VIX arbitrages
- 6 Exact joint calibration of SPX and VIX smiles: VIX-constrained martingale Schrödinger problems/bridges

Bounds on options on SPX and VIX: an optimal transport approach

Setting and notation



- For simplicity: zero interest rates, repos, and dividends.
- μ_1 = risk-neutral distribution of $S_1 \longleftrightarrow$ market smile of SPX at T_1 .
- μ_V = risk-neutral distribution of $V \longleftrightarrow$ market smile of VIX at T_1 .
- μ_2 = risk-neutral distribution of $S_2 \longleftrightarrow$ market smile of SPX at T_2 .
- F_V : value at time 0 of VIX future maturing at T_1 .
- We denote $\mathbb{E}^i := \mathbb{E}^{\mu_i}$, $\mathbb{E}^V := \mathbb{E}^{\mu_V}$ and assume

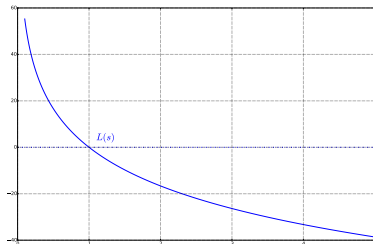
$$\mathbb{E}^i[S_i] = S_0, \quad \mathbb{E}^i[|\ln S_i|] < \infty, \quad i \in \{1, 2\}; \quad \mathbb{E}^V[V] = F_V, \quad \mathbb{E}^V[V^2] < \infty.$$

- No calendar arbitrage $\iff \mu_1 \leq_c \mu_2$ (convex order)

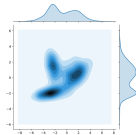
Setting and notation

$$V^2 := (\text{VIX}_{T_1})^2 := -\frac{2}{\tau} \text{Price}_{T_1} \left[\ln \left(\frac{S_2}{S_1} \right) \right] = \text{Price}_{T_1} \left[L \left(\frac{S_2}{S_1} \right) \right]$$

- $\tau := 30$ days.
- $L(x) := -\frac{2}{\tau} \ln x$: convex, decreasing.



Example: superreplication of forward-starting options



- The knowledge of μ_1 and μ_2 gives little information on the prices $\mathbb{E}^\mu[g(S_1, S_2)]$, e.g., prices of forward-starting options $\mathbb{E}^\mu[f(S_2/S_1)]$.
- Computing upper and lower bounds of these prices:
Optimal transport (Monge, 1781; Kantorovich, 1942)
- Adding the no-arbitrage constraint that (S_1, S_2) is a martingale leads to more precise bounds, as this provides information on the conditional average of S_2/S_1 given S_1 :
Martingale optimal transport (Henry-Labordère, 2017)
- When $S = \text{SPX}$: Adding VIX market data information produces even more precise bounds, as it gives information on the conditional dispersion of S_2/S_1 , which is controlled by the VIX V :

Dispersion-constrained martingale optimal transport

Classical optimal transport

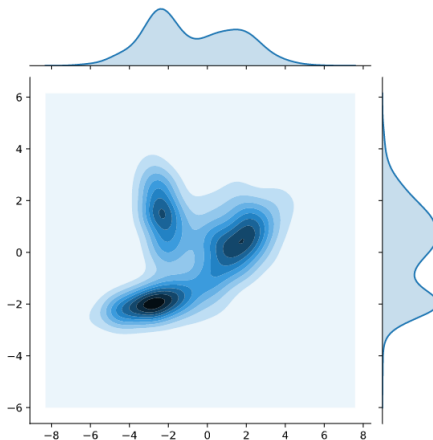


Figure: Example of a transport plan. Source: Wikipedia

Superreplication: primal problem

Fundamental principle: Upper bound for the price of payoff $f(S_1, V, S_2) =$ smallest price at time 0 of a superreplicating portfolio.

Following De Marco-Henry-Labordère (2015), G.-Menegaux-Nutz (2017), the available instruments for superreplication are:

■ At time 0:

- $u_1(S_1)$: SPX vanilla payoff maturity T_1 (including cash)
- $u_2(S_2)$: SPX vanilla payoff maturity T_2
- $u_V(V)$: **VIX vanilla payoff maturity T_1**

$$\begin{aligned} \text{Cost: } & \text{MktPrice}[u_1(S_1)] + \text{MktPrice}[u_2(S_2)] + \text{MktPrice}[u_V(V)] \\ &= \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^2[u_2(S_2)] + \mathbb{E}^V[u_V(V)] \end{aligned}$$

■ At time T_1 :

- $\Delta_S(S_1, V)(S_2 - S_1)$: delta hedge
- $\Delta_L(S_1, V)(L(S_2/S_1) - V^2)$: buy $\Delta_L(S_1, V)$ log-contracts

Cost: 0

Shorthand notation:

$$\Delta^{(S)}(s_1, v, s_2) := \Delta(s_1, v)(s_2 - s_1), \quad \Delta^{(L)}(s_1, v, s_2) := \Delta(s_1, v) \left(L\left(\frac{s_2}{s_1}\right) - v^2 \right)$$

Superreplication: primal problem

- The model-independent no-arbitrage upper bound for the derivative with payoff $f(S_1, V, S_2)$ is the smallest price at time 0 of a superreplicating portfolio:

$$P_f := \inf_{\mathcal{U}_f} \left\{ \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \right\}.$$

- $\mathcal{U}_f$: set of superreplicating portfolios, i.e., the set of all functions $(u_1, u_V, u_2, \Delta_S, \Delta_L)$ that satisfy the superreplication constraint:

$$u_1(s_1) + u_V(v) + u_2(s_2) + \Delta_S^{(S)}(s_1, v, s_2) + \Delta_L^{(L)}(s_1, v, s_2) \geq f(s_1, v, s_2).$$

- Shorthand notation:

$$(u_1 \oplus u_V \oplus u_2)(s_1, v, s_2) := u_1(s_1) + u_V(v) + u_2(s_2)$$

Superreplication constraint:

$$u_1 \oplus u_V \oplus u_2 + \Delta_S^{(S)} + \Delta_L^{(L)} \geq f$$

- Linear program

Superreplication: dual problem

- $\mathcal{P}(\mu_1, \mu_V, \mu_2)$: set of all the probability measures μ on $\mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0}$ such that

$$S_1 \sim \mu_1, \quad V \sim \mu_V, \quad S_2 \sim \mu_2, \quad \mathbb{E}^\mu[S_2|S_1, V] = S_1, \quad \mathbb{E}^\mu\left[L\left(\frac{S_2}{S_1}\right)\middle|S_1, V\right] = V^2.$$

- Dual problem:

$$D_f := \sup_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)} \mathbb{E}^\mu[f(S_1, V, S_2)].$$

- **Dispersion-constrained martingale optimal transport problem.**
- $\mathbb{E}^\mu[S_2|S_1, V] = S_1$: martingality condition of the SPX index, condition on the average of the distribution of S_2 given S_1 and V .
- $\mathbb{E}^\mu[L(S_2/S_1)|S_1, V] = V^2$: consistency condition, condition on dispersion around the average.

Strong duality for the VIX-constrained martingale optimal transport (Monge-Kantorovich) problem

Theorem (G. 2020)

Let $f : \mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0} \rightarrow \mathbb{R}$ be upper semicontinuous and satisfy

$$|f(s_1, v, s_2)| \leq C(1 + s_1 + s_2 + |L(s_1)| + |L(s_2)| + v^2)$$

for some constant $C > 0$. Then

$$\begin{aligned} P_f &:= \inf_{\mathcal{U}_f} \left\{ \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \right\} \\ &= \sup_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)} \mathbb{E}^\mu[f(S_1, V, S_2)] =: D_f. \end{aligned}$$

Moreover, $D_f \neq -\infty$ if and only if $\mathcal{P}(\mu_1, \mu_V, \mu_2) \neq \emptyset$, and in that case the supremum is attained.

Superreplication of forward-starting options

- The knowledge of μ_1 and μ_2 gives little information on the prices $\mathbb{E}^\mu[g(S_1, S_2)]$, e.g., prices of forward-starting options $\mathbb{E}^\mu[f(S_2/S_1)]$.
- Computing upper and lower bounds of these prices:
Optimal transport (Monge, 1781; Kantorovich, 1942)
- Adding the no-arbitrage constraint that (S_1, S_2) is a martingale leads to more precise bounds, as this provides information on the conditional average of S_2/S_1 given S_1 :
Martingale optimal transport (Henry-Labordère, 2017)
- When $S = \text{SPX}$: Adding VIX market data information produces even more precise bounds, as it information on the conditional dispersion of S_2/S_1 , which is controlled by the VIX V :
Dispersion-constrained martingale optimal transport (This talk)
- **Adding VIX market data may possibly reveal a joint SPX/VIX arbitrage. Corresponds to $\mathcal{P}(\mu_1, \mu_V, \mu_2) = \emptyset$** (see next slides).
- In the limiting case where $\mathcal{P}(\mu_1, \mu_V, \mu_2) = \{\mu_0\}$ is a singleton, the joint SPX/VIX market data information completely specifies the joint distribution of (S_1, S_2) , hence the price of forward starting options.

Contents

- 1 The particle method for smile calibration
- 2 The multi-asset case: calibrating local correlation models
- 3 VIX and VIX derivatives
- 4 Robust hedging of derivatives on SPX and/or VIX: VIX-constrained martingale optimal transport
- 5 Joint SPX/VIX arbitrages
- 6 Exact joint calibration of SPX and VIX smiles: VIX-constrained martingale Schrödinger problems/bridges

Joint SPX/VIX arbitrage: definition, characterization

Joint SPX/VIX arbitrage

- $\mathcal{U}_0$ = the portfolios $(u_1, u_2, u_V, \Delta^S, \Delta^L)$ superreplicating 0:

$$u_1(s_1) + u_2(s_2) + u_V(v) + \Delta_S(s_1, v)(s_2 - s_1) + \Delta_L(s_1, v) \left(L \left(\frac{s_2}{s_1} \right) - v^2 \right) \geq 0$$

- A joint SPX/VIX arbitrage, or (S_1, S_2, V) -arbitrage, is an element of $\mathcal{U}_0$ with negative price:

$$\text{MktPrice}[u_1(S_1)] + \text{MktPrice}[u_2(S_2)] + \text{MktPrice}[u_V(V)] < 0$$

- Equivalently, there is an (S_1, S_2, V) -arbitrage if and only if

$$\inf_{\mathcal{U}_0} \{ \text{MktPrice}[u_1(S_1)] + \text{MktPrice}[u_2(S_2)] + \text{MktPrice}[u_V(V)] \} = -\infty$$

Consistent extrapolation of SPX and VIX smiles

- If $\mathbb{E}^V[V^2] \neq \mathbb{E}^2[L(S_2)] - \mathbb{E}^1[L(S_1)]$, there is a trivial (S_1, S_2, V) -arbitrage. For instance, if $\mathbb{E}^V[V^2] < \mathbb{E}^2[L(S_2)] - \mathbb{E}^1[L(S_1)]$, pick

$$u_1(s_1) = L(s_1), \quad u_2(s_2) = -L(s_2), \quad u_V(v) = v^2, \quad \Delta_S(s_1, v) = 0, \quad \Delta_L(s_1, v) = 1.$$

- $\implies$ We assume that

$$\mathbb{E}^V[V^2] = \mathbb{E}^2[L(S_2)] - \mathbb{E}^1[L(S_1)]. \quad (5.1)$$

- Violations of (5.1) in the market have been reported, suggesting arbitrage opportunities, see, e.g., Section 7.7.4 in Bergomi (2016).
- However, the quantities in (5.1) do not purely depend on market data. They depend on smile extrapolations.
- The reported violations of (5.1) actually rely on some arbitrary smile extrapolations.
- G. (2018) explains how to build **consistent extrapolations of the VIX and SPX smiles** so that (5.1) holds.

Characterization of joint SPX/VIX arbitrage

Theorem (G. 2020)

The following assertions are equivalent:

- (i) *The market is free of (S_1, S_2, V) -arbitrage,*
- (ii) $\mathcal{P}(\mu_1, \mu_V, \mu_2) \neq \emptyset$,
- (iii) *There exists a coupling ν of μ_1 and μ_V such that $\text{Law}_\nu(S_1, L(S_1) + V^2)$ and $\text{Law}_{\mu_2}(S_2, L(S_2))$ are in convex order, i.e., for any convex function $f : \mathbb{R}_{>0} \times \mathbb{R} \rightarrow \mathbb{R}$,*

$$\mathbb{E}^\nu[f(S_1, L(S_1) + V^2)] \leq \mathbb{E}^2[f(S_2, L(S_2))].$$

Joint SPX/VIX arbitrage

- (i) The market is free of (S_1, S_2, V) -arbitrage,
 - (ii) $\mathcal{P}(\mu_1, \mu_V, \mu_2) \neq \emptyset$,
 - (iii) There exists a coupling ν of μ_1 and μ_V such that $\text{Law}_\nu(S_1, L(S_1) + V^2)$ and $\text{Law}_{\mu_2}(S_2, L(S_2))$ are in convex order.
- Directly solving the linear problem associated to (i) is not easy as one needs to try all possible $(u_1, u_V, u_2, \Delta_S, \Delta_V)$ and check the superreplication constraints for all $s_1, s_2 > 0$ and $v \geq 0$.
 - Checking (iii) numerically is difficult as, in dimension two, the extreme rays of the convex cone of convex functions are dense in the cone (Johansen 1974), contrary to the case of dimension one where the extreme rays are the call and put payoffs (Blaschke-Pick 1916).
 - Instead, we verify absence of (S_1, S_2, V) -arbitrage by building – numerically, but with high accuracy – an element of $\mathcal{P}(\mu_1, \mu_V, \mu_2)$, thus checking (ii).

Contents

- 1 The particle method for smile calibration
- 2 The multi-asset case: calibrating local correlation models
- 3 VIX and VIX derivatives
- 4 Robust hedging of derivatives on SPX and/or VIX: VIX-constrained martingale optimal transport
- 5 Joint SPX/VIX arbitrages
- 6 Exact joint calibration of SPX and VIX smiles: VIX-constrained martingale Schrödinger problems/bridges

Motivation

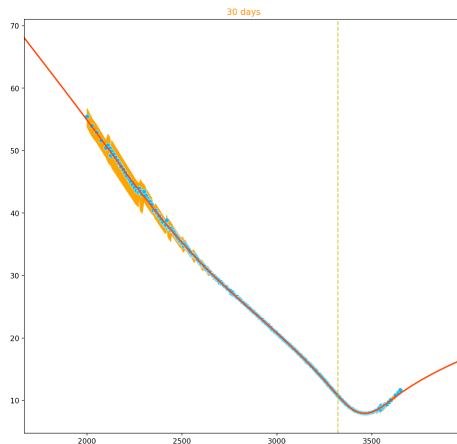


Figure: SPX smile as of January 22, 2020, $T = 30$ days

Motivation

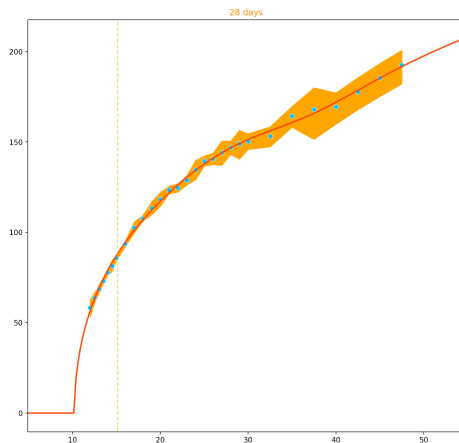


Figure: VIX smile as of January 22, 2020, $T = 28$ days

Motivation

■ ATM skew:

$$\text{Definition: } \mathcal{S}_T = \left. \frac{d\sigma_{BS}(K, T)}{\frac{dK}{K}} \right|_{K=F_T}$$

$$\text{SPX, small } T: \mathcal{S}_T \approx -1.5$$

$$\text{Classical one-factor SV model: } \mathcal{S}_T \xrightarrow{T \rightarrow 0} \frac{1}{2} \times \text{spot-vol correl} \times \text{vol-of-vol}$$

■ Calibration to short-term ATM SPX skew $\implies$

$$\text{vol-of-vol} \geq 3 = 300\% \gg \text{short-term ATM VIX implied vol}$$

The **very large negative skew of short-term SPX options**, which in classical continuous SV models implies a very large volatility of volatility, **seems inconsistent with the comparatively low levels of VIX implied volatilities**.

Gatheral (2008)

Consistent Modeling of SPX and VIX options

Consistent Modeling of SPX and VIX options

Jim Gatheral



The Fifth World Congress of the Bachelier Finance Society
London, July 18, 2008

Consistent Modeling of SPX and VIX options

Variance curve models

Double CEV dynamics and consistency

Double CEV dynamics

- Buehler's affine variance curve functional is consistent with double mean reverting dynamics of the form:

$$\begin{aligned}\frac{dS}{S} &= \sqrt{v} dW \\ dv &= -\kappa (v - v') dt + \eta_1 v^\alpha dZ_1 \\ dv' &= -c (v' - z_3) dt + \eta_2 v'^\beta dZ_2\end{aligned}\quad (2)$$

for any choice of $\alpha, \beta \in [1/2, 1]$.

- We will call the case $\alpha = \beta = 1/2$ *Double Heston*,
 - the case $\alpha = \beta = 1$ *Double Lognormal*,
 - and the general case *Double CEV*.
- All such models involve a short term variance level v that reverts to a moving level v' at rate κ . v' reverts to the long-term level z_3 at the slower rate $c < \kappa$.

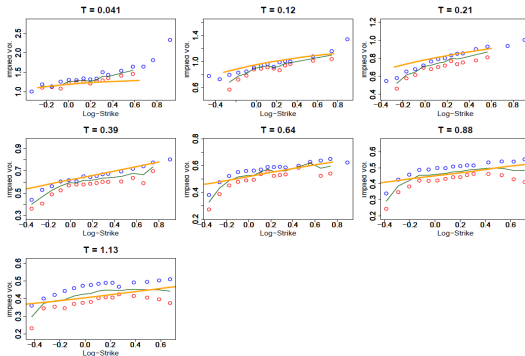
Consistent Modeling of SPX and VIX options

The Double CEV model

Calibration of ξ_1 , ξ_2 to VIX option prices

Double CEV fit to VIX options as of 03-Apr-2007

Setting the correlation ρ between volatility factors z_1 and z_2 to its historical average (see later) and iterating on the volatility of volatility parameters ξ_1 and ξ_2 to minimize the differences between model and market VIX option prices, we obtain the following fits (orange lines):



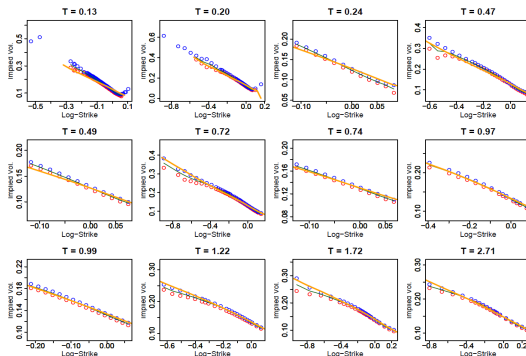
Consistent Modeling of SPX and VIX options

The Double CEV model

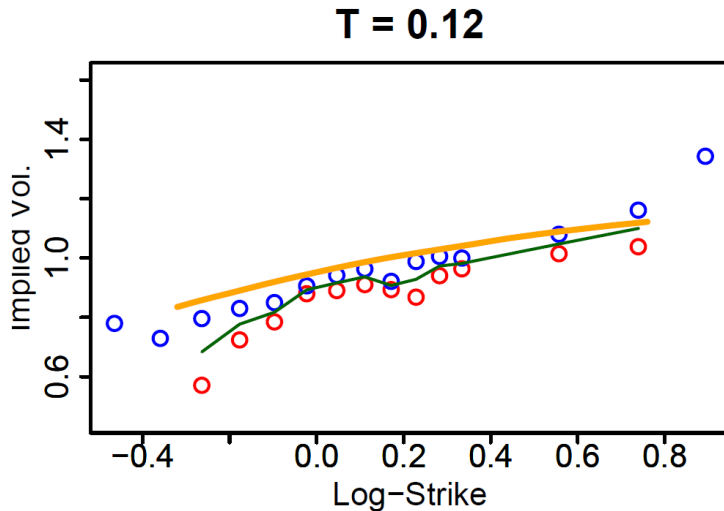
Calibration of ρ_1 and ρ_2 to SPX option prices

Double CEV fit to SPX options as of 03-Apr-2007

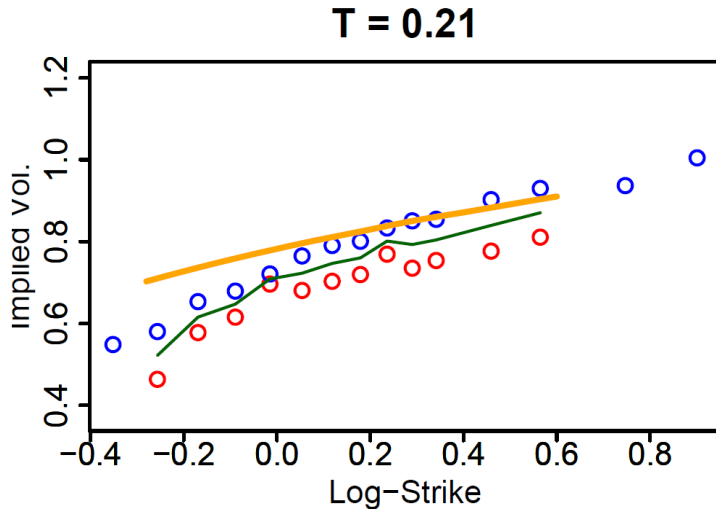
Minimizing the differences between model and market SPX option prices, we find $\rho_1 = -0.9$, $\rho_2 = -0.7$ and obtain the following fits to SPX option prices (orange lines):



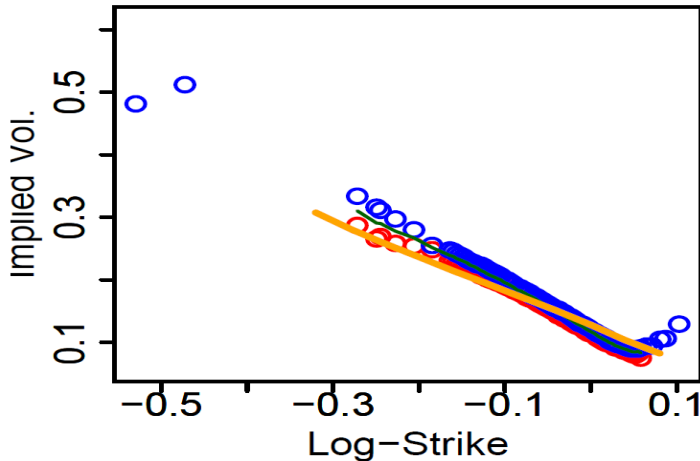
Fit to VIX options



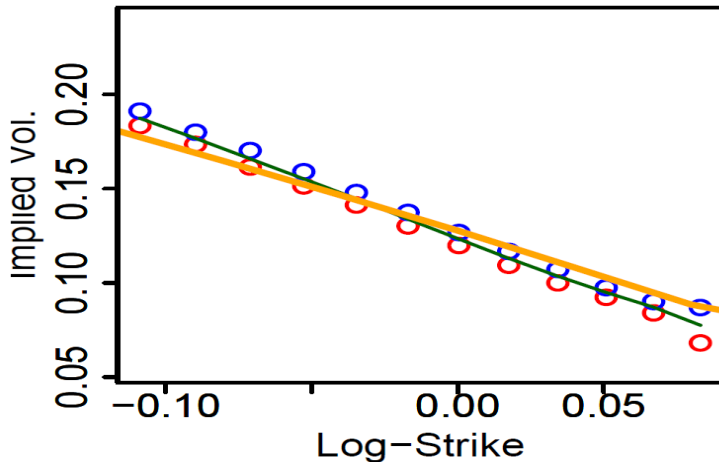
Fit to VIX options



Fit to SPX options

 $T = 0.13$ 

Fit to SPX options

 $T = 0.24$ 

Similar experiments with other models

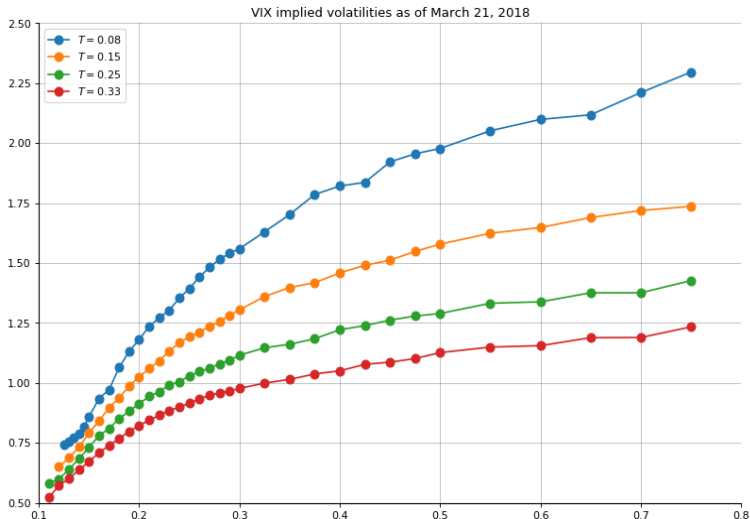
- Skewed 2-factor Bergomi model (Bergomi 2008)
- Skewed rough Bergomi model (G. 2018, De Marco 2018):

$$\sigma_t^2 = \xi_0^t \left((1 - \lambda) \mathcal{E} \left(\nu_0 \int_0^t (t - s)^{H - \frac{1}{2}} dZ_s \right) + \lambda \mathcal{E} \left(\nu_1 \int_0^t (t - s)^{H - 1/2} dZ_s \right) \right)$$

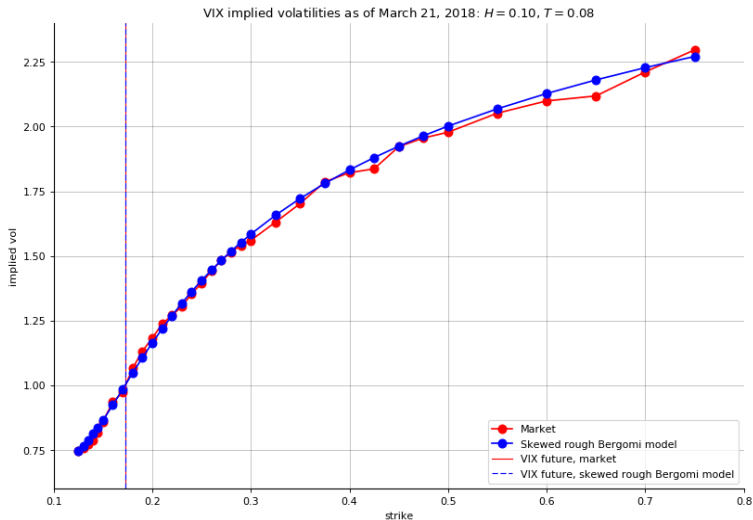
with $\lambda \in [0, 1]$.

- Quadratic rough Heston model (Gatheral Jusselin Rosenbaum 2020)
- **VIX smile well calibrated $\implies$ not enough SPX ATM skew**

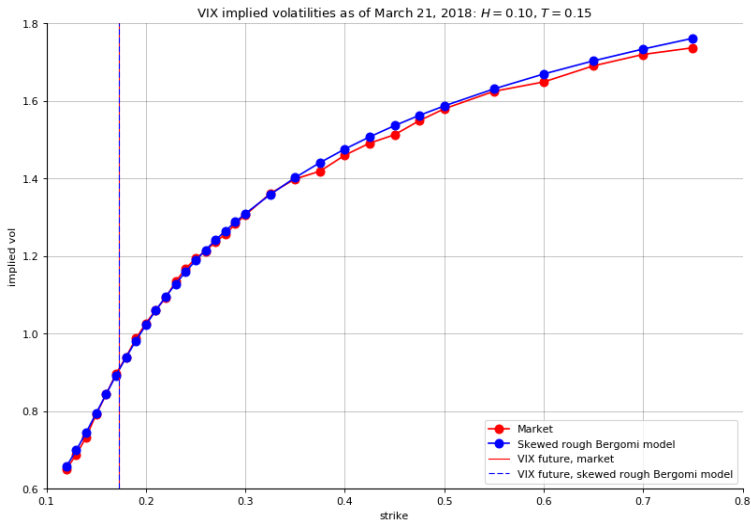
Skewed rough Bergomi: Calibration to VIX futures and options (G. 2018)



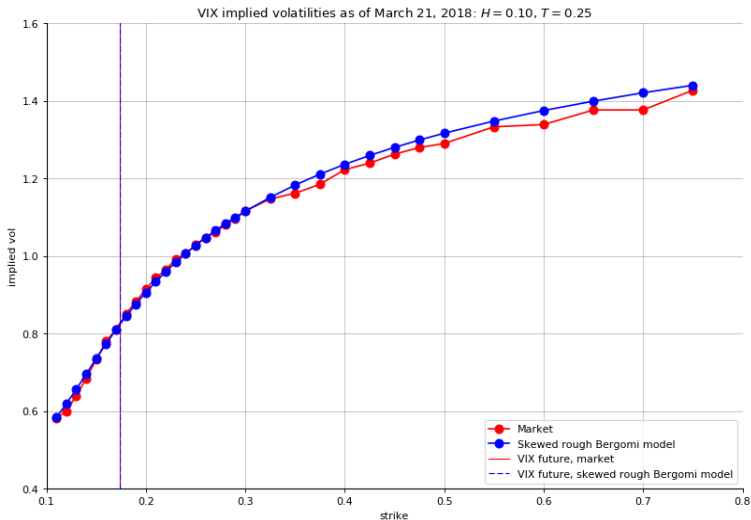
Skewed rough Bergomi: Calibration to VIX futures and options (G. 2018)



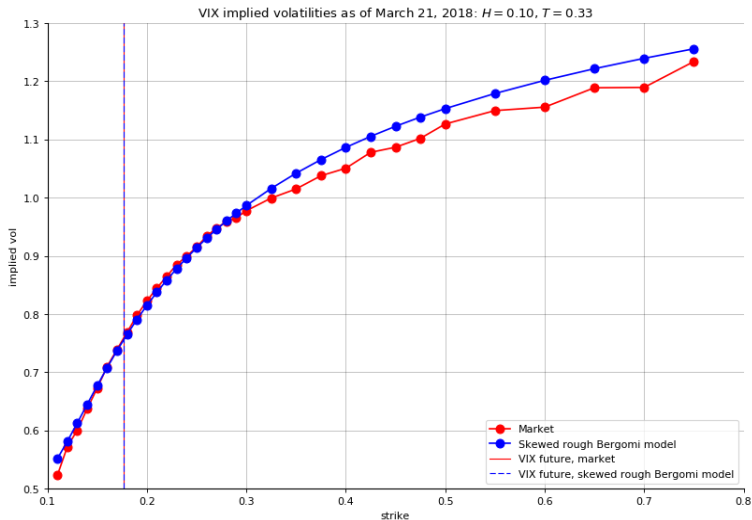
Skewed rough Bergomi: Calibration to VIX futures and options (G. 2018)



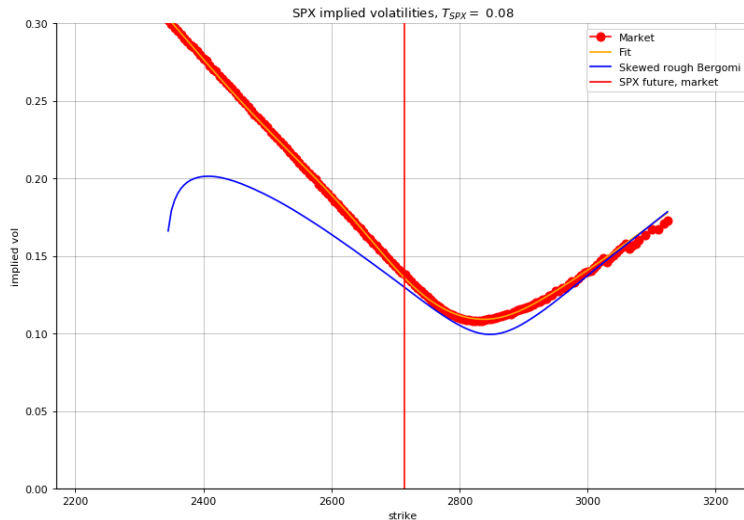
Skewed rough Bergomi: Calibration to VIX futures and options (G. 2018)



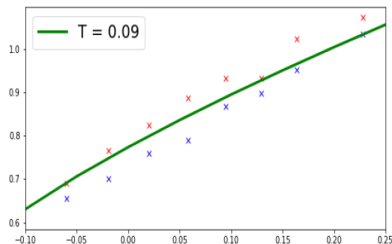
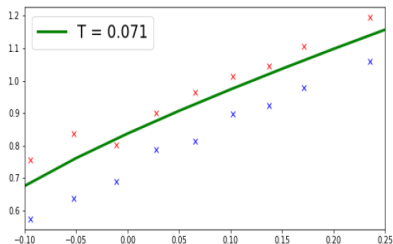
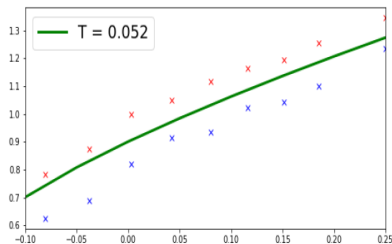
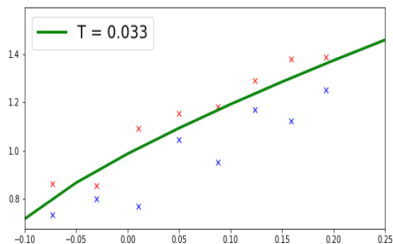
Skewed rough Bergomi: Calibration to VIX future and options (G. 2018)



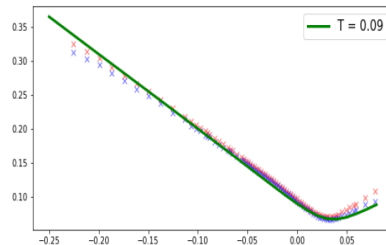
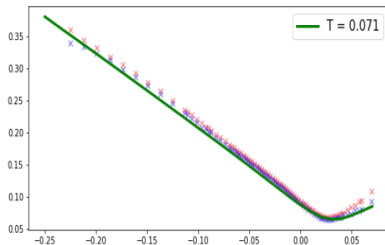
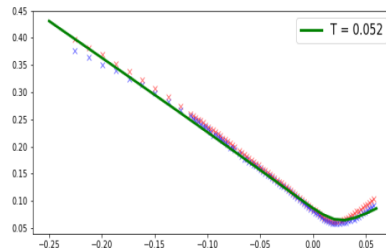
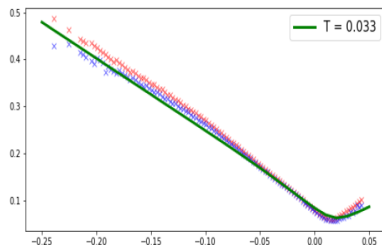
Skewed rough Bergomi calibrated to VIX: SPX smile (G. 2018)



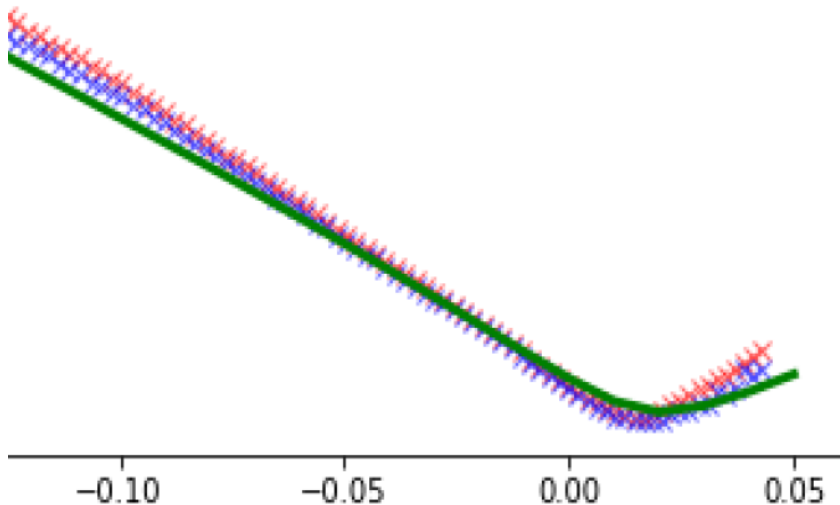
Quadratic rough Heston model (Gatheral Jusselin Rosenbaum 2020)



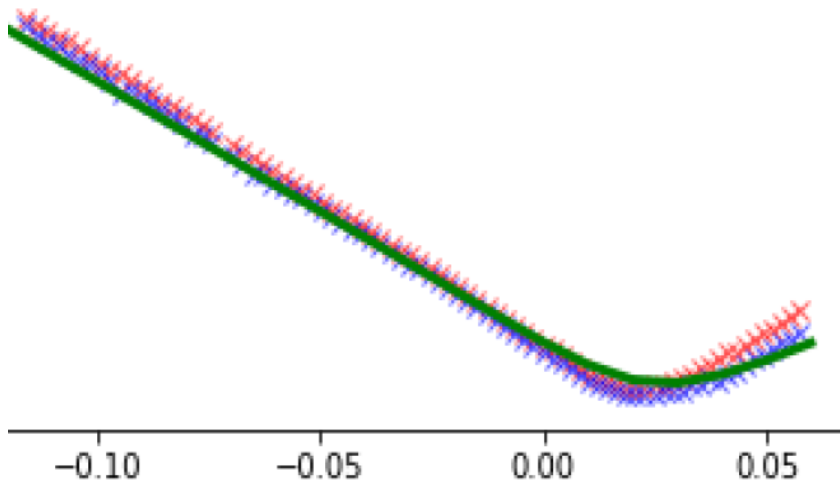
Quadratic rough Heston model (Gatheral Jusselin Rosenbaum 2020)



Quadratic rough Heston model (Gatheral Jusselin Rosenbaum 2020)



Quadratic rough Heston model (Gatheral Jusselin Rosenbaum 2020)



Joint calibration of 2-factor Bergomi model to term-structure of SPX ATM skew and VIX² implied vol (G. 2020)

THE VIX FUTURE IN BERGOMI MODELS: ANALYTIC EXPANSIONS AND JOINT CALIBRATION WITH S&P 500 SKEW

JULIEN GUYON
QUANTITATIVE RESEARCH, BLOOMBERG L.P.

ABSTRACT. We derive the expansion of the price of a VIX future in various Bergomi models at order 6 in small volatility-of-volatility. We introduce the notion of volatility of the VIX squared implied by the VIX future, which we call “VIX² implied volatility”, expand this quantity at order 5, and show that the implied volatility expansion converges much faster than the price expansion. We cover the one-factor, two-factor, and skewed two-factor Bergomi models and allow for maturity-dependent and/or time-dependent parameters. The expansions allow us to precisely pinpoint the roles of all the model parameters (volatility-of-volatility, mean reversion, correlations, mixing fraction) in the formation of the prices of VIX futures in Bergomi models. The derivation of the expansion naturally involves the (classical or dual bivariate) Hermite polynomials and exploits their orthogonality properties. When the initial term-structure of variance swaps is flat, the expansion is a closed-form expression; otherwise, it involves one-dimensional integrals which are extremely fast to compute. The VIX² implied volatility expansion is extremely precise for both the one-factor model and the two-factor model with independent factors, even for the very large values of volatility-of-volatility that are usual in equity derivatives markets, and can virtually be considered an exact formula in those cases. We use the new expansion together with the Bergomi-Guyon expansion of the S&P 500 smile to (instantaneously) calibrate the two-factor Bergomi model jointly to the term-structures of S&P 500 at-the-money skew and VIX² implied volatility. Our tests and the new expansion shed more light on the inability of traditional stochastic volatility models to jointly fit S&P 500 and VIX market data. The (imperfect but decent) joint fit requires much larger values of volatility-of-volatility and fast mean reversion than the ones previously reported in [10, 14].

Keywords. VIX, VIX futures, Bergomi models, VIX² implied volatility, analytic expansion, small volatility-of-volatility, at-the-money skew, S&P 500/VIX joint calibration, Hermite polynomials.

1. INTRODUCTION

Closed-form approximations are always very useful in mathematical modeling. They give insights on the structural properties of the models and the precise role of model parameters. They are computed in no time

Joint calibration of 2-factor Bergomi model to term-structure of SPX ATM skew and VIX² implied vol (G. 2020)

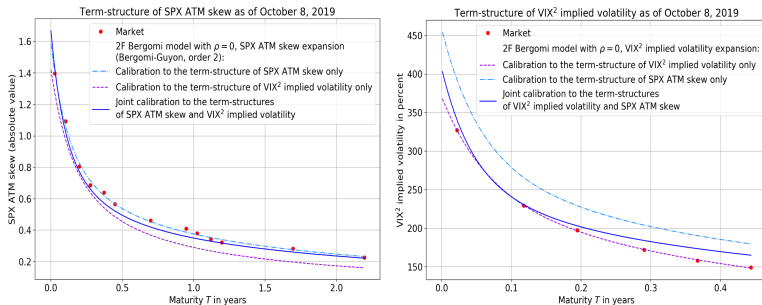


Figure: Left: ATM skew of SPX options as a function of maturity. Right: implied volatility of the squared VIX as a function of maturity. Calibration of the Bergomi-Guyon expansion of the SPX ATM skew and a newly derived expansion of the VIX² implied volatility, either jointly or separately. Calibration as of October 8, 2019

Joint calibration of 2-factor Bergomi model on October 8, 2019

Parameter	ω	k_1	k_2	θ_1	ρ	ρ_{S1}	ρ_{S2}
Calib to SPX ATM skew only	6.71	18.20	1.20	0.77	0.89	-0.92	-0.99
Calib to VIX ² implied vol first	6.24	19.25	1.59	0.75	0.65	-0.92	-0.89
Joint calibration	6.66	22.01	1.04	0.78	0.96	-0.99	-0.99

- **Large vol of vol**
- **Large fast mean-reversion**
- **All correlations ± 1** : a unique Brownian motion driving everything
 $\implies$ **0-factor SV model** $\longleftrightarrow$ **path-dependent volatility model**
- ω : vol of instantaneous variance
- k_1, k_2 : fast and slow mean-reversions
- θ_1 : mixing parameter
- ρ : long vol-short vol correlation
- ρ_{S1}, ρ_{S2} : spot-vol correlations

Related works with continuous models on the SPX

- Fouque-Saporito (2018), Heston with stochastic vol-of-vol. Problem: their approach does not apply to short maturities (below 4 months), for which VIX derivatives are most liquid and the joint calibration is most difficult.
- Goutte-Ismail-Pham (2017), Heston with parameters driven by a Hidden Markov jump process.
- Jacquier-Martini-Muguruza, *On the VIX futures in the rough Bergomi model* (2017):

"Interestingly, we observe a 20% difference between the [vol-of-vol] parameter obtained through VIX calibration and the one obtained through SPX. This suggests that the volatility of volatility in the SPX market is 20% higher when compared to VIX, revealing potential data inconsistencies (arbitrage?)."
- Guo-Loeper-Obłój-Wang (2020): joint calibration via semimartingale nonlinear optimal transport. **Closely related to VIX-constrained martingale Schrödinger bridges.**

Motivation

- To try to jointly fit the SPX and VIX smiles, many authors have incorporated **jumps** in the dynamics of the SPX: Sepp, Cont-Kokholm, Papanicolaou-Sircar, Baldeaux-Badran, Pacati-Pompa-Renò, Kokholm-Stisen, Bardgett-Gourier-Leippold...
- Jumps offer extra degrees of freedom to partly decouple the ATM SPX skew and the ATM VIX implied volatility.
- So far all the attempts at solving the joint SPX/VIX smile calibration problem only produced an **approximate fit**.

Exact joint SPX/VIX smile calibration: a dispersion-constrained martingale Schrödinger problem approach

(G. 2019)

Exact joint calibration as a dispersion-constrained martingale Schrödinger problem (G. 2019)

- **A completely different approach**: instead of postulating a parametric continuous-time (jump-)diffusion model on the SPX, we build a **nonparametric discrete-time model**.
- Given a VIX future maturity T_1 , we build a **joint probability measure on (S_1, V, S_2)** which is **perfectly calibrated** to the SPX smiles at T_1 and $T_2 = T_1 + 30$ days, and the VIX future and VIX smile at T_1 .
- S_1 : SPX at T_1 , V : VIX at T_1 , S_2 : SPX at T_2 .
- Our model satisfies:
 - **Martingality constraint** on the SPX;
 - **Consistency condition**: the VIX at T_1 is the implied volatility of the 30-day log-contract on the SPX.
- Our model is cast as the solution of a **dispersion-constrained martingale Schrödinger problem** which is solved using the **Sinkhorn algorithm**, in the spirit of De March and Henry-Labordère (2019).

Risk, April 2020

The joint S&P 500/Vix smile calibration puzzle solved

Since Vix options started trading in 2006, many researchers have tried to build a model that jointly and exactly calibrates to the prices of Standard & Poor's 500 options, Vix futures and Vix options. In this article, Julien Guyon solves this long-standing puzzle by casting it as a discrete-time dispersion-constrained martingale transport problem, which he solves in a non-parametric way using Sinkhorn's algorithm

Volatility indexes, such as the Vix index, do not just serve as market-implied indicators of volatility. Futures and options on these indexes are also widely used as risk management tools to hedge the volatility exposure of options portfolios. The existence of a liquid market for these futures and options has led to the need for models that jointly calibrate to the prices of options on the underlying asset and the prices of volatility derivatives. Without such models, financial institutions could possibly arbitrage each other: even market-making desks within the same institution could do so, eg, the Vix desk could arbitrage the S&P 500 (SPX) desk. By using models that fail to correctly incorporate the prices of the hedging instruments, such as SPX options, Vix futures and Vix options, exotic desks may misprice options, especially (but not only) those with payoffs that involve both the underlying and its volatility index.

For this reason, since Vix options began trading in 2006, many researchers and practitioners have tried to build a model that jointly and exactly calibrates to the prices of SPX futures, SPX options, Vix futures and Vix options. This is known to be a very challenging problem, especially for short maturities. In particular, the very large negative skew of short-term SPX options,

and Vix smiles: that the distribution of the Dupire market local variance be smaller than the distribution of the (instantaneous) Vix squared in the convex order, at all times. He also reported that for short maturities the distribution of the true Vix squared in the market local volatility model is actually larger than the market-implied distribution of the true Vix squared in the convex order. Guyon showed numerically that when the (typically negative) spot-vol correlation is large enough in absolute value, both (a) traditional stochastic volatility models with large mean reversion and (b) rough volatility models with a small Hurst exponent can reproduce this inversion of convex ordering. Acciaio & Guyon (2020) provide a mathematical proof that the inversion of convex ordering can be produced by continuous models. However, the inversion of convex ordering is only a necessary condition for the joint SPX/Vix calibration of continuous models; it is not sufficient.

Since it looks to be very difficult to jointly calibrate the SPX and Vix smiles with continuous models, many authors have incorporated jumps in the dynamics of the SPX: see references in Guyon (2019a). Jumps offer extra degrees of freedom to partly decouple the ATM SPX skew and the ATM Vix implied volatility. However, the attempts so far have only produced

Preprint, May 2021

DISPERSION-CONSTRAINED MARTINGALE SCHRÖDINGER PROBLEMS
AND THE EXACT JOINT S&P 500/VIX SMILE CALIBRATION PUZZLE*JULIEN GUYON
QUANTITATIVE RESEARCH, BLOOMBERG L.P.

ABSTRACT. We solve for the first time* a longstanding puzzle of quantitative finance that has often been described as the Holy Grail of volatility modeling: build a model that jointly and exactly calibrates to the prices of S&P 500 (SPX) options, VIX futures, and VIX options. So far the best attempts, which used parametric continuous-time (jump-)diffusion models on the SPX, only produced approximate fits. We use a very different, nonparametric and discrete-time approach. Given a VIX future maturity T_1 , we consider the set $\mathcal{P}$ of all joint probability measures on the SPX at T_1 , the VIX at T_1 , and the SPX at $T_2 = T_1 + 30$ days which are perfectly calibrated to the full SPX smiles at T_1 and T_2 , and the full VIX smile at T_1 , and which also satisfy the martingality constraint on the SPX as well as the requirement that the VIX is the implied volatility of the 30-day log-contract on the SPX.

We first consider robust hedging in this setting. By casting the superreplication problem as what we call a *dispersion-constrained martingale optimal transport problem*, we establish a strong duality theorem and, as a result, prove that the absence of joint SPX/VIX arbitrage is equivalent to the set $\mathcal{P}$ of jointly calibrating models being nonempty. Should they arise, joint arbitrages are identified using classical linear programming. In the absence of joint arbitrage, we then provide a solution to the joint calibration puzzle by solving what we call a *dispersion-constrained martingale Schrödinger problem*: we choose a reference measure and build the unique jointly calibrating model that minimizes the relative entropy. We establish several dual versions of the problem, one of which has a natural financial interpretation in terms of exponential utility indifference pricing, and prove absence of duality gaps. The minimum entropy jointly calibrating model is explicit in terms of what we call the dual *Schrödinger portfolio*, i.e., the maximizer of the dual problems, should it exist. We numerically compute this Schrödinger portfolio using an extension of the Sinkhorn algorithm, in the spirit of De March and Henry-Labordère (2019). Our numerical experiments show that the algorithm performs very well in both low and high volatility regimes.

Along the way, we provide new variants, as well as a new proof, of strong duality theorems for the classical Schrödinger problem and for a mixed *Schrödinger-Monge-Kantorovich problem* (also known as *entropic optimal transport problem*) that has recently attracted a lot of attention in the optimal transport community, which are interesting in themselves. Our methodology applies not only to the VIX, but also to any index computed as a function of the price of an option on some underlying asset.

**Solve the joint calibration puzzle:
build a model in $\mathcal{P}(\mu_1, \mu_V, \mu_2)$
as the solution of a
dispersion-constrained martingale
Schrödinger problem**

Build a model in $\mathcal{P}(\mu_1, \mu_V, \mu_2)$

- Recall $\mathcal{P}(\mu_1, \mu_V, \mu_2) :=$ probability measures on $\mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0}$ s.t.

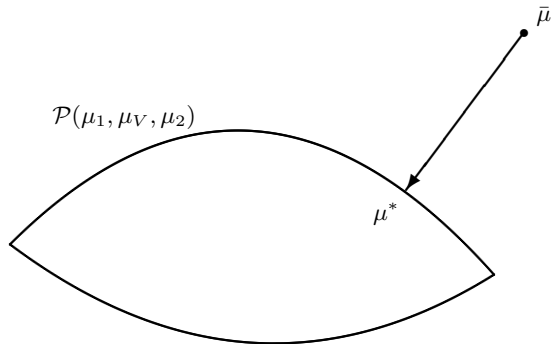
$$S_1 \sim \mu_1, \quad V \sim \mu_V, \quad S_2 \sim \mu_2, \quad \mathbb{E}^\mu[S_2|S_1, V] = S_1, \quad \mathbb{E}^\mu \left[L \left(\frac{S_2}{S_1} \right) \middle| S_1, V \right] = V^2.$$

- Assume absence of joint SPX/VIX arbitrage: $\mathcal{P}(\mu_1, \mu_V, \mu_2) \neq \emptyset$.
- Build a model $\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)$ = solve the joint calibration puzzle.**
- Our strategy is inspired by Avellaneda (1998, 2001) and De March and Henry-Labordère (2019).
- To build an element $\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)$, we choose a **reference probability measure $\bar{\mu}$** on $\mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0}$ and look for the measure $\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)$ that **minimizes the relative entropy $H(\mu|\bar{\mu})$** of μ w.r.t. $\bar{\mu}$, also known as the Kullback-Leibler divergence:

$$D_{\bar{\mu}} := \inf_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)} H(\mu|\bar{\mu}), \quad H(\mu|\bar{\mu}) := \begin{cases} \mathbb{E}^\mu \left[\ln \frac{d\mu}{d\bar{\mu}} \right] = \mathbb{E}^{\bar{\mu}} \left[\frac{d\mu}{d\bar{\mu}} \ln \frac{d\mu}{d\bar{\mu}} \right] & \text{if } \mu \ll \bar{\mu}, \\ +\infty & \text{otherwise.} \end{cases}$$

- This is a **strictly convex problem that can be solved after dualization using, e.g., Sinkhorn's fixed point iteration** (Sinkhorn, 1967).

Build a model in $\mathcal{P}(\mu_1, \mu_V, \mu_2)$



Reminder on Lagrange multipliers

$$\begin{aligned} \inf_{g(x,y)=c} f(x,y) &= \inf_{x,y} \sup_{\lambda \in \mathbb{R}} \{f(x,y) - \lambda(g(x,y) - c)\} \\ &= \sup_{\lambda \in \mathbb{R}} \inf_{x,y} \{f(x,y) - \lambda(g(x,y) - c)\} \end{aligned}$$

- To compute the **inner inf over x, y unconstrained**, simply solve $\nabla f(x, y) = \lambda \nabla g(x, y)$: easy!
- Then **maximize the result over λ unconstrained**: easy!
- Constraint $g(x, y) = c \iff \frac{\partial}{\partial \lambda} \{f(x, y) - \lambda(g(x, y) - c)\} = 0$.

$$\inf_{\mu \text{ s.t. } S_1 \sim \mu_1} H(\mu | \bar{\mu}) = \inf_{\mu} \sup_{u_1(\cdot)} \left\{ H(\mu | \bar{\mu}) + \mathbb{E}^1[u_1(S_1)] - \mathbb{E}^\mu[u_1(S_1)] \right\}$$

$$\inf_{\mu \text{ s.t. } \mathbb{E}^\mu[S_2 | S_1, V] = S_1} H(\mu | \bar{\mu}) = \inf_{\mu} \sup_{\Delta_S(\cdot, \cdot)} \left\{ H(\mu | \bar{\mu}) - \mathbb{E}^\mu[\Delta_S(S_1, V)(S_2 - S_1)] \right\}$$

$$\inf_{\mu \text{ s.t. } \mathbb{E}^\mu \left[L \left(\frac{S_2}{S_1} \right) \middle| S_1, V \right] = V^2} H(\mu | \bar{\mu}) = \inf_{\mu} \sup_{\Delta_L(\cdot, \cdot)} \left\{ H(\mu | \bar{\mu}) - \mathbb{E}^\mu \left[\Delta_L(S_1, V) \left(L \left(\frac{S_2}{S_1} \right) - V^2 \right) \right] \right\}$$

Reminder on Lagrange multipliers

$$\begin{aligned} \inf_{g(x,y)=c} f(x,y) &= \inf_{x,y} \sup_{\lambda \in \mathbb{R}} \{f(x,y) - \lambda(g(x,y) - c)\} \\ &= \sup_{\lambda \in \mathbb{R}} \inf_{x,y} \{f(x,y) - \lambda(g(x,y) - c)\} \end{aligned}$$

- To compute the inner inf over x, y unconstrained, simply solve $\nabla f(x, y) = \lambda \nabla g(x, y)$: easy!
- Then maximize the result over λ unconstrained: easy!
- Constraint $g(x, y) = c \iff \frac{\partial}{\partial \lambda} \{f(x, y) - \lambda(g(x, y) - c)\} = 0$.

$$\inf_{\mu \text{ s.t. } S_1 \sim \mu_1} H(\mu|\bar{\mu}) = \inf_{\mu} \sup_{u_1(\cdot)} \left\{ H(\mu|\bar{\mu}) + \mathbb{E}^1[u_1(S_1)] - \mathbb{E}^\mu[u_1(S_1)] \right\}$$

$$\inf_{\mu \text{ s.t. } \mathbb{E}^\mu[S_2|S_1, V]=S_1} H(\mu|\bar{\mu}) = \inf_{\mu} \sup_{\Delta_S(\cdot, \cdot)} \left\{ H(\mu|\bar{\mu}) - \mathbb{E}^\mu[\Delta_S(S_1, V)(S_2 - S_1)] \right\}$$

$$\inf_{\mu \text{ s.t. } \mathbb{E}^\mu\left[L\left(\frac{S_2}{S_1}\right) \middle| S_1, V\right]=V^2} H(\mu|\bar{\mu}) = \inf_{\mu} \sup_{\Delta_L(\cdot, \cdot)} \left\{ H(\mu|\bar{\mu}) - \mathbb{E}^\mu\left[\Delta_L(S_1, V) \left(L\left(\frac{S_2}{S_1}\right) - V^2\right)\right] \right\}$$

Build a model in $\mathcal{P}(\mu_1, \mu_V, \mu_2)$

- $\mathcal{M}_1$: set of probability measures on $\mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0}$: **unconstrained**
- $\mathcal{U}$: set of portfolios $u = (u_1, u_V, u_2, \Delta_S, \Delta_L)$: **Lagrange multipliers**

$$\begin{aligned}
 D_{\bar{\mu}} &:= \inf_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)} H(\mu | \bar{\mu}) \\
 &= \inf_{\mu \in \mathcal{M}_1} \sup_{u \in \mathcal{U}} \left\{ H(\mu | \bar{\mu}) + \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \right. \\
 &\quad \left. - \mathbb{E}^\mu \left[u_1(S_1) + u_V(V) + u_2(S_2) + \Delta_S^{(S)}(S_1, V, S_2) + \Delta_L^{(L)}(S_1, V, S_2) \right] \right\} \\
 &= \sup_{u \in \mathcal{U}} \inf_{\mu \in \mathcal{M}_1} \left\{ H(\mu | \bar{\mu}) + \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \right. \\
 &\quad \left. - \mathbb{E}^\mu \left[u_1(S_1) + u_V(V) + u_2(S_2) + \Delta_S^{(S)}(S_1, V, S_2) + \Delta_L^{(L)}(S_1, V, S_2) \right] \right\}
 \end{aligned}$$

Build a model in $\mathcal{P}(\mu_1, \mu_V, \mu_2)$

- $\mathcal{M}_1$: set of probability measures on $\mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0}$: **unconstrained**
- $\mathcal{U}$: set of portfolios $u = (u_1, u_V, u_2, \Delta_S, \Delta_L)$: **Lagrange multipliers**

$$\begin{aligned}
 D_{\bar{\mu}} &:= \inf_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)} H(\mu | \bar{\mu}) \\
 &= \inf_{\mu \in \mathcal{M}_1} \sup_{u \in \mathcal{U}} \left\{ H(\mu | \bar{\mu}) + \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \right. \\
 &\quad \left. - \mathbb{E}^\mu \left[u_1(S_1) + u_V(V) + u_2(S_2) + \Delta_S^{(S)}(S_1, V, S_2) + \Delta_L^{(L)}(S_1, V, S_2) \right] \right\} \\
 &= \sup_{u \in \mathcal{U}} \inf_{\mu \in \mathcal{M}_1} \left\{ H(\mu | \bar{\mu}) + \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \right. \\
 &\quad \left. - \mathbb{E}^\mu \left[u_1(S_1) + u_V(V) + u_2(S_2) + \Delta_S^{(S)}(S_1, V, S_2) + \Delta_L^{(L)}(S_1, V, S_2) \right] \right\}
 \end{aligned}$$

Build a model in $\mathcal{P}(\mu_1, \mu_V, \mu_2)$

- $\mathcal{M}_1$: set of probability measures on $\mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0}$: **unconstrained**
- $\mathcal{U}$: set of portfolios $u = (u_1, u_V, u_2, \Delta_S, \Delta_L)$: **Lagrange multipliers**

$$\begin{aligned}
 D_{\bar{\mu}} &:= \inf_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)} H(\mu | \bar{\mu}) \\
 &= \inf_{\mu \in \mathcal{M}_1} \sup_{u \in \mathcal{U}} \left\{ H(\mu | \bar{\mu}) + \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \right. \\
 &\quad \left. - \mathbb{E}^\mu \left[u_1(S_1) + u_V(V) + u_2(S_2) + \Delta_S^{(S)}(S_1, V, S_2) + \Delta_L^{(L)}(S_1, V, S_2) \right] \right\} \\
 &= \sup_{u \in \mathcal{U}} \inf_{\mu \in \mathcal{M}_1} \left\{ H(\mu | \bar{\mu}) + \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \right. \\
 &\quad \left. - \mathbb{E}^\mu \left[u_1(S_1) + u_V(V) + u_2(S_2) + \Delta_S^{(S)}(S_1, V, S_2) + \Delta_L^{(L)}(S_1, V, S_2) \right] \right\}
 \end{aligned}$$

Build a model in $\mathcal{P}(\mu_1, \mu_V, \mu_2)$

- $\mathcal{M}_1$: set of probability measures on $\mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0}$: **unconstrained**
- $\mathcal{U}$: set of portfolios $u = (u_1, u_V, u_2, \Delta_S, \Delta_L)$: **Lagrange multipliers**

$$\begin{aligned}
 D_{\bar{\mu}} &:= \inf_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)} H(\mu | \bar{\mu}) \\
 &= \inf_{\mu \in \mathcal{M}_1} \sup_{u \in \mathcal{U}} \left\{ H(\mu | \bar{\mu}) + \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \right. \\
 &\quad \left. - \mathbb{E}^\mu \left[u_1(S_1) + u_V(V) + u_2(S_2) + \Delta_S^{(S)}(S_1, V, S_2) + \Delta_L^{(L)}(S_1, V, S_2) \right] \right\} \\
 &= \sup_{u \in \mathcal{U}} \inf_{\mu \in \mathcal{M}_1} \left\{ H(\mu | \bar{\mu}) + \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \right. \\
 &\quad \left. - \mathbb{E}^\mu \left[u_1(S_1) + u_V(V) + u_2(S_2) + \Delta_S^{(S)}(S_1, V, S_2) + \Delta_L^{(L)}(S_1, V, S_2) \right] \right\}
 \end{aligned}$$

Build a model in $\mathcal{P}(\mu_1, \mu_V, \mu_2)$

- $\mathcal{M}_1$: set of probability measures on $\mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0}$: **unconstrained**
- $\mathcal{U}$: set of portfolios $u = (u_1, u_V, u_2, \Delta_S, \Delta_L)$: **Lagrange multipliers**

$$\begin{aligned}
 D_{\bar{\mu}} &:= \inf_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)} H(\mu | \bar{\mu}) \\
 &= \inf_{\mu \in \mathcal{M}_1} \sup_{u \in \mathcal{U}} \left\{ H(\mu | \bar{\mu}) + \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \right. \\
 &\quad \left. - \mathbb{E}^\mu \left[u_1(S_1) + u_V(V) + u_2(S_2) + \Delta_S^{(S)}(S_1, V, S_2) + \Delta_L^{(L)}(S_1, V, S_2) \right] \right\} \\
 &= \sup_{u \in \mathcal{U}} \inf_{\mu \in \mathcal{M}_1} \left\{ H(\mu | \bar{\mu}) + \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \right. \\
 &\quad \left. - \mathbb{E}^\mu \left[u_1(S_1) + u_V(V) + u_2(S_2) + \Delta_S^{(S)}(S_1, V, S_2) + \Delta_L^{(L)}(S_1, V, S_2) \right] \right\}
 \end{aligned}$$

Build a model in $\mathcal{P}(\mu_1, \mu_V, \mu_2)$

- $\mathcal{M}_1$: set of probability measures on $\mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0}$: **unconstrained**
- $\mathcal{U}$: set of portfolios $u = (u_1, u_V, u_2, \Delta_S, \Delta_L)$: **Lagrange multipliers**

$$\begin{aligned}
 D_{\bar{\mu}} &:= \inf_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)} H(\mu | \bar{\mu}) \\
 &= \inf_{\mu \in \mathcal{M}_1} \sup_{u \in \mathcal{U}} \left\{ H(\mu | \bar{\mu}) + \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \right. \\
 &\quad \left. - \mathbb{E}^\mu \left[u_1(S_1) + u_V(V) + u_2(S_2) + \Delta_S^{(S)}(S_1, V, S_2) + \Delta_L^{(L)}(S_1, V, S_2) \right] \right\} \\
 &= \sup_{u \in \mathcal{U}} \inf_{\mu \in \mathcal{M}_1} \left\{ H(\mu | \bar{\mu}) + \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \right. \\
 &\quad \left. - \mathbb{E}^\mu \left[u_1(S_1) + u_V(V) + u_2(S_2) + \Delta_S^{(S)}(S_1, V, S_2) + \Delta_L^{(L)}(S_1, V, S_2) \right] \right\}
 \end{aligned}$$

Build a model in $\mathcal{P}(\mu_1, \mu_V, \mu_2)$

- $\mathcal{M}_1$: set of probability measures on $\mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0}$: **unconstrained**
- $\mathcal{U}$: set of portfolios $u = (u_1, u_V, u_2, \Delta_S, \Delta_L)$: **Lagrange multipliers**

$$\begin{aligned}
 D_{\bar{\mu}} &:= \inf_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)} H(\mu | \bar{\mu}) \\
 &= \inf_{\mu \in \mathcal{M}_1} \sup_{u \in \mathcal{U}} \left\{ H(\mu | \bar{\mu}) + \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \right. \\
 &\quad \left. - \mathbb{E}^\mu \left[u_1(S_1) + u_V(V) + u_2(S_2) + \Delta_S^{(S)}(S_1, V, S_2) + \Delta_L^{(L)}(S_1, V, S_2) \right] \right\} \\
 &= \sup_{u \in \mathcal{U}} \inf_{\mu \in \mathcal{M}_1} \left\{ H(\mu | \bar{\mu}) + \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \right. \\
 &\quad \left. - \mathbb{E}^\mu \left[u_1(S_1) + u_V(V) + u_2(S_2) + \Delta_S^{(S)}(S_1, V, S_2) + \Delta_L^{(L)}(S_1, V, S_2) \right] \right\}
 \end{aligned}$$

Build a model in $\mathcal{P}(\mu_1, \mu_V, \mu_2)$

$$D_{\bar{\mu}} = \sup_{u \in \mathcal{U}} \inf_{\mu \in \mathcal{M}_1} \left\{ H(\mu | \bar{\mu}) + \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \right. \\ \left. - \mathbb{E}^\mu \left[u_1(S_1) + u_V(V) + u_2(S_2) + \Delta_S^{(S)}(S_1, V, S_2) + \Delta_L^{(L)}(S_1, V, S_2) \right] \right\}$$

- **Remarkable fact: The inner infimum can be exactly computed:**

$$\inf_{\mu \in \mathcal{M}_1} \{ H(\mu | \bar{\mu}) - \mathbb{E}^\mu[X] \} = -\ln \mathbb{E}^{\bar{\mu}} [e^X]$$

and **the infimum is attained at $\mu = \bar{\mu}_X$ defined by** (Gibbs type)

$$\frac{d\bar{\mu}_X}{d\bar{\mu}} = \frac{e^X}{\mathbb{E}^{\bar{\mu}}[e^X]}.$$

- That is why we like (and chose) the “distance” $H(\mu | \bar{\mu})$!

Build a model in $\mathcal{P}(\mu_1, \mu_V, \mu_2)$

$$D_{\bar{\mu}} := \inf_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)} H(\mu | \bar{\mu}) = \sup_{u \in \mathcal{U}} J_{\bar{\mu}}^{\text{ln}}(u) =: P_{\bar{\mu}}$$

$$J_{\bar{\mu}}^{\text{ln}}(u) := \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] \\ - \ln \mathbb{E}^{\bar{\mu}} \left[e^{u_1(S_1) + u_V(V) + u_2(S_2) + \Delta_S^{(S)}(S_1, V, S_2) + \Delta_L^{(L)}(S_1, V, S_2)} \right].$$

- $\inf_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)}$: **constrained** optimization, **difficult**.
- $\sup_{u \in \mathcal{U}}$: **unconstrained** optimization, **easy!** If sup is attained, to find the optimum $u^* = (u_1^*, u_V^*, u_2^*, \Delta_S^*, \Delta_L^*)$, simply cancel the gradient of $J_{\bar{\mu}}^{\text{ln}}$.
- Most important, $\inf_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)} H(\mu | \bar{\mu})$ is reached at

$$\mu^*(ds_1, dv, ds_2) = \bar{\mu}(ds_1, dv, ds_2) \frac{e^{u_1^*(s_1) + u_V^*(v) + u_2^*(s_2) + \Delta_S^{*(S)}(s_1, v, s_2) + \Delta_L^{*(L)}(s_1, v, s_2)}}{\mathbb{E}^{\bar{\mu}} \left[e^{u_1^*(S_1) + u_V^*(V) + u_2^*(S_2) + \Delta_S^{*(S)}(S_1, V, S_2) + \Delta_L^{*(L)}(S_1, V, S_2)} \right]}.$$

- **Problem solved:** $\mu^* \in \mathcal{P}(\mu_1, \mu_V, \mu_2)$!

Strong duality for the VIX-constrained martingale Schrödinger problem

$$D_{\bar{\mu}} := \inf_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)} H(\mu | \bar{\mu}) = \sup_{u \in \mathcal{U}} J_{\bar{\mu}}^{\text{In}}(u) =: P_{\bar{\mu}}$$

$$J_{\bar{\mu}}^{\text{In}}(u) := \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] - \text{In} \mathbb{E}^{\bar{\mu}} \left[e^{(u_1 \oplus u_V \oplus u_2 + \Delta_S^{(S)} + \Delta_L^{(L)})(S_1, V, S_2)} \right]$$

Generalization:

$$\mathcal{G} := \{g : (0, +\infty] \rightarrow (-\infty, +\infty] \mid \forall x \in (0, +\infty], g(x) \geq \ln x \text{ and } g(1) = 0\}$$

For $g \in \mathcal{G}$ we define $J_{\bar{\mu}}^g : \mathcal{U} = L^1 B \rightarrow [-\infty, +\infty)$ by

$$J_{\bar{\mu}}^g(u) := \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] - g \left(\mathbb{E}^{\bar{\mu}} \left[e^{(u_1 \oplus u_V \oplus u_2 + \Delta_S^{(S)} + \Delta_L^{(L)})(S_1, V, S_2)} \right] \right)$$

$$Z_{\bar{\mu}}(u) := \mathbb{E}^{\bar{\mu}} \left[e^{(u_1 \oplus u_V \oplus u_2 + \Delta_S^{(S)} + \Delta_L^{(L)})(S_1, V, S_2)} \right]$$

For $E \in \{L^1 B, CC_b, C_b C_b\}$, we denote $E_{\text{exp}} := \{u \in E \mid Z_{\bar{\mu}}(u) < +\infty\}$.

Strong duality for the VIX-constrained martingale Schrödinger problem

Theorem (G. 2020)

Let $\bar{\mu} \in \mathcal{M}_1$ and $g \in \mathcal{G}$. The following equality holds in $[0, +\infty]$:

$$\sup_{u \in CC_b} J_{\bar{\mu}}^g(u) = \sup_{u \in L^1 B} J_{\bar{\mu}}^g(u) = \inf_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)} H(\mu | \bar{\mu}).$$

Moreover:

- 1 The stronger $C_b C_b$ -duality $\sup_{u \in C_b C_b} J_{\bar{\mu}}^g(u) = \inf_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)} H(\mu | \bar{\mu})$ holds if for all $(\Delta_S, \Delta_L) \in C_b(\mathcal{X}_1 \times \mathcal{X}_V)^2$,
$$\mathbb{E}^{\bar{\mu}} \left[e^{(\Delta_S^{(S)} + \Delta_L^{(L)})(S_1, V, S_2)} \right] < +\infty.$$
- 2 If $\mathcal{P}(\mu_1, \mu_V, \mu_2) \neq \emptyset$, then the infimum is attained.
- 3 If the problem is finite, then $\mathcal{P}(\mu_1, \mu_V, \mu_2) \neq \emptyset$ and the infimum is uniquely attained. We then denote by $\mu^* := \arg \min_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)} H(\mu | \bar{\mu})$ the minimum entropy jointly calibrating model.

Strong duality for the VIX-constrained martingale Schrödinger problem

Theorem (G. 2020, cont'd)

4 Let $E \in \{L^1 B, CC_b\}$ and $u^* \in E$. The following assertions are equiv.:

- (i) $J_{\bar{\mu}}^{\text{ln}}(u^*) = \sup_{u \in E} J_{\bar{\mu}}^{\text{ln}}(u)$.
- (ii) The problem is finite, $u^* \in E_{\text{exp}}$, and

$$\frac{d\mu^*}{d\bar{\mu}} = Z_{\bar{\mu}}(u^*)^{-1} e^{(u_1^* \oplus u_V^* \oplus u_2^* + \Delta_S^{*(S)} + \Delta_L^{*(L)})(S_1, V, S_2)} \quad \bar{\mu}\text{-a.s.}$$

In this case, let $u^\dagger := (u_1^* - \ln Z_{\bar{\mu}}(u^*), u_V^*, u_2^*, \Delta_S^*, \Delta_L^*)$. Then $u^\dagger \in E$ and $u^\dagger$ satisfies the three equivalent assertions below.

5 Let $E \in \{L^1 B, CC_b\}$ and $u^\dagger \in E$. The following assertions are equivalent:

- (i) $J_{\bar{\mu}}^{\text{id}-1}(u^\dagger) = \sup_{u \in E} J_{\bar{\mu}}^{\text{id}-1}(u)$.
- (ii) For all $g \in \mathcal{G}$, $J_{\bar{\mu}}^g(u^\dagger) = \sup_{u \in E} J_{\bar{\mu}}^g(u)$.
- (iii) The problem is finite and

$$\frac{d\mu^*}{d\bar{\mu}} = e^{(u_1^\dagger \oplus u_V^\dagger \oplus u_2^\dagger + \Delta_S^{\dagger(S)} + \Delta_L^{\dagger(L)})(S_1, V, S_2)} \quad \bar{\mu}\text{-a.s.}$$

In this case, $Z_{\bar{\mu}}(u^\dagger) = 1$.

Strong duality for the VIX-constrained martingale Schrödinger problem

$$\begin{aligned} \frac{d\mu^*}{d\bar{\mu}} &= Z_{\bar{\mu}}(u^*)^{-1} e^{\left(u_1^* \oplus u_V^* \oplus u_2^* + \Delta_S^{*(S)} + \Delta_L^{*(L)}\right)(S_1, V, S_2)} \quad \bar{\mu}\text{-a.s.} \\ &= e^{\left(u_1^\dagger \oplus u_V^\dagger \oplus u_2^\dagger + \Delta_S^{\dagger(S)} + \Delta_L^{\dagger(L)}\right)(S_1, V, S_2)} \quad \bar{\mu}\text{-a.s.} \end{aligned}$$

- We call maximizers $(u_1^*, u_V^*, u_2^*, \Delta_S^*, \Delta_L^*)$ and $(u_1^\dagger, u_V^\dagger, u_2^\dagger, \Delta_S^\dagger, \Delta_L^\dagger)$ **Schrödinger potentials** (if they exist).
- We call the corresponding portfolios $\pi_{u^*} := u_1^* \oplus u_V^* \oplus u_2^* + \Delta_S^{*(S)} + \Delta_L^{*(L)}$ and $\pi_{u^\dagger} := u_1^\dagger \oplus u_V^\dagger \oplus u_2^\dagger + \Delta_S^{\dagger(S)} + \Delta_L^{\dagger(L)}$ **Schrödinger portfolios**.
- The Schrödinger portfolio is essentially unique: two Schrödinger portfolios π_{u^*} and $\pi_{u^\dagger}$ are $\bar{\mu}$ -a.e. equal up to an additive constant.
- We call $\pi_{u^\dagger}$ the **standard Schrödinger portfolio**.

Strong duality for the VIX-constrained martingale Schrödinger problem

Sketch of the proof of strong duality:

- 1 Prove strong duality for the classical Schrödinger problem (**marginal constraints only**, $E = L^1$ or C or C_b)

$$\inf_{\mu \in \Pi(\mu_1, \mu_V, \mu_2)} H(\mu | \bar{\mu}) = \sup_{(u_1, u_V, u_2) \in E} J_{\bar{\mu}}^{S, g}(u_1, u_V, u_2)$$

using Fenchel-Rockafellar convex duality theorem. Be careful: in general $\mathcal{M}(\mathcal{X}) \subsetneq C_b(\mathcal{X})^*$ when $\mathcal{X}$ is not compact!

- 2 Extend to the mixed Schrödinger-Monge-Kantorovich problem (or entropy-regularized optimal transport problem, $E = L^1$ or C)

$$\inf_{\mu \in \Pi(\mu_1, \mu_V, \mu_2)} \{H(\mu | \bar{\mu}) - \mathbb{E}^\mu[f(S_1, V, S_2)]\} = \sup_{(u_1, u_V, u_2) \in E} J_{\bar{\mu}, f}^{SMK, g}(u_1, u_V, u_2).$$

$$J_{\bar{\mu}}^{S, g}(u_1, u_V, u_2) := \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] - g\left(\mathbb{E}^{\bar{\mu}}\left[e^{(u_1 \oplus u_V \oplus u_2)(S_1, V, S_2)}\right]\right)$$

$$J_{\bar{\mu}, f}^{SMK, g}(u_1, u_V, u_2) := \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] - g\left(\mathbb{E}^{\bar{\mu}}\left[e^{(u_1 \oplus u_V \oplus u_2 + f)(S_1, V, S_2)}\right]\right)$$

Strong duality for the VIX-constrained martingale Schrödinger problem

Sketch of the proof:

- 1 Prove strong duality for the classical Schrödinger problem (marginal constraints only, $E = L^1$ or C or C_b)

$$\inf_{\mu \in \Pi(\mu_1, \mu_V, \mu_2)} H(\mu | \bar{\mu}) = \sup_{(u_1, u_V, u_2) \in E} J_{\bar{\mu}}^{S, g}(u_1, u_V, u_2)$$

using Fenchel-Rockafellar convex duality theorem. Be careful: in general $\mathcal{M}(\mathcal{X}) \subsetneq C_b(\mathcal{X})^*$ when $\mathcal{X}$ is not compact!

- 2 Extend to the mixed Schrödinger-Monge-Kantorovich problem (or entropy-regularized optimal transport problem, $E = L^1$ or C)

$$\inf_{\mu \in \Pi(\mu_1, \mu_V, \mu_2)} \{H(\mu | \bar{\mu}) - \mathbb{E}^\mu[f(S_1, V, S_2)]\} = \sup_{(u_1, u_V, u_2) \in E} J_{\bar{\mu}, f}^{SMK, g}(u_1, u_V, u_2).$$

$$J_{\bar{\mu}}^{S, g}(u_1, u_V, u_2) := \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] - g\left(\mathbb{E}^{\bar{\mu}}\left[e^{(u_1 \oplus u_V \oplus u_2)(S_1, V, S_2)}\right]\right)$$

$$J_{\bar{\mu}, f}^{SMK, g}(u_1, u_V, u_2) := \mathbb{E}^1[u_1(S_1)] + \mathbb{E}^V[u_V(V)] + \mathbb{E}^2[u_2(S_2)] - g\left(\mathbb{E}^{\bar{\mu}}\left[e^{(u_1 \oplus u_V \oplus u_2 + f)(S_1, V, S_2)}\right]\right)$$

Strong duality for the VIX-constrained martingale Schrödinger problem

Sketch of proof:

- 1 Prove strong duality for the classical Schrödinger problem

$$\inf_{\mu \in \Pi(\mu_1, \mu_V, \mu_2)} H(\mu | \bar{\mu}) = \sup_{(u_1, u_V, u_2) \in E} J_{\bar{\mu}}^{S, g}(u_1, u_V, u_2)$$

- 2 Extend to the mixed Schrödinger-Monge-Kantorovich problem

$$\inf_{\mu \in \Pi(\mu_1, \mu_V, \mu_2)} \{H(\mu | \bar{\mu}) - \mathbb{E}^\mu[f(S_1, V, S_2)]\} = \sup_{(u_1, u_V, u_2) \in E} J_{\bar{\mu}, f}^{SMK, g}(u_1, u_V, u_2).$$

- 3 Back to the dispersion-constrained martingale Schrödinger problem:

- 1 Dualize the martingality and dispersion constraints:

$$\inf_{\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)} H(\mu | \bar{\mu}) = \inf_{\mu \in \Pi(\mu_1, \mu_V, \mu_2)} \sup_{(\Delta_S, \Delta_L) \in C_b} \left\{ H(\mu | \bar{\mu}) - \mathbb{E}^\mu \left[\left(\Delta_S^{(S)} + \Delta_L^{(L)} \right) (S_1, V, \right. \right.$$

- 2 Use the weak compactness of $\Pi(\mu_1, \mu_V, \mu_2)$ and Sion's minimax theorem to swap inf and sup.
- 3 Apply the SMK strong duality.

Build a model in $\mathcal{P}(\mu_1, \mu_V, \mu_2)$

$$\mu^*(ds_1, dv, ds_2) = \bar{\mu}(ds_1, dv, ds_2) \frac{e^{u_1^*(s_1) + u_V^*(v) + u_2^*(s_2) + \Delta_S^{*(S)}(s_1, v, s_2) + \Delta_L^{*(L)}(s_1, v, s_2)}}{\mathbb{E}^{\bar{\mu}} \left[e^{u_1^*(S_1) + u_V^*(V) + u_2^*(S_2) + \Delta_S^{*(S)}(S_1, V, S_2) + \Delta_L^{*(L)}(S_1, V, S_2)} \right]}$$

- $J_{\bar{\mu}}^{\text{ln}}$ is invariant by translation of u_1 , u_V , and u_2 (c = cash position):

$$J_{\bar{\mu}}^{\text{ln}}(u_1 + c_1, u_V + c_V, u_2 + c_2, \Delta_S, \Delta_L) = J_{\bar{\mu}}^{\text{ln}}(u_1, u_V, u_2, \Delta_S, \Delta_L)$$

$\implies$ We always work with a normalized version of $u^* \in \mathcal{U}$ s.t. the normalizing factor

$$Z_{\bar{\mu}}(u^*) = 1 \quad \rightarrow u^\dagger, J_{\bar{\mu}}^{\text{id}-1}$$

- **By duality, the initial, difficult problem of minimizing over $\mu \in \mathcal{P}(\mu_1, \mu_V, \mu_2)$ (constrained) has been reduced to the simpler problem of maximizing the strictly concave function $J_{\bar{\mu}}^{\text{id}-1}$ over $u \in \mathcal{U}$ (unconstrained).** If it exists, the optimum $u^\dagger$ cancels the gradient of

$$J_{\bar{\mu}}^{\text{id}-1}: \frac{\partial J_{\bar{\mu}}^{\text{id}-1}}{\partial u_1(s_1)} = \frac{\partial J_{\bar{\mu}}^{\text{id}-1}}{\partial u_V(v)} = \frac{\partial J_{\bar{\mu}}^{\text{id}-1}}{\partial u_2(s_2)} = \frac{\partial J_{\bar{\mu}}^{\text{id}-1}}{\partial \Delta_S(s_1, v)} = \frac{\partial J_{\bar{\mu}}^{\text{id}-1}}{\partial \Delta_L(s_1, v)} = 0.$$

The Schrödinger equations (a.k.a. Schrödinger system)

$$\frac{\partial J_{\bar{\mu}}^{\text{ln}}}{\partial u_1(s_1)} = 0 : \quad \forall s_1 > 0, \quad u_1(s_1) = \Phi_1(s_1; u_V, u_2, \Delta_S, \Delta_L)$$

$$\frac{\partial J_{\bar{\mu}}^{\text{ln}}}{\partial u_V(v)} = 0 : \quad \forall v \geq 0, \quad u_V(v) = \Phi_V(v; u_1, u_2, \Delta_S, \Delta_L)$$

$$\frac{\partial J_{\bar{\mu}}^{\text{ln}}}{\partial u_2(s_2)} = 0 : \quad \forall s_2 > 0, \quad u_2(s_2) = \Phi_2(s_2; u_1, u_V, \Delta_S, \Delta_L)$$

$$\frac{\partial J_{\bar{\mu}}^{\text{ln}}}{\partial \Delta_S(s_1, v)} = 0 : \quad \forall s_1 > 0, \forall v \geq 0, \quad 0 = \Phi_{\Delta_S}(s_1, v; u_2, \Delta_S(s_1, v), \Delta_L(s_1, v))$$

$$\frac{\partial J_{\bar{\mu}}^{\text{ln}}}{\partial \Delta_L(s_1, v)} = 0 : \quad \forall s_1 > 0, \forall v \geq 0, \quad 0 = \Phi_{\Delta_L}(s_1, v; u_2, \Delta_S(s_1, v), \Delta_L(s_1, v))$$

- If Schrödinger potentials exist, we know that μ^* is of the form

$$\mu(ds_1, dv, ds_2) = \bar{\mu}(ds_1, dv, ds_2) e^{u_1(s_1) + u_V(v) + u_2(s_2) + \Delta_S^{(S)}(s_1, v, s_2) + \Delta_L^{(L)}(s_1, v, s_2)}.$$

- That $\mu^* \in \mathcal{P}(\mu_1, \mu_V, \mu_2)$ translates into the 5 above equations.
- The system of equations is solved using **Sinkhorn's algorithm**.

The Schrödinger equations (a.k.a. Schrödinger system)

$$\forall s_1 > 0, \quad u_1(s_1) = \Phi_1(s_1; u_V, u_2, \Delta_S, \Delta_L)$$

$$\forall v \geq 0, \quad u_V(v) = \Phi_V(v; u_1, u_2, \Delta_S, \Delta_L)$$

$$\forall s_2 > 0, \quad u_2(s_2) = \Phi_2(s_2; u_1, u_V, \Delta_S, \Delta_L)$$

$$\forall s_1 > 0, \forall v \geq 0, \quad 0 = \Phi_{\Delta_S}(s_1, v; \Delta_S(s_1, v), \Delta_L(s_1, v))$$

$$\forall s_1 > 0, \forall v \geq 0, \quad 0 = \Phi_{\Delta_L}(s_1, v; \Delta_S(s_1, v), \Delta_L(s_1, v))$$

$$\Phi_1(s_1; u_V, \Delta_S, \Delta_L) := \ln \mu_1(s_1) - \ln \left(\int \bar{\mu}(s_1, dv, ds_2) e^{u_V(v) + u_2(s_2) + \Delta_S^{(S)}(s_1, v, s_2) + \Delta_L^{(L)}(s_1, v, s_2)} \right)$$

$$\Phi_V(v; u_1, \Delta_S, \Delta_L) := \ln \mu_V(v) - \ln \left(\int \bar{\mu}(ds_1, v, ds_2) e^{u_1(s_1) + u_2(s_2) + \Delta_S^{(S)}(s_1, v, s_2) + \Delta_L^{(L)}(s_1, v, s_2)} \right)$$

$$\Phi_2(s_2; u_1, u_V, \Delta_S, \Delta_L) := \ln \mu_2(s_2) - \ln \left(\int \bar{\mu}(ds_1, dv, s_2) e^{u_1(s_1) + u_V(v) + \Delta_S^{(S)}(s_1, v, s_2) + \Delta_L^{(L)}(s_1, v, s_2)} \right)$$

$$\Phi_{\Delta_S}(s_1, v; u_2, \delta_S, \delta_L) := \int \bar{\mu}(s_1, v, ds_2) (s_2 - s_1) e^{u_2(s_2) + \delta_S(s_2 - s_1) + \delta_L \left(L \left(\frac{s_2}{s_1} \right) - v^2 \right)}$$

$$\Phi_{\Delta_L}(s_1, v; u_2, \delta_S, \delta_L) := \int \bar{\mu}(s_1, v, ds_2) \left(L \left(\frac{s_2}{s_1} \right) - v^2 \right) e^{u_2(s_2) + \delta_S(s_2 - s_1) + \delta_L \left(L \left(\frac{s_2}{s_1} \right) - v^2 \right)}.$$

Sinkhorn's algorithm

- Sinkhorn's algorithm (1967) is a coordinate ascent method which was first used in the context of optimal transport by Cuturi (2013). It performs **alternating projections**.
- In our context: **Fixed point method that alternates maximizations in the different directions (one per Lagrange multiplier) to approximate the maximizer $u^\dagger$** .
- Start from initial guess $u^{(0)} = (u_1^{(0)}, u_V^{(0)}, u_2^{(0)}, \Delta_S^{(0)}, \Delta_L^{(0)})$, recursively define $u^{(n+1)}$ knowing $u^{(n)}$ by

$$\forall s_1 > 0, \quad u_1^{(n+1)}(s_1) = \Phi_1(s_1; u_V^{(n)}, u_2^{(n)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall v \geq 0, \quad u_V^{(n+1)}(v) = \Phi_V(v; u_1^{(n+1)}, u_2^{(n)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall s_2 > 0, \quad u_2^{(n+1)}(s_2) = \Phi_2(s_2; u_1^{(n+1)}, u_V^{(n+1)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall s_1 > 0, \forall v \geq 0, \quad 0 = \Phi_{\Delta_S}(s_1, v; u_2^{(n+1)}, \Delta_S^{(n+1)}(s_1, v), \Delta_L^{(n)}(s_1, v))$$

$$\forall s_1 > 0, \forall v \geq 0, \quad 0 = \Phi_{\Delta_L}(s_1, v; u_2^{(n+1)}, \Delta_S^{(n+1)}(s_1, v), \Delta_L^{(n+1)}(s_1, v))$$

until convergence.

- Each of the above 5 lines corresponds to a Bregman projection in the space of measures.

Sinkhorn's algorithm

- Sinkhorn's algorithm (1967) is a coordinate ascent method which was first used in the context of optimal transport by Cuturi (2013). It performs **alternating projections**.
- In our context: **Fixed point method that alternates maximizations in the different directions (one per Lagrange multiplier) to approximate the maximizer $u^\dagger$** .
- Start from initial guess $u^{(0)} = (u_1^{(0)}, u_V^{(0)}, u_2^{(0)}, \Delta_S^{(0)}, \Delta_L^{(0)})$, recursively define $u^{(n+1)}$ knowing $u^{(n)}$ by

$$\forall s_1 > 0, \quad u_1^{(n+1)}(s_1) = \Phi_1(s_1; u_V^{(n)}, u_2^{(n)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall v \geq 0, \quad u_V^{(n+1)}(v) = \Phi_V(v; u_1^{(n+1)}, u_2^{(n)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall s_2 > 0, \quad u_2^{(n+1)}(s_2) = \Phi_2(s_2; u_1^{(n+1)}, u_V^{(n+1)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall s_1 > 0, \forall v \geq 0, \quad 0 = \Phi_{\Delta_S}(s_1, v; u_2^{(n+1)}, \Delta_S^{(n+1)}(s_1, v), \Delta_L^{(n)}(s_1, v))$$

$$\forall s_1 > 0, \forall v \geq 0, \quad 0 = \Phi_{\Delta_L}(s_1, v; u_2^{(n+1)}, \Delta_S^{(n+1)}(s_1, v), \Delta_L^{(n+1)}(s_1, v))$$

until convergence.

- Each of the above 5 lines corresponds to a Bregman projection in the space of measures.

Sinkhorn's algorithm

- Sinkhorn's algorithm (1967) is a coordinate ascent method which was first used in the context of optimal transport by Cuturi (2013). It performs **alternating projections**.
- In our context: **Fixed point method that alternates maximizations in the different directions (one per Lagrange multiplier) to approximate the maximizer $u^\dagger$.**
- Start from initial guess $u^{(0)} = (u_1^{(0)}, u_V^{(0)}, u_2^{(0)}, \Delta_S^{(0)}, \Delta_L^{(0)})$, recursively define $u^{(n+1)}$ knowing $u^{(n)}$ by

$$\forall s_1 > 0, \quad u_1^{(n+1)}(s_1) = \Phi_1(s_1; u_V^{(n)}, u_2^{(n)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall v \geq 0, \quad u_V^{(n+1)}(v) = \Phi_V(v; u_1^{(n+1)}, u_2^{(n)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall s_2 > 0, \quad u_2^{(n+1)}(s_2) = \Phi_2(s_2; u_1^{(n+1)}, u_V^{(n+1)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall s_1 > 0, \forall v \geq 0, \quad 0 = \Phi_{\Delta_S}(s_1, v; u_2^{(n+1)}, \Delta_S^{(n+1)}(s_1, v), \Delta_L^{(n)}(s_1, v))$$

$$\forall s_1 > 0, \forall v \geq 0, \quad 0 = \Phi_{\Delta_L}(s_1, v; u_2^{(n+1)}, \Delta_S^{(n+1)}(s_1, v), \Delta_L^{(n+1)}(s_1, v))$$

until convergence.

- Each of the above 5 lines corresponds to a Bregman projection in the space of measures.

Sinkhorn's algorithm

- Sinkhorn's algorithm (1967) is a coordinate ascent method which was first used in the context of optimal transport by Cuturi (2013). It performs **alternating projections**.
- In our context: **Fixed point method that alternates maximizations in the different directions (one per Lagrange multiplier) to approximate the maximizer $u^\dagger$** .
- Start from initial guess $u^{(0)} = (u_1^{(0)}, u_V^{(0)}, u_2^{(0)}, \Delta_S^{(0)}, \Delta_L^{(0)})$, recursively define $u^{(n+1)}$ knowing $u^{(n)}$ by

$$\forall s_1 > 0, \quad u_1^{(n+1)}(s_1) = \Phi_1(s_1; u_V^{(n)}, u_2^{(n)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall v \geq 0, \quad u_V^{(n+1)}(v) = \Phi_V(v; u_1^{(n+1)}, u_2^{(n)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall s_2 > 0, \quad u_2^{(n+1)}(s_2) = \Phi_2(s_2; u_1^{(n+1)}, u_V^{(n+1)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall s_1 > 0, \forall v \geq 0, \quad 0 = \Phi_{\Delta_S}(s_1, v; u_2^{(n+1)}, \Delta_S^{(n+1)}(s_1, v), \Delta_L^{(n)}(s_1, v))$$

$$\forall s_1 > 0, \forall v \geq 0, \quad 0 = \Phi_{\Delta_L}(s_1, v; u_2^{(n+1)}, \Delta_S^{(n+1)}(s_1, v), \Delta_L^{(n+1)}(s_1, v))$$

until convergence.

- Each of the above 5 lines corresponds to a Bregman projection in the space of measures.

Sinkhorn's algorithm

- Sinkhorn's algorithm (1967) is a coordinate ascent method which was first used in the context of optimal transport by Cuturi (2013). It performs **alternating projections**.
- In our context: **Fixed point method that alternates maximizations in the different directions (one per Lagrange multiplier) to approximate the maximizer $u^\dagger$.**
- Start from initial guess $u^{(0)} = (u_1^{(0)}, u_V^{(0)}, u_2^{(0)}, \Delta_S^{(0)}, \Delta_L^{(0)})$, recursively define $u^{(n+1)}$ knowing $u^{(n)}$ by

$$\forall s_1 > 0, \quad u_1^{(n+1)}(s_1) = \Phi_1(s_1; u_V^{(n)}, u_2^{(n)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall v \geq 0, \quad u_V^{(n+1)}(v) = \Phi_V(v; u_1^{(n+1)}, u_2^{(n)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall s_2 > 0, \quad u_2^{(n+1)}(s_2) = \Phi_2(s_2; u_1^{(n+1)}, u_V^{(n+1)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall s_1 > 0, \forall v \geq 0, \quad 0 = \Phi_{\Delta_S}(s_1, v; u_2^{(n+1)}, \Delta_S^{(n+1)}(s_1, v), \Delta_L^{(n)}(s_1, v))$$

$$\forall s_1 > 0, \forall v \geq 0, \quad 0 = \Phi_{\Delta_L}(s_1, v; u_2^{(n+1)}, \Delta_S^{(n+1)}(s_1, v), \Delta_L^{(n+1)}(s_1, v))$$

until convergence.

- Each of the above 5 lines corresponds to a Bregman projection in the space of measures.

Sinkhorn's algorithm

- Sinkhorn's algorithm (1967) is a coordinate ascent method which was first used in the context of optimal transport by Cuturi (2013). It performs **alternating projections**.
- In our context: **Fixed point method that alternates maximizations in the different directions (one per Lagrange multiplier) to approximate the maximizer $u^\dagger$** .
- Start from initial guess $u^{(0)} = (u_1^{(0)}, u_V^{(0)}, u_2^{(0)}, \Delta_S^{(0)}, \Delta_L^{(0)})$, recursively define $u^{(n+1)}$ knowing $u^{(n)}$ by

$$\forall s_1 > 0, \quad u_1^{(n+1)}(s_1) = \Phi_1(s_1; u_V^{(n)}, u_2^{(n)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall v \geq 0, \quad u_V^{(n+1)}(v) = \Phi_V(v; u_1^{(n+1)}, u_2^{(n)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall s_2 > 0, \quad u_2^{(n+1)}(s_2) = \Phi_2(s_2; u_1^{(n+1)}, u_V^{(n+1)}, \Delta_S^{(n)}, \Delta_L^{(n)})$$

$$\forall s_1 > 0, \forall v \geq 0, \quad 0 = \Phi_{\Delta_S}(s_1, v; u_2^{(n+1)}, \Delta_S^{(n+1)}(s_1, v), \Delta_L^{(n)}(s_1, v))$$

$$\forall s_1 > 0, \forall v \geq 0, \quad 0 = \Phi_{\Delta_L}(s_1, v; u_2^{(n+1)}, \Delta_S^{(n+1)}(s_1, v), \Delta_L^{(n+1)}(s_1, v))$$

until convergence.

- Each of the above 5 lines corresponds to a **Bregman projection** in the space of measures.

Sinkhorn's algorithm

- If the algorithm diverges, then $P_{\bar{\mu}} = +\infty$, so $D_{\bar{\mu}} = +\infty$, i.e.,

$$\mathcal{P}(\mu_1, \mu_V, \mu_2) \cap \{\mu \in \mathcal{M}_1 | H(\mu | \bar{\mu}) < +\infty\} = \emptyset.$$

- In practice, when $\bar{\mu}$ has full support, this is a sign that **there likely exists a joint SPX/VIX arbitrage**.
- One should directly check if $\mathcal{P}(\mu_1, \mu_V, \mu_2) = \emptyset$ (linear program).
- We have never experienced this situation in our numerical tests, which covered both low and high volatility regimes.

Numerical experiments

Implementation details

- Choice of $\bar{\mu}$:
 - $S_1 \sim \mu_1$ and $V \sim \mu_V$ independent;
 - Conditional on (S_1, V) , S_2 lognormal with mean S_1 and variance V .Under $\bar{\mu}$, $S_2 \not\sim \mu_2$.
- Instead of abstract payoffs u_1, u_V, u_2 , we work with market strikes and market prices of vanilla options on S_1 , V , and S_2 .
- Canceling the gradient of $J_{\bar{\mu}}^{\text{ln}} \rightarrow$ system of equations solved using Sinkhorn's algorithm.
- Enough accuracy is typically reached after ≈ 100 iterations.

Implementation details

Practically, we consider market strikes $\mathcal{K} := (\mathcal{K}_1, \mathcal{K}_V, \mathcal{K}_2)$ and market prices (C_K^1, C_K^V, C_K^2) of vanilla options on S_1 , V , and S_2 , and we build the model

$$\mu_{\mathcal{K}}^*(ds_1, dv, ds_2) = \bar{\mu}(ds_1, dv, ds_2) e^{c^\dagger + \Delta_S^{0\dagger} s_1 + \Delta_V^{0\dagger} v + \sum_{K \in \mathcal{K}_1} a_K^{1\dagger} (s_1 - K)_+ + \sum_{K \in \mathcal{K}_V} a_K^{V\dagger} (v - K)_+ + \sum_{K \in \mathcal{K}_2} a_K^{2\dagger} (s_2 - K)_+ + \Delta_S^{\dagger(S)}(s_1, v, s_2) + \Delta_L^{\dagger(L)}(s_1, v, s_2)}$$

where $\theta^\dagger := (c^\dagger, \Delta_S^{0\dagger}, \Delta_V^{0\dagger}, a^{1\dagger}, a^{V\dagger}, a^{2\dagger}, \Delta_S^\dagger, \Delta_L^\dagger)$ maximizes

$$J_{\bar{\mu}, \mathcal{K}}^{\text{id}-1}(\theta) := c + \Delta_S^0 S_0 + \Delta_V^0 F_V + \sum_{K \in \mathcal{K}_1} a_K^1 C_K^1 + \sum_{K \in \mathcal{K}_V} a_K^V C_K^V + \sum_{K \in \mathcal{K}_2} a_K^2 C_K^2$$

$$- \mathbb{E}^{\bar{\mu}} \left[e^{c + \Delta_S^0 S_1 + \Delta_V^0 V + \sum_{K \in \mathcal{K}_1} a_K^1 (S_1 - K)_+ + \sum_{K \in \mathcal{K}_V} a_K^V (V - K)_+ + \sum_{K \in \mathcal{K}_2} a_K^2 (S_2 - K)_+ + \Delta_S^{(S)}(\dots) + \Delta_L^{(L)}(\dots)} \right]$$

over the set Θ of portfolios $\theta := (c, \Delta_S^0, \Delta_V^0, a^1, a^V, a^2, \Delta_S, \Delta_L)$ such that $c, \Delta_S^0, \Delta_V^0 \in \mathbb{R}$, $a^1 \in \mathbb{R}^{\mathcal{K}_1}$, $a^V \in \mathbb{R}^{\mathcal{K}_V}$, $a^2 \in \mathbb{R}^{\mathcal{K}_2}$, and $\Delta_S, \Delta_L : \mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ are measurable functions of (s_1, v) .

Implementation details

- This corresponds to solving the entropy minimization problem

$$P_{\bar{\mu}, \mathcal{K}} := \inf_{\mu \in \mathcal{P}(\mathcal{K})} H(\mu | \bar{\mu}) = \sup_{\theta \in \Theta} J_{\bar{\mu}, \mathcal{K}}^{\text{id}-1}(\theta) =: D_{\bar{\mu}, \mathcal{K}}$$

where $\mathcal{P}(\mathcal{K})$ denotes the set of probability measures μ on $\mathbb{R}_{>0} \times \mathbb{R}_{\geq 0} \times \mathbb{R}_{>0}$ such that

$$\begin{aligned} \mathbb{E}^\mu[S_1] &= S_0, \quad \mathbb{E}^\mu[V] = F_V, \quad \forall K \in \mathcal{K}_1, \quad \mathbb{E}^\mu[(S_1 - K)_+] = C_K^1, \\ \forall K \in \mathcal{K}_V, \quad \mathbb{E}^\mu[(V - K)_+] &= C_K^V, \quad \forall K \in \mathcal{K}_2, \quad \mathbb{E}^\mu[(S_2 - K)_+] = C_K^2, \\ \mathbb{E}^\mu[S_2 | S_1, V] &= S_1, \quad \mathbb{E}^\mu \left[L \left(\frac{S_2}{S_1} \right) \middle| S_1, V \right] = V^2. \end{aligned}$$

- One can directly check that model $\mu_{\mathcal{K}}^*$ is an arbitrage-free model that jointly calibrates the prices of SPX futures, options, VIX future, and VIX options. Indeed, if $J_{\bar{\mu}, \mathcal{K}}^{\text{id}-1}$ reaches its maximum at $\theta^\dagger$, then $\theta^\dagger$ is solution to $\frac{\partial J_{\bar{\mu}, \mathcal{K}}^{\text{id}-1}}{\partial \theta_i}(\theta) = 0$:

Implementation details

$$J_{\bar{\mu}, \mathcal{K}}^{\text{id}-1}(\theta) := c + \Delta_S^0 S_0 + \Delta_V^0 F_V + \sum_{K \in \mathcal{K}_1} a_K^1 C_K^1 + \sum_{K \in \mathcal{K}_V} a_K^V C_K^V + \sum_{K \in \mathcal{K}_2} a_K^2 C_K^2$$

$$- \mathbb{E}^{\bar{\mu}} \left[e^{c + \Delta_S^0 S_1 + \Delta_V^0 V + \sum_{\mathcal{K}_1} a_K^1 (S_1 - K)_+ + \sum_{\mathcal{K}_V} a_K^V (V - K)_+ + \sum_{\mathcal{K}_2} a_K^2 (S_2 - K)_+ + \Delta_S^{(S)}(\dots) + \Delta_L^{(L)}(\dots)} \right]$$

$$\frac{\partial J_{\bar{\mu}, \mathcal{K}}^{\text{id}-1}}{\partial c} = 0 : \mathbb{E}^{\bar{\mu}} \left[\frac{d\mu_{\mathcal{K}}^*}{d\bar{\mu}} \right] = 1$$

$$\frac{\partial J_{\bar{\mu}, \mathcal{K}}^{\text{id}-1}}{\partial \Delta_S^0} = 0 : \mathbb{E}^{\bar{\mu}} \left[S_1 \frac{d\mu_{\mathcal{K}}^*}{d\bar{\mu}} \right] = S_0$$

$$\frac{\partial J_{\bar{\mu}, \mathcal{K}}^{\text{id}-1}}{\partial \Delta_V^0} = 0 : \mathbb{E}^{\bar{\mu}} \left[V \frac{d\mu_{\mathcal{K}}^*}{d\bar{\mu}} \right] = F_V$$

$$\frac{\partial J_{\bar{\mu}, \mathcal{K}}^{\text{id}-1}}{\partial a_K^1} = 0 : \mathbb{E}^{\bar{\mu}} \left[(S_1 - K)_+ \frac{d\mu_{\mathcal{K}}^*}{d\bar{\mu}} \right] = C_K^1$$

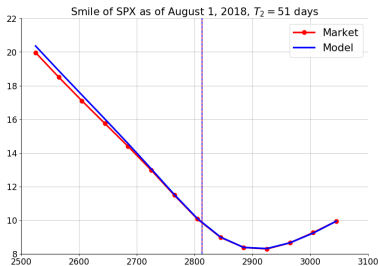
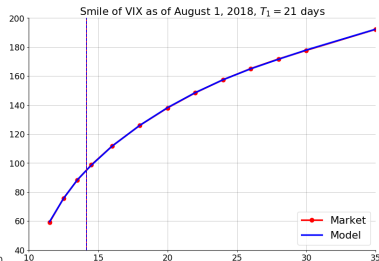
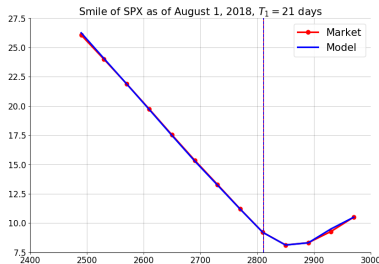
$$\frac{\partial J_{\bar{\mu}, \mathcal{K}}^{\text{id}-1}}{\partial a_K^V} = 0 : \mathbb{E}^{\bar{\mu}} \left[(V - K)_+ \frac{d\mu_{\mathcal{K}}^*}{d\bar{\mu}} \right] = C_K^V$$

$$\frac{\partial J_{\bar{\mu}, \mathcal{K}}^{\text{id}-1}}{\partial a_K^2} = 0 : \mathbb{E}^{\bar{\mu}} \left[(S_2 - K)_+ \frac{d\mu_{\mathcal{K}}^*}{d\bar{\mu}} \right] = C_K^2$$

$$\frac{\partial J_{\bar{\mu}, \mathcal{K}}^{\text{id}-1}}{\partial \Delta_S(s_1, v)} = 0 : \mathbb{E}^{\bar{\mu}} \left[(S_2 - S_1) \frac{d\mu_{\mathcal{K}}^*}{d\bar{\mu}} \middle| S_1 = s_1, V = v \right] = 0, \quad \forall s_1 \geq 0, v > 0$$

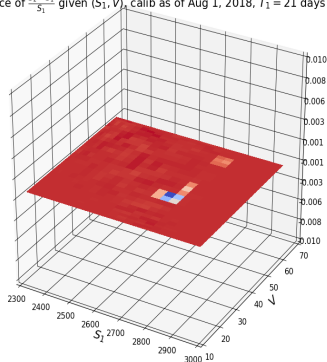
$$\frac{\partial J_{\bar{\mu}, \mathcal{K}}^{\text{id}-1}}{\partial \Delta_L(s_1, v)} = 0 : \mathbb{E}^{\bar{\mu}} \left[\left(L \left(\frac{S_2}{S_1} \right) - V^2 \right) \frac{d\mu_{\mathcal{K}}^*}{d\bar{\mu}} \middle| S_1 = s_1, V = v \right] = 0, \quad \forall s_1 \geq 0, v > 0$$

August 1, 2018, $T_1 = 21$ days

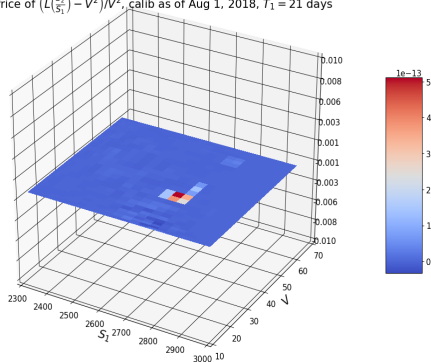


August 1, 2018, $T_1 = 21$ days

Price of $\frac{S_2 - S_1}{S_1}$ given (S_1, V) , calib as of Aug 1, 2018, $T_1 = 21$ days

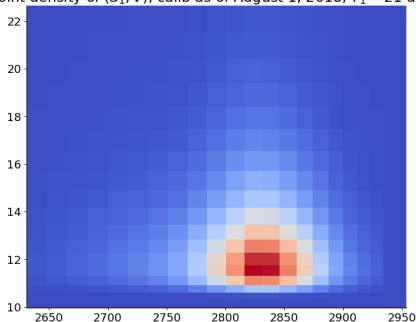


Price of $(L(\frac{S_2}{S_1}) - V^2)/V^2$, calib as of Aug 1, 2018, $T_1 = 21$ days



August 1, 2018, $T_1 = 21$ days

Joint density of (S_1, V) , calib as of August 1, 2018, $T_1 = 21$ days



Local VIX, calibration as of August 1, 2018, $T_1 = 21$ days

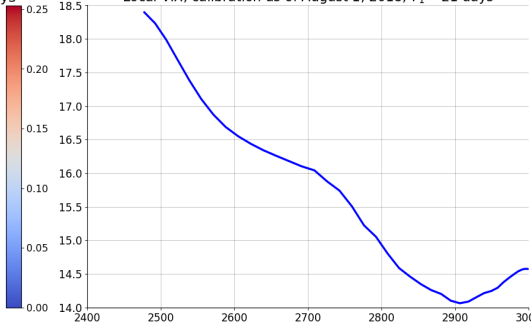


Figure: Joint distribution of (S_1, V) and local VIX function $VIX_{\text{loc}}(s_1)$

$$VIX_{\text{loc}}^2(S_1) := \mathbb{E}^{\mu^*} [V^2 | S_1]$$

August 1, 2018, $T_1 = 21$ days

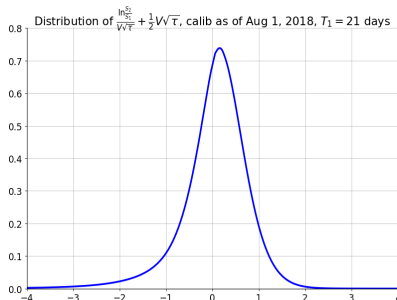
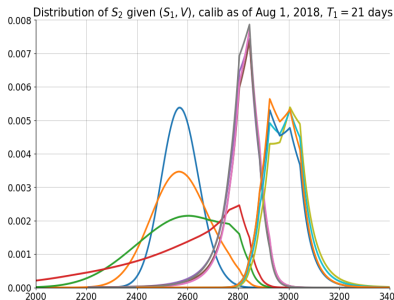
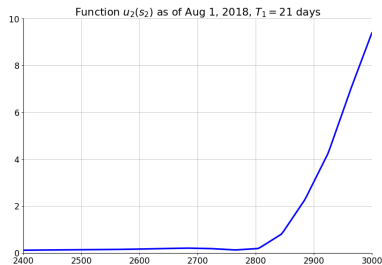
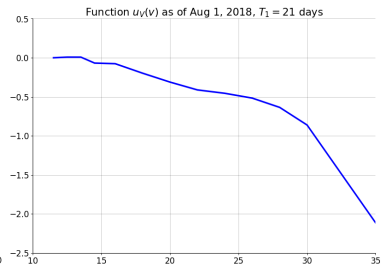
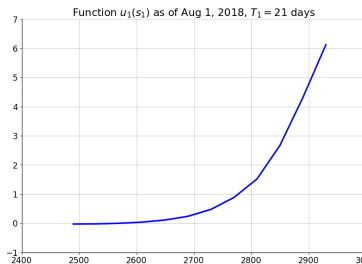


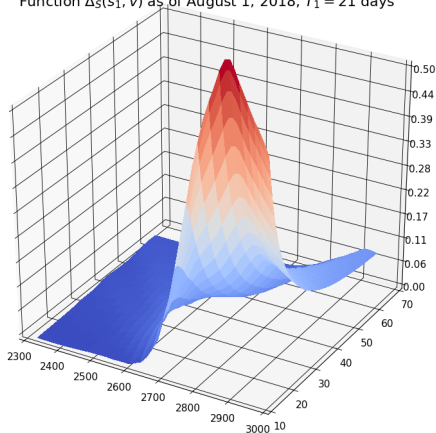
Figure: Conditional distribution of S_2 given (s_1, v) under μ^* for different vales of (s_1, v) : $s_1 \in \{2571, 2808, 3000\}$, $v \in \{10.10, 15.30, 23.20, 35.72\}\%$, and distribution of the normalized return $R := \frac{\ln(S_2/S_1)}{V\sqrt{T}} + \frac{1}{2}V\sqrt{T}$

August 1, 2018, $T_1 = 21$ days



August 1, 2018, $T_1 = 21$ days

Function $\Delta_S(s_1, v)$ as of August 1, 2018, $T_1 = 21$ days



Function $\Delta_L(s_1, v)$ as of August 1, 2018, $T_1 = 21$ days

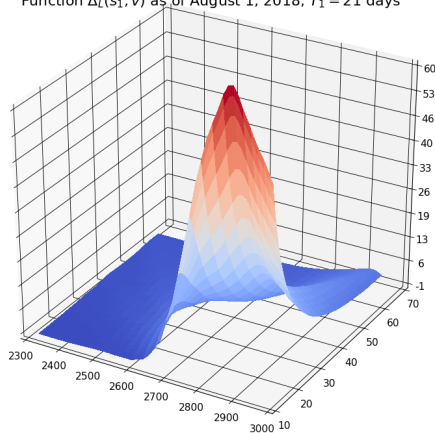
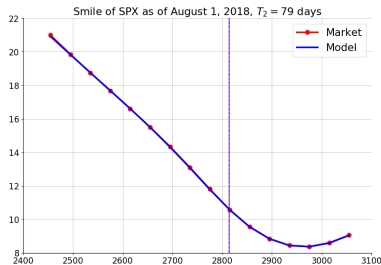
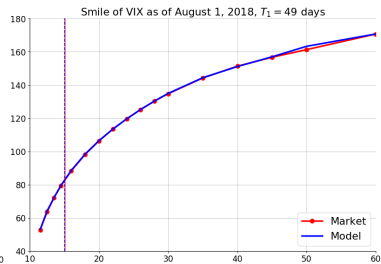
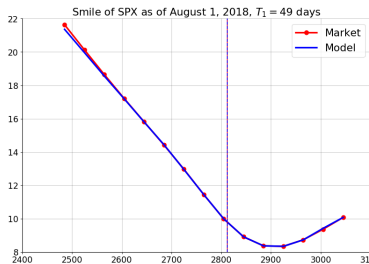


Figure: Optimal functions $\Delta_S^*(s_1, v)$ and $\Delta_L^*(s_1, v)$ for (s_1, v) in the quadrature grid

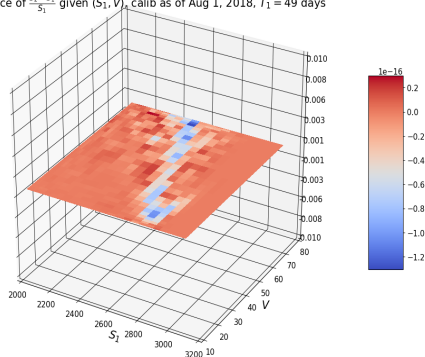


August 1, 2018, $T_1 = 49$ days

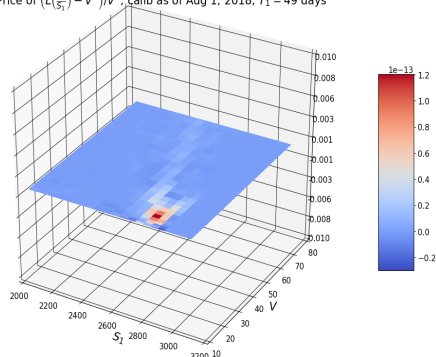


August 1, 2018, $T_1 = 49$ days

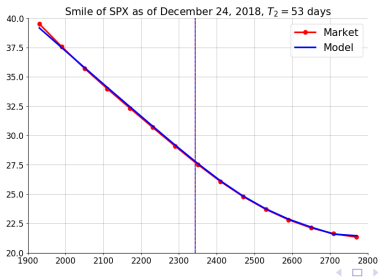
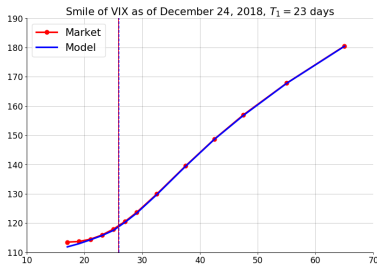
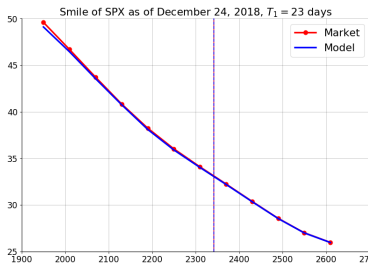
Price of $\frac{S_2 - S_1}{S_1}$ given (S_1, V) , calib as of Aug 1, 2018, $T_1 = 49$ days



Price of $(L(\frac{S_2}{S_1}) - V^2)/V^2$, calib as of Aug 1, 2018, $T_1 = 49$ days

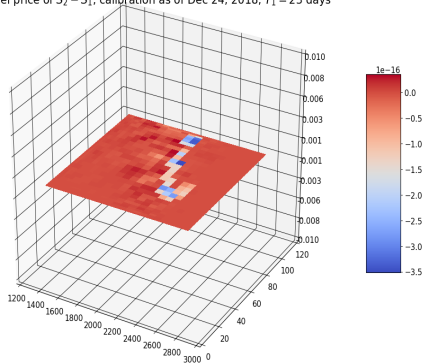


December 24, 2018, $T_1 = 23$ days: large VIX, $F_V \approx 26\%$

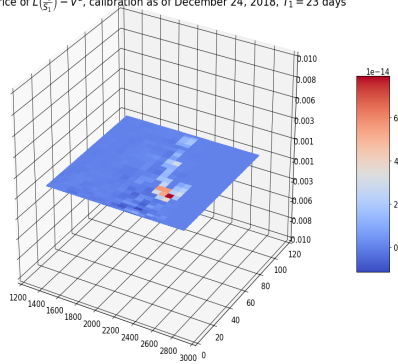


December 24, 2018, $T_1 = 23$ days

Model price of $S_2 - S_1$, calibration as of Dec 24, 2018, $T_1 = 23$ days

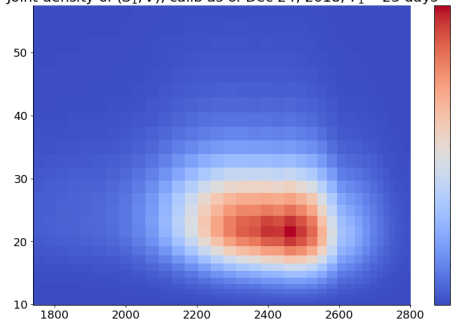


Model price of $L\left(\frac{S_2}{S_1}\right) - V^2$, calibration as of December 24, 2018, $T_1 = 23$ days



December 24, 2018, $T_1 = 23$ days

Joint density of (S_1, V) , calib as of Dec 24, 2018, $T_1 = 23$ days



Local VIX, calibration as of December 24, 2018, $T_1 = 23$ days

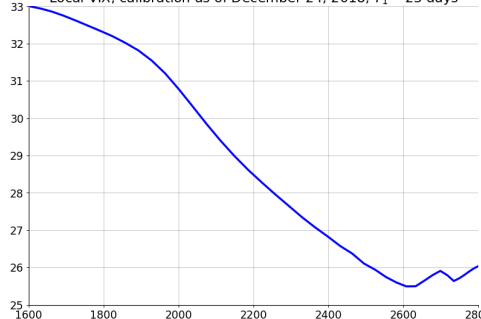


Figure: Joint distribution of (S_1, V) and local VIX function $VIX_{loc}(s_1)$

December 24, 2018, $T_1 = 23$ days

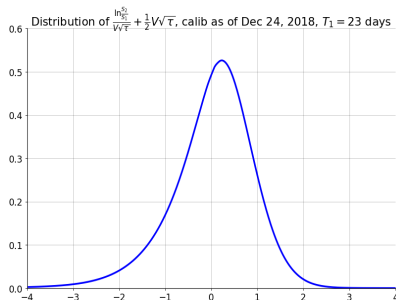
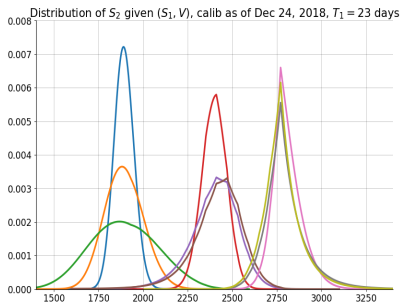


Figure: Conditional distribution of S_2 given (s_1, v) under μ^* for different vales of (s_1, v) : $s_1 \in \{2571, 2808, 3000\}$, $v \in \{10.10, 15.30, 23.20, 35.72\}\%$, and distribution of the normalized return $R := \frac{\ln(S_2/S_1)}{V\sqrt{\tau}} + \frac{1}{2}V\sqrt{\tau}$

December 24, 2018, $T_1 = 23$ days

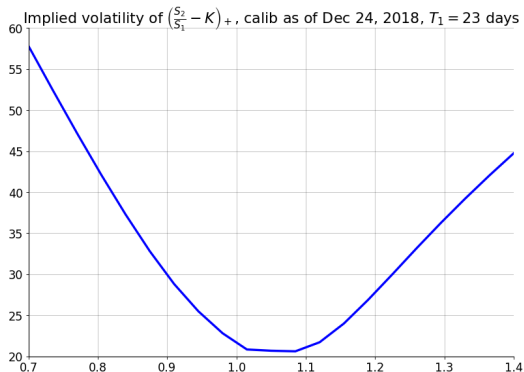
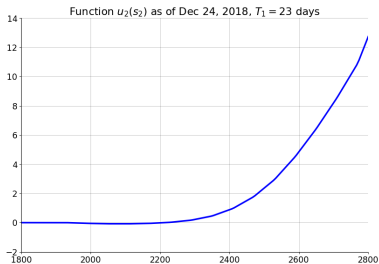
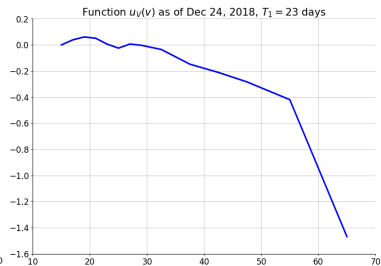
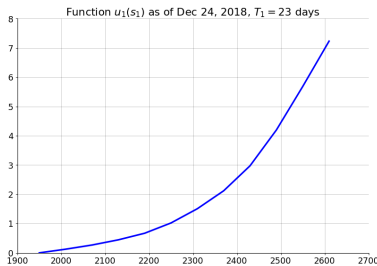


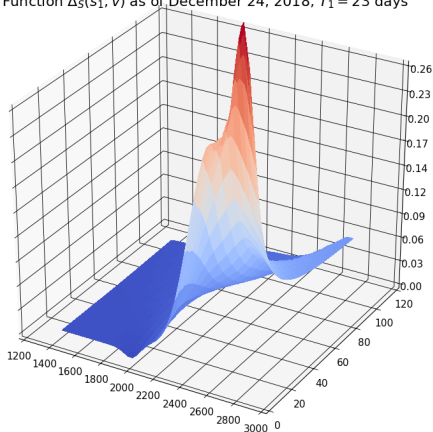
Figure: Smile of forward starting call options $(S_2/S_1 - K)_+$

December 24, 2018, $T_1 = 23$ days



December 24, 2018, $T_1 = 23$ days

Function $\Delta_S(s_1, v)$ as of December 24, 2018, $T_1 = 23$ days



Function $\Delta_L(s_1, v)$ as of December 24, 2018, $T_1 = 23$ days

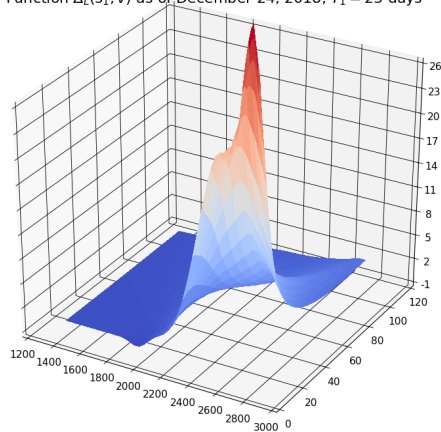


Figure: Optimal functions $\Delta_S^*(s_1, v)$ and $\Delta_L^*(s_1, v)$ for (s_1, v) in the quadrature grid



Maturity issue

- SPX options maturity $T'_1 = T_1 + 2$ days or $T_1 - 5$ days.
- Rigorous treatment: introduce S'_1 representing the value of the SPX index at time T'_1 . If T'_1 is two days after T_1 , we consider the primal portfolios

$$u_1(s'_1) + u_V(v) + u_2(s_2) + \Delta_S(s_1, v)(s'_1 - s_1) + \Delta'_S(s_1, v, s'_1)(s_2 - s'_1) + \Delta_L(s_1, v) \left(L \left(\frac{s_2}{s_1} \right) - v \right)$$

and the dual risk-neutral probability measures $V \sim \mu_V, S'_1 \sim \mu_1, S_2 \sim \mu_2$,

$$\mathbb{E}^\mu [S'_1 | S_1, V] = S_1, \quad \mathbb{E}^\mu [S_2 | S_1, V, S'_1] = S'_1, \quad \mathbb{E}^\mu \left[L \left(\frac{S_2}{S_1} \right) \middle| S_1, V \right] = V^2.$$

- If T'_1 is five days before T_1 , the primal portfolios are

$$u_1(s'_1) + u_V(v) + u_2(s_2) + \Delta'_S(s'_1)(s_1 - s'_1) + \Delta_S(s'_1, s_1, v)(s_2 - s_1) + \Delta_L(s'_1, s_1, v) \left(L \left(\frac{s_2}{s_1} \right) - v \right)$$

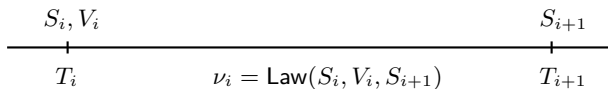
and the dual risk-neutral probability measures $V \sim \mu_V, S'_1 \sim \mu_1, S_2 \sim \mu_2$,

$$\mathbb{E}^\mu [S_1 | S'_1] = S'_1, \quad \mathbb{E}^\mu [S_2 | S'_1, S_1, V] = S_1, \quad \mathbb{E}^\mu \left[L \left(\frac{S_2}{S_1} \right) \middle| S'_1, S_1, V \right] = V^2.$$

- Approx: assume SPX options mature exactly at T_1 ; maturity interpolation of SPX data.

Extension to the multi-maturity case

Extension to the multi-maturity case



- Assume for simplicity that monthly SPX options and VIX futures maturities T_i perfectly coincide: $T_{i+1} - T_i = \tau$ for all $i \geq 1$.
- For each i we build a jointly calibrating model ν_i using the Sinkhorn algorithm.
- Glue the models together:** Build a calibrated model on $(S_i, V_i)_{i \geq 1}$ as follows: $(S_1, V_1, S_2) \sim \nu_1$; recursively define the distribution of (V_{i+1}, S_{i+2}) given $(S_1, V_1, S_2, V_2, \dots, S_i, V_i, S_{i+1})$ as the conditional distribution of (V_{i+1}, S_{i+2}) given S_{i+1} under ν_{i+1} .
- The resulting model ν is arbitrage-free, consistent, and calibrated to all the SPX and VIX monthly market smiles μ_{S_i} and μ_{V_i} : for all $i \geq 1$,

$$S_i \sim \mu_{S_i}, \quad V_i \sim \mu_{V_i}, \quad \mathbb{E}^\nu [S_{i+1} | (S_j, V_j)_{1 \leq j \leq i}] = S_i, \quad \mathbb{E}^\nu \left[L \left(\frac{S_{i+1}}{S_i} \right) \middle| (S_j, V_j)_{1 \leq j \leq i} \right] = V_i^2.$$